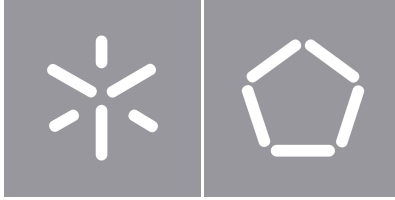


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BEAM: A Web Playground for personalized training of Attention



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Master's Dissertation in Informatics Engineering

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Statement of Integrity

I hereby declare having conducted this academic work with integrity.

I confirm that I have not used plagiarism or any form of undue use of information or falsification of results along the process leading to its elaboration.

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University of Minho, Braga, november 2025

José Miguel Ferreira Barbosa

Abstract

Attention is a fundamental cognitive ability that plays a vital role in daily life and is increasingly a focus of concern in modern society. With the growing demands of the digital age, attention-related challenges have become more prominent, affecting individuals of all ages and professions. Research has shown that attention can be developed through structured exercises, making attention training a rapidly expanding field.

Consequently, the main goal of the Master's Project here reported was to create a platform named **BEAM**, designed to assist therapists in training the attention of their patients through targeted exercises and techniques.

To achieve this, a comprehensive analysis of attention and its subdomains was conducted, along with a review of established attention training methods. Based on this foundation, the **BEAM** platform was implemented as a web-based tool that allows therapists to personalize, assign, and monitor cognitive tasks. The system supports a growing collection of interactive exercises, each targeting specific cognitive abilities. Therapists can create customized configurations for each task, assign them to patients, and track progress over time using detailed performance metrics and visual dashboards.

The platform includes over 10 configurable tasks, many of which were inspired by clinically validated exercises and literature-reviewed models. A recommendation system allows professionals to guide training plans, while a dedicated statistics module supports detailed analysis. The platform was developed in close collaboration with therapists and tested with young users in preliminary sessions. Initial feedback from professionals emphasized **BEAM's** potential as a flexible and intuitive tool for guided cognitive training. This Master's Dissertation documents the full development of the platform, including its technical implementation, the early design of the platform and the current look of **BEAM**. It also discusses future improvements, clinical evaluation and testing.

Keywords Attention, Attention Training, Cognition, Web-based educational tools

Resumo

A atenção é uma capacidade cognitiva fundamental que desempenha um papel vital na vida quotidiana e constitui uma preocupação na sociedade moderna. Com as crescentes exigências da era digital, os desafios relacionados com a atenção tornaram-se mais proeminentes, afetando indivíduos de todas as idades e profissões. A investigação tem demonstrado que a atenção pode ser desenvolvida através de exercícios estruturados, o que torna o treino de atenção um campo em rápida expansão.

Consequentemente, o principal objetivo deste projeto de mestrado foi criar uma plataforma denominada **BEAM**, concebida para apoiar terapeutas no treino da atenção dos seus pacientes através de exercícios e técnicas direcionadas.

Para atingir este objetivo, foi realizada uma análise abrangente da atenção e dos seus subdomínios, bem como uma revisão dos métodos consolidados de treino cognitivo. Com base nesta fundação, foi implementada a plataforma **BEAM** como uma ferramenta web que permite aos terapeutas atribuir, personalizar e monitorizar as tarefas. O sistema suporta uma coleção crescente de exercícios interativos, cada um focado em diferentes competências cognitivas. Os terapeutas podem criar configurações personalizadas para cada tarefa, atribuí-las aos pacientes e acompanhar a sua evolução ao longo do tempo, recorrendo a métricas de desempenho detalhadas e painéis visuais.

A plataforma integra mais de 10 tarefas configuráveis, muitas das quais inspiradas em exercícios clinicamente validados ou encontrados na literatura. Um sistema de recomendações permite orientar os planos de treino, enquanto um módulo dedicado a estatísticas oferece suporte a análises detalhadas. O desenvolvimento da plataforma contou com a colaboração próxima de terapeutas e foi testado, em fases preliminares, com utilizadores mais jovens. Os primeiros feedbacks dos profissionais destacaram o potencial da **BEAM** como uma ferramenta flexível e intuitiva para treino cognitivo orientado.

Esta Dissertação de Mestrado documenta todo o processo de desenvolvimento da plataforma, incluindo a sua implementação técnica, o design inicial e o aspeto atual da **BEAM**. São ainda discutidas melhorias futuras, avaliação clínica e testes.

Palavras-chave Atenção, Treino de Atenção, Cognição, Ferramentas Educacionais Web

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Acronyms

ACC Anterior Cingulate Cortex.

ADHD Attention Deficit Hyperactivity Disorder.

AST Attention State Training.

AT Attention Training.

ATT Attention Training Technique.

AX-CPT AX Continuous Performance Task.

BEAM Brain Engagement and Attention Management.

CEN Central Executive Network.

dACC dorsal Anterior Cingulate Cortex.

DLPFC Dorsolateral Prefrontal Cortex.

DMN Default Mode Network.

EEG Electroencephalography.

ERPs Event-related potentials.

fMRI Functional magnetic resonance imaging.

FPN FrontoParietal Network.

GABA Gamma-Aminobutyric Acid.

IA Internet Addiction.

L-FPN lateral-Frontoparietal Network.

PFC Prefrontal Cortex.

PPC Posterior Parietal Cortex.

PTSD Post-traumatic Stress Disorder.

SART Sustained Attention to Response Task.

SMA Social Media Addiction.

SN Salience Network.

SPA Smartphone Addiction.

TVA Theory of Visual Attention.

VR Virtual Reality.

Chapter 1

Introduction

From an early age, parents and educators emphasize the importance of focusing on specific areas of knowledge and tasks that require attention. Whether in the process of learning new concepts or completing important activities, the ability to maintain focus plays a crucial role in both personal and academic development. Without it, achieving long-term success and reaching meaningful goals becomes significantly more challenging. However, the concept of attention itself is often overlooked. While phrases such as “pay attention” or “stay focused” are commonly used, the precise meaning of attention is rarely examined. At its core, attention refers to the concentration of awareness on a particular phenomenon while filtering out other stimuli.

The importance of attention across all stages of life makes it even more concerning that many people, both young and old, are increasingly struggling to maintain it, which causes significant impacts on daily life, including academic, professional, and personal settings. This challenge may sometimes be associated with **Attention Deficit Hyperactivity Disorder (ADHD)** or similar conditions, but it often represents a skill that can be cultivated and improved. The COVID-19 pandemic has amplified this issue, affecting people of all ages. Children, in particular, experienced a decline in attention span and self-regulation due to extended periods of remote learning and isolation. These children, now entering adolescence, face additional challenges during a critical stage of cognitive and emotional development.

The idea of training attention is not new; it has been discussed for over a century in different contexts. Today, it is increasingly recognized as a vital skill that can be improved through various methods. Techniques like mindfulness meditation, cognitive exercises, and even certain physical activities have been shown to enhance the ability to concentrate, manage distractions, and sustain focus for longer periods. As the understanding of the brain and cognitive functions evolves, the significance of **Attention Training (AT)** becomes clearer, underscoring its essential role in achieving personal and academic success.

Building on these insights, this project aims to develop an accessible web-based tool that provides targeted exercises to improve attention. Studies have demonstrated that attention can be trained through personal-

ized methods, drawing from both medical and educational practices. By creating a customizable platform, this initiative aims to enhance attention capacity in individuals, benefiting not only the general population but also those in school and clinical environments, thus addressing a growing concern in today's society.

1.1 Objectives

The main goal of this Master's Project is to develop a web-based tool designed to improve users' attention through personalized training exercises. The focus is on addressing the growing challenges of sustaining attention in both young and adult populations, which may stem from conditions like **ADHD** or simply reflect a skill that requires further development. To achieve this, several key objectives have been outlined:

- Review investigations on the underlying causes of increasing attention difficulties in various populations, focusing on both clinical and non-clinical contexts.
- Examine existing methods and solutions used to improve attention, evaluating their effectiveness and identifying best practices.
- Select appropriate diagnostic and training tests, determining which types of attention and test formats are most relevant for the target audience.
- Develop the web platform, ensuring it is user-friendly, adaptable, and scalable.
- Conduct experiments with users to measure the effectiveness of the training exercises in improving attention span and focus.
- Facilitate future customization, allowing the platform to integrate new or different types of tests with relative ease.

The expected outcome is a functional, adaptable web platform that not only enhances users' attention but also provides valuable insights into **AT**, with potential applications in general, educational, and clinical settings.

1.2 Research Hypothesis

The purpose of this Master's project is to demonstrate, through the development of the web platform, that it is possible to effectively train attention using targeted challenges and exercises that are both engaging and effective.

1.3 Research Method

To achieve the objectives outlined in this project, a structured methodology will be followed, focusing on the development, implementation, and testing of a web-based tool for **AT**. The methodology will be divided into the following steps:

- Conduct an extensive literature review about: attention, **Attention Training** methods, and related technologies.
- Evaluate different techniques used to train attention, such as cognitive exercises and diagnostic tests, with a focus on their applicability to a digital platform.
- Design the web platform.
- Incorporate diagnostic tests and training exercises in the web platform, selecting those most relevant for the target audience.
- Conduct experiments with a group of users to assess the platform, the tests and the exercises.
- Analyze the results of the experiments, refine and optimize the system.

This methodology ensures a thorough and iterative approach, combining theoretical research with practical implementation and testing. Each phase is designed to support the development of a robust and effective **AT** tool that addresses the growing challenges of sustaining focus in both young and adult populations.

1.4 Document structure

This document is divided into seven chapters. Chapter 1 introduces the topic of the Master's project reported in this Master's dissertation, defines the research hypothesis, and outlines the methodology, providing the groundwork for the study.

Chapter 2 explores the project's domain, focusing on attention as a cognitive process, its types, associated disorders, and challenges in the modern world. It examines how attention is shaped by brain function and investigates **AT** methods and exercises relevant to the application. This state-of-the-art review offers essential insights to support the platform's design.

Chapter 3 presents the system proposal, detailing the application's design through diagrams of key components and their interactions. It includes the domain definition, use cases, requirements, mockups, system architecture, and the technologies selected for implementation.

Chapter 4 documents the technical development of the platform, describing the backend and frontend architecture, the use of Phoenix LiveView, the integration of configurable cognitive tasks, and the implementation of the system's features.

Chapter 5 serves as a visual map of the platform, presenting the main screens and explaining their structure and functionality. It details each stage of user interaction, from registration and login to task execution and result analysis. Additionally, this chapter describes all the cognitive exercises implemented, illustrating how each one fits into the platform's flow and contributes to the training objectives.

Chapter 6 outlines the testing phase of the platform, including the collaboration with therapists and early trials. It presents the feedback obtained from healthcare professionals and discusses the observed engagement and results of young users during cognitive tasks.

Chapter 7 provides a summary of the information stated in this document and offers a reflection on the work accomplished so far and the developments planned for the future.

Chapter 2

State of the Art

Attention is the mental process that enables us to focus on specific stimuli while filtering out distractions, allowing us to make sense of our complex surroundings. It has to do with the immediate experience of an individual, being a state of current awareness (Araujo, 2022). This chapter will provide a detailed literature review covering the brain, particularly how attention is influenced by brain function. Following this, attention will be explored in depth, including its types, disorders, and the new challenges it faces in the modern world. Finally, the chapter will conclude by examining **Attention Training (AT)**.

2.1 The Brain

The Brain is the organ that commands the Human Body. In this Master's dissertation, attention and how one can improve and train it is the main focus of our study, but before “deep diving” into attention as a whole, understanding how attention is shaped by brain function is essential. To gain a clearer perspective on this theme various topics will be explored, for example, the neural substrates of attention, neural network theories, and role of neurotransmitters.

2.1.1 Neurotransmitters

One crucial factor in someone's ability to focus is the brain's neurotransmitters and neuromodulators (subset of the transmitters). These chemical messengers play distinct but interconnected roles in managing attention-related tasks. The most common neurotransmitters, such as **Glutamate** and **Gamma-Aminobutyric Acid (GABA)**, help communication in neural networks. Glutamate mediates excitatory signals (+), driving neural activity, while **GABA** mediates inhibitory signals (-). This balance between excitation and inhibition is essential for the precise functioning of the neural networks involved in attention (Thiele and Bellgrove, 2018; Khetrapal, 2021; Herlenius and Lagercrantz, 2010).

In addition to this foundational system, various regions in the brainstem release neuromodulators that

adjust how neural networks process these excitatory and inhibitory signals. Neuromodulators, including *dopamine*, *norepinephrine*, *serotonin*, and *acetylcholine*, act on postsynaptic neurons to alter their responsiveness. Unlike glutamate and **GABA**, which act locally and directly, neuromodulators are distributed diffusely across the brain. They influence broad areas, such as the frontal eye fields, posterior cingulate cortex and **Dorsolateral Prefrontal Cortex (DLPFC)** (Thiele and Bellgrove, 2018).

Dopamine (or DA) plays a pivotal role in motivation, attention, and reward processing. Known as the “feel-good” hormone, it not only provides a sense of pleasure, but also the motivation to engage in pleasurable activities. This neurotransmitter helps regulate how to perceive and respond to stimuli, influencing ones’ ability to stay focused on tasks. Dopamine is central to the brain’s reward system, which evolved to reward essential survival behaviors, such as eating, drinking, competing, and reproducing. Insufficient dopamine levels can hurt concentration, organization, and time management, leading to challenges in sustaining attention on tasks and also break an individual’s reward processing system (Cleveland Clinic, 2022b; Thiele and Bellgrove, 2018).

In a level of how everything works inside the brain, dopamine’s effects are mediated through five receptor subtypes: D1, D2, D3, D4, and D5. These receptors are often categorized into two families based on their structural and functional similarities, the D1 family (D1 and D5) and the D2 family (D2, D3, and D4). D1 receptors are located mainly in the striatum and the cerebral cortex while D2 receptors, are primarily located in the striatum, the substantia nigra, and the hypothalamus (Celtarys, 2021; Thiele and Bellgrove, 2018).

Dopamine’s influence on attention and working memory is particularly pronounced in areas like the **DLPFC** and the frontal eye fields. D1 receptors enhance spatial attention by improving input and recurrent connection efficacy, which sharpens attentional focus. In contrast, D2 receptors modulate motor-choice activity, supporting decision-making processes during attention-demanding tasks (Noudoost and Moore, 2011). Similarly, for working memory D1 receptor activity in the **DLPFC** is crucial for maintaining task-relevant information, as it strengthens spatial tuning by reducing noise from non-preferred neural activity. D2 receptors, on the other hand, primarily influence motor-related outputs, which are critical for implementing decisions derived from working memory (Thiele and Bellgrove, 2018).

Norepinephrine (or NA) affects alertness and arousal. The noradrenergic system, originating in the locus coeruleus, influences the cortex and was initially thought to have nonspecific projections and functions, primarily regulating arousal across sleep-wake states. It helps the brain stay alert and responsive to stimuli, which is essential for maintaining focus. It has been shown to alter network activity in attention and working memory through arousal activation of the **FrontoParietal Network (FPN)**, which is critical

for sustaining attention and alertness (Esterman and Rothlein, 2019; Thiele and Bellgrove, 2018).

Similarly to DA, NA actions in regulating attention follow an inverted U shape, with both low and very high activity correlating with poor cognitive performance while moderate levels optimize task performance (Thiele and Bellgrove, 2018). This is developed in more detail in the section about Sustained Attention.

Norepinephrine exerts its effects through adrenergic receptors, of which the more relevant for attention are $\alpha 1$ and $\alpha 2$. Each one plays a distinct role in modulating neural activity and cognitive processes. The $\alpha 1$ receptors primarily excite neurons in the sensory cortex, but have the opposite effect in the **Pre-frontal Cortex (PFC)**, therefore facilitating bottom-up processing and enhancing responses to external stimuli. Conversely, $\alpha 2$ receptors inhibit neuronal activity, being crucial in top-down control by reducing distractibility and improving working memory, particularly in the sensory cortex, but increase neuronal firing in the **PFC**, thereby playing a vital role. This balance highlights how receptor-specific actions enable NA to fine-tune both sensory processing and cognitive control (Thiele and Bellgrove, 2018).

The locus coeruleus operates in two modes: phasic and tonic. Phasic activity, characterized by brief bursts of NA in response to task-relevant stimuli, enhances focus and performance. Tonic activity, marked by a higher baseline release of NA, supports exploratory behavior and vigilance but can increase distractibility and reduce task-specific focus. The cognitive effects of NA are highly dose-dependent. Moderate levels of NA via $\alpha 2$ receptor activation enhance working memory and attentional performance, while excessive NA levels, activating $\alpha 1$ receptors, impair these functions. This dynamic LC-NA system allows for adaptive regulation of cortical network activity, balancing task engagement with the need to respond flexibly to environmental demands (Thiele and Bellgrove, 2018).

Serotonin (or 5HT) impacts mood, impulse regulation, and overall cognitive function, often referred to as the “body’s natural feel-good chemical”. It contributes to managing emotional responses and maintaining steady focus. Beyond mood, serotonin influences learning, memory, and affection, while also playing roles in regulating body temperature, sleep, sexual behavior, and hunger. Fluctuations in serotonin levels can disrupt emotional stability and focus, potentially leading to increased distractibility and impulsivity (Cleveland Clinic, 2022c; Thiele and Bellgrove, 2018).

The serotonergic system plays an indirect but significant role in top-down attention and spatial working memory. It’s role in attention is somewhat nuanced, largely shaped by receptor subtypes and regional specificity. For instance, 5-HT_{2A} receptor activation enhances spatial tuning in **DLPFC** neurons, improving processing for preferred spatial targets. However, excessive signaling, such as that induced by 5-HT_{1A/2A} agonists, can impair attentional tracking by reducing the brain’s ability to suppress distracting stimuli. These findings highlight the dual role of serotonin in modulating both focus and distractibility,

depending on receptor engagement and signaling intensity. Furthermore, serotonergic modulation of attention and impulsivity is highly context dependent. While systemic 5-HT_{2A} activation can impair attention and heighten impulsivity, localized antagonism in the medial prefrontal cortex reduces impulsivity without sacrificing accuracy (Thiele and Bellgrove, 2018).

Serotonin also plays a pivotal role in cognitive flexibility and reversal learning. By influencing the orbitofrontal cortex and **DLPFC**, serotonin supports the inhibition of previously reinforced responses, facilitating new stimulus-reward associations and enabling adaptive shifts in behavior. High 5-HT levels enhance cognitive flexibility, promoting exploratory behaviors that are essential for adapting to changing environments, but may increase distractibility. Conversely, lower 5-HT levels favor exploitation and task-focused behaviors. Mechanistically, this balance is achieved through serotonergic regulation of neural noise and activity thresholds, particularly in prefrontal circuits. The serotonergic system, therefore, acts as a critical mediator of behavioral adaptability, balancing attentional focus with the flexibility to explore alternative strategies (Thiele and Bellgrove, 2018).

Acetylcholine (or ACh) governs memory, learning, motivation and attention. Synthesized from choline, acetylcholine is the chief neurotransmitter of the parasympathetic nervous system and has a significant impact on muscle function, sleep, cognitive processes and mental performance. ACh is also released in the brain, not only within each region, originating from local interneurons, for instance in the striatum and accumbens, but also at the distance, originating from projecting neurons of the cholinergic nuclei, namely in the brainstem (pedunculopontine and laterodorsal tegmental areas), the medial habenula and the basal forebrain complex (including the medial septum) (Picciotto et al., 2012). Changes in acetylcholine levels can affect short-term memory, processing speed, and recall, potentially causing learning difficulties and symptoms like mental fog (Cleveland Clinic, 2022a; Thiele and Bellgrove, 2018).

Diving deeper into its role in the brain, the basal forebrain cholinergic system is a primary source of ACh for cortical regions. While earlier theories questioned its precision in modulating top-down attentional control, recent evidence supports its role in spatially and temporally specific processes. This evidence points to ACh's dual functionality: tonic activity that is associated with global state changes and general network tuning and phasic release that is linked to reinforcement learning, reward signaling, and cue detection in attention tasks (Thiele and Bellgrove, 2018).

Acetylcholine, with localized ACh release via NMDA receptor activation and glutamate, stabilizes neuronal firing rates and reduces variability, important for efficient neural dynamics. By balancing excitatory and inhibitory inputs, ACh minimizes distractions and ensures reliable encoding of task-relevant information. This stabilizing effect, described as the formation of attractor dynamics, allows the brain to maintain focus

and resist cognitive drift over extended periods. Moreover, acetylcholine plays a key role in desynchronizing neural oscillatory activity, particularly in attention-demanding states. For example, ACh reduces noise correlations between neurons, improving population coding and neural decoding efficiency. This neuro-modulator plays a vital role in fine-tuning the brain's neural circuits for optimal attention, task performance, and overall cognitive stability (Thiele and Bellgrove, 2018).

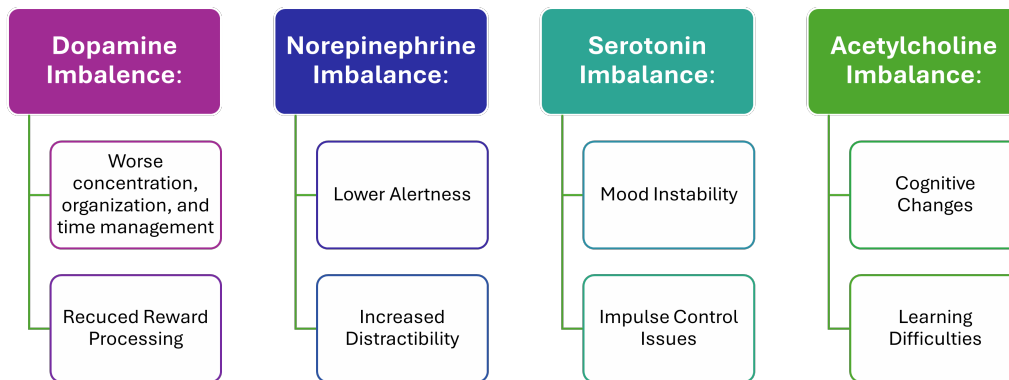


Figure 1: Issues caused by Neurotransmitter Fluctuations

Maintaining healthy neurotransmitter levels is essential not only for attention and focus but for overall brain health. Ensuring balance in these neurotransmitters can help prevent the attention and focus issues discussed above (seen in Figure 1).

2.1.2 The Triple Network Model

It's widely accepted that the brain's networks are adaptive and hierarchically organized, allowing rapid reorganization based on environmental stimuli, like responding to threats or complex problems. Functional connectivity within these networks relies on structural pathways but also evolves with individual experiences, affecting behaviors and cognitive traits, including fluid intelligence and personality (Schimmelpfennig et al., 2023).

The *triple network model* is a foundational framework in neuroscience that describes how three primary brain networks, the **Default Mode Network (DMN)**, **FrontoParietal Network (FPN)**, and **Salience Network (SN)**, inter-

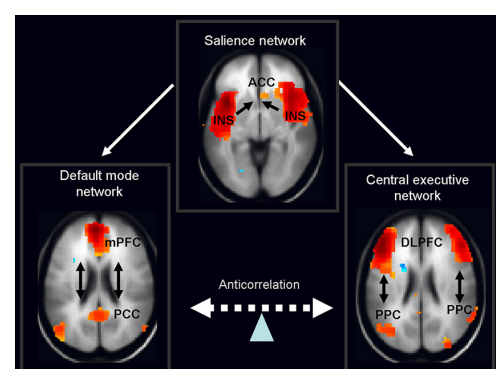


Figure 2: Connectivity between the networks in the Triple Network Model (from Wikipedia)

act dynamically to support core cognitive functions and mental health (Figure 2). This model highlights the flexibility of the brain, which is not structured as isolated modules but as interdependent, adaptable networks capable of reconfiguring in response to environmental demands. The connectivity within and between these networks enables the brain to shift seamlessly between internal and external tasks, regulate attention, and engage in self-referential thought. Studies in functional connectivity reveal that interactions among these networks are critical to mental resilience, while dysregulation often correlate with psychiatric symptoms, which really puts in perspective the role of these “canonical networks” in psychological well-being. This model explains how deviations in functional organization or network communication may contribute to psychopathology. Each network is essential in their own way (Schimmelpfennig et al., 2023).

- **Default Mode Network (DMN):** Active during rest and introspective tasks, it supports memory, empathy, and self-referential thinking.
- **FrontoParietal Network (FPN):** Engaged in goal-directed tasks, it facilitates attention, decision-making, and executive control.
- **Salience Network (SN):** Acts as a switch, identifying relevant stimuli to prioritize attention between internal reflection (**DMN**) and external tasks (**FPN**).

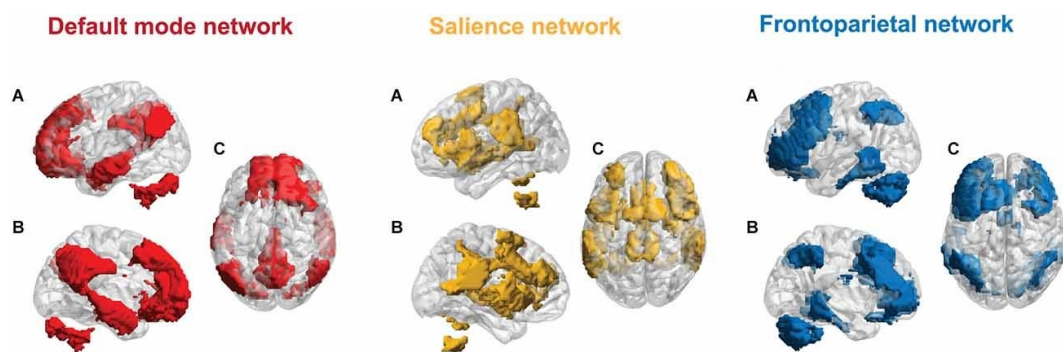


Figure 3: Three canonical networks that constitute the triple network model (from Schimmelpfennig et al. (2023)). A - Lateral view B - Sagittal view C - Superior view

When the brain is not actively engaged in a task, it falls into one of a number of resting states. The most common of these involves the **DMN** (Carter et al., 2019). The **DMN** is a collection of interconnected brain regions that becomes more active during periods of rest or when attention is directed away from external stimuli. It is involved in internally focused thought processes, such as self-reflection, mind-wandering, recalling past experiences, and envisioning future events. When an individual is not actively engaged with

external tasks or stimuli, the **DMN** defaults to these internal processes. This network, which includes key regions like the medial prefrontal cortex and posterior cingulate cortex, is typically suppressed when focused attention is required, allowing for more effective concentration on external demands. The **DMN**, therefore, represents the brain's natural resting state, where attention shifts from the outer world to inner thoughts and self-referential processes. Some say that the mind-wandering state is the default state of the human brain because of this fact (Esterman and Rothlein, 2019; Menon, 2023; Carter et al., 2019).

From the perspective of attention, the **DMN** has a direct impact on cognitive processes. The network plays a critical role in generating an ongoing internal narrative, integrating episodic memory, language, and semantic memory to maintain a cohesive sense of self. This internal narrative influences the brain's allocation of attention, as moments of mind-wandering or introspection are often associated with lapses in external focus. The **DMN**'s activity can lead to inefficient suppression of internal thoughts when attention is required for an external task, resulting in distractions or shifts in focus. However, this mind-wandering state is also thought to play a role in cognitive processes such as memory consolidation and creativity, highlighting a complex relationship between the **DMN** and attention (Menon, 2023).

Additionally, the **DMN**'s role is linked to periods of more stable brain behavior, suggesting that it may contribute to both memory retrieval and consolidation processes (Kucyi et al., 2016). Although mind-wandering is typically associated with a lack of focus on the external environment, it serves essential functions in cognitive processing. This dynamic highlights that attention is not simply a matter of sustained focus on external tasks but is also shaped by the interaction between task-related networks and the internally focused processes, many of which governed by the **DMN**.

The **FPN**, also known as the **Central Executive Network (CEN)** or **lateral-Frontoparietal Network (L-FPN)**, is composed primarily of the rostral lateral and **DLPFC** particularly the middle frontal gyrus, and the anterior inferior parietal lobule. Additional areas include the middle cingulate gyrus, dorsal precuneus, posterior inferior temporal lobe, dorsomedial thalamus, and the head of the caudate nucleus. This network's unique structure places it at the center of cognitive control, where it plays an essential role in goal-oriented behavior and cognitive flexibility.

Functionally, the **FPN** operates as a flexible "hub" within the brain, facilitating rapid and accurate cognitive responses by coordinating various brain networks in a distributed manner. Its connectivity is not limited to its main Triple Network Model counterparts, the **DMN** and the **SN**, but extends to other regions across the brain, allowing it to support a wide range of high-level cognitive functions. This extensive integration with other networks has been shown to correlate positively with fluid intelligence and superior cognitive abilities, as the **FPN**'s connectivity underpins adaptive problem-solving and real-time adjustments to complex

mental demands (Marek and Dosenbach, 2018).

Evidence from studies demonstrates that the **FPN** adjusts its connectivity patterns more swiftly and flexibly than any other network, including other control networks, when tasked with complex, novel instructions. These dynamic shifts are not arbitrary but appear to be context-specific, suggesting that the **FPN** not only adapts to external demands but also anticipates and coordinates effectively with other brain networks. This rapid adaptability makes the **FPN** integral to mental flexibility, providing the brain with a functional backbone for communication and coordination across various cognitive tasks (Marek and Dosenbach, 2018).

The final network, and the “moderator” of the Triple network model, is the **SN**. The primary components can be seen in Figure 4, with them being the anterior insula (in orange) and the **dorsal Anterior Cingulate Cortex (dACC)** (in green). This network also includes additional nodes such as the amygdala, hypothalamus, ventral striatum, thalamus, and specific brainstem nuclei, making it one of the most distributed yet cohesive networks in the brain (Crawford, 2022; Schimmelpfennig et al., 2023).

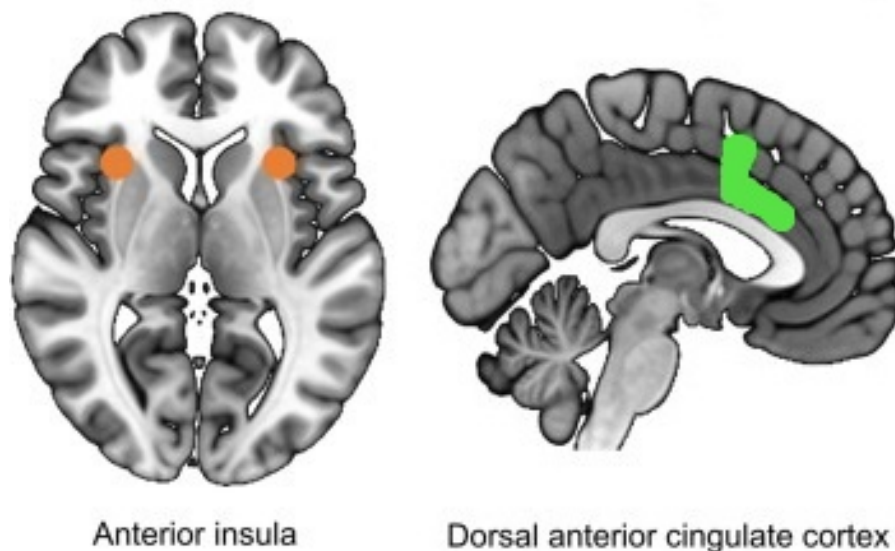


Figure 4: Primary Components of the Salience Network (Adapted from Kim et al. (2023)).

These regions work in unison to fulfill the **SN**’s role of scanning and evaluating sensory and cognitive information, prioritizing relevant stimuli for further processing. The anterior insula, for instance, integrates bodily signals with external inputs, while the **dACC** filters these inputs to amplify those that require additional cortical analysis, making the **SN** essential for guiding focus, processing reward signals, and modulating responses to pain and emotion. Within the brain’s triple-network model, the **SN** has a unique responsibility: to seamlessly toggle between the **DMN**, and the **FPN**. This switching function is central

to human adaptive cognition, as the **DMN** and **FPN** are rarely active at the same time in healthy brains. When an important event or stimulus appears, the **SN** quickly assesses its relevance, activating either the **DMN** or the **FPN** depending on what response is needed. The **SN** “moderates traffic” ensuring the brain focuses on the most pertinent information without distraction, by continuously aligning neural resources with situational demands (Crawford, 2022; Schimmelpfennig et al., 2023).

Beyond its role in moderating network activity, the **SN** is also critical in linking emotional responses with reward, motivation, and pain processing. As the brain’s primary assessor of relevance, the **SN** fine-tunes attention allocation, determining which signals receive heightened awareness based on their potential impact on well-being and motivation. In essence, the **SN** is crucial for aligning internal states with external actions, maintaining harmony between introspective processes and task-oriented behavior, all while remaining highly responsive to emotional and motivational demands (Crawford, 2022; Schimmelpfennig et al., 2023).

As mentioned, the networks in the Triple Network Model all play vital roles in managing cognitive control, emotional regulation, and attentional processes. So it should come as no surprise that disruptions within or between these networks are linked to a range of mental health conditions, for example when they affect the networks’ connectivity and functional dynamics. Such disruptions are implicated in conditions including depression, anxiety, **Post-traumatic Stress Disorder (PTSD)**, schizophrenia, and bipolar disorder, where each condition can present unique connectivity patterns and impairments across the model’s networks (Crawford, 2022; Schimmelpfennig et al., 2023; Marek and Dosenbach, 2018).

In schizophrenia, abnormalities within **SN** and **FPN** highlight significant deficits in cognitive and emotional processing. Schizophrenia’s “aberrant salience” model illustrates how **SN** dysfunction can cause the individual to assign undue importance to irrelevant stimuli, resulting in symptoms like hallucinations and delusions. At the same time, lower activity and reduced connectivity in the **FPN**, responsible for adaptive cognitive control, lead to cognitive inflexibility, further impairing one’s ability to manage thoughts and behaviors in context (Marek and Dosenbach, 2018; Schimmelpfennig et al., 2023).

Bipolar disorder also presents distinct connectivity disruptions across the Triple Network Model, with variances in network behavior depending on the phase of the disorder. During euthymic (stable mood) phases, individuals with this disorder can show hyperconnectivity within the **FPN**, combined with reduced connectivity between the **SN** and the **DMN** and **FPN**. This pattern points to impaired integration between these networks, possibly due to neurotransmitter signaling irregularities across different BD stages, which can impact both emotional and cognitive regulation (Schimmelpfennig et al., 2023).

Overall, disruptions in the Triple Network Model, whether primary (directly affecting cognitive control in

conditions like schizophrenia) or secondary (as in anxiety or depression, where cognitive control deficits may exacerbate symptoms), underscore the critical role of these networks in maintaining mental health and highlight the need to balance connectivity among them for optimal brain function.

2.1.3 Neural Substrates of Attention

A neural substrate refers to the biological foundation or physical structure that underlies neural processes in the brain. In other words, a neural substrate is a term used in neuroscience to indicate the part of the central nervous system (brain and spinal cord, connected by the brainstem) that is behind certain behaviors, cognitive process, or psychological state. The scientific community has good ideas on which parts of the brain influence attention, but the way this happens is still unclear (Wein, 2015; Mayer et al., 2007; Johnson and Zatorre, 2006).

While various parts of the brain impact attention, the two regions with the most significant influence are the **Prefrontal Cortex (PFC)** and the **Posterior Parietal Cortex (PPC)**. These areas play central roles in modulating attention, with the **PFC** contributing to higher-order executive functions and the **PPC** involved in spatial and sensory integration. Together, they form a foundation for attention control processes. They can be seen in Figure 5 (Wein, 2015). They are also the main components of the **FPN** discussed above.

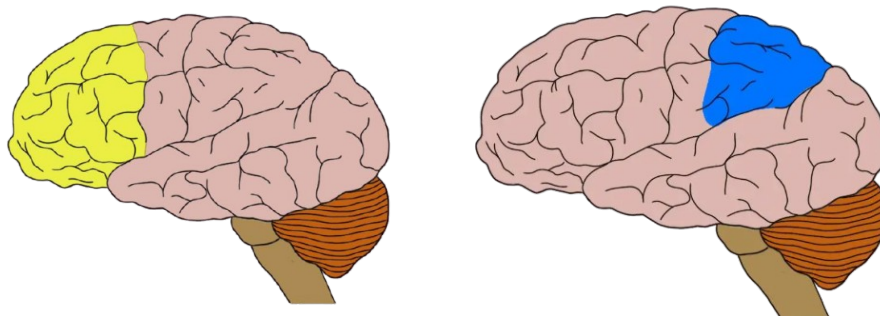


Figure 5: Prefrontal Cortex (Left) Posterior Parietal Cortex (Right) (both from Neuroscientifically Challenged)

The **PPC** is pivotal in attentional control, particularly in its ability to modulate activity in this sensory regions. Damage to the right **PPC**, for example, often results in unilateral visual neglect, which is thought to be a disorder in the ability to deploy spatial attention. Key subregions such as the lateral intraparietal area and the superior parietal lobule are instrumental in tasks requiring sustained attention to spatial locations. The intraparietal sulcus, for instance, exhibits spatial maps of attention and dynamically represents attended locations in the visual field, facilitating both voluntary and stimulus-driven attention (Yantis, 2008).

When it comes to detecting and processing sensory stimuli, the occipital and temporal lobes are also major

contributors. The visual cortices, situated in the occipital lobe, play a critical role in processing visual information, with their activation increasing in response to more complex visual stimuli. These areas collaborate with the lateral occipital complex along the ventral pathway, which specializes in identifying visual objects, the superior parietal lobes (in the **PPC**) along the dorsal pathway, which focus on integrating visual information in both passive and attention-driven contexts and the frontal eye field (in the **PFC**) which is important for the control of voluntary eye movements. The temporal lobe also houses structures involved in auditory attention. The primary auditory cortex, located in Heschl's gyri, and the secondary auditory processing regions, such as the planum temporale and the superior temporal gyri, contribute to the interpretation of auditory stimuli. These areas are particularly essential for discerning and prioritizing auditory cues within an environment (Mayer et al., 2007; Johnson and Zatorre, 2006).

Within the **PFC**, distinct regions specialize in managing different attentional demands. The right **DLPFC** is particularly responsive to attentional demands, playing a critical role in tasks involving visual search and visuospatial orientation. By contrast, the left **DLPFC** demonstrates a strong relationship with working memory, selectively activating in response to increased cognitive load. Under high attentional demands, this region serves as a compensatory mechanism, ramping up its activation to efficiently allocate limited neural resources (Mayer et al., 2007).

2.2 Attention

As previously mentioned, attention is the cognitive process that allows individuals to concentrate on specific stimuli while ignoring irrelevant distractions, helping to navigate and interpret the complexities of their environment. It relates closely to an individual's immediate experience and current awareness, serving as a foundation for learning, decision-making, and executing tasks that demand focus (Araujo, 2022).

Attention can be divided into two main forms: voluntary and reflexive. Voluntary attention, also known as endogenous attention, is a top-down, goal-directed process where individuals consciously choose what to focus on based on their goals, expectations, or tasks at hand. On the other hand, reflexive or exogenous attention is a bottom-up, stimulus-driven process, where external sensory events, such as sudden sounds or movements, involuntarily capture our attention (Araujo, 2022).

2.2.1 Attention Types

Beyond these forms, attention also includes different types. According to Carter et al. (2019), Araujo (2022) and Bouchard et al. (2025) there are 5 different types of attention: sustained attention, selective

attention, focused attention, alternating attention and divided attention (Figure 6). The brain's intricate networks play a central role in managing attention, influencing how individuals' maintain focus and shift between tasks. Depending on the type of attention required, various regions of the brain are engaged, highlighting the critical connection between brain function and our ability to concentrate effectively.

	Sustained attention:	Ability to maintain concentration on a particular object, activity, or task for a continuous period of time
	Selective attention:	Ability to resist shifting attention from a selected target or goal
	Focused attention:	Ability to single out one object in one's environment and respond to it while being able to <i>tune out</i> all the other elements around.
	Alternating attention:	Ability to quickly shift focus from one stimulus to another
	Divided attention:	Ability to attend to two different stimuli at the same time

Figure 6: Types of Attention

Sustained Attention

Sustained attention (or vigilance) is the ability to maintain concentration on a particular object, activity, or task for a continuous period of time (Araujo, 2022; Carter et al., 2019; Mcavinue et al., 2012a; Simon et al., 2023; Esterman and Rothlein, 2019). For example, when reading a book or listening to a long lecture, the ability to stay focused without getting distracted demonstrates sustained attention.

Attention is prone to wandering, and sustained attention does not remain consistent throughout time. Studies show that children have a lower capability of sustaining attention, with this ability growing and peaking in young adulthood before gradually declining as people reach older age (Mcavinue et al., 2012a; Simon et al., 2023). This change throughout the lifespan is observable using both common methods for measuring sustained attention and other metrics. Traditional measures typically focus on response time metrics, such as the average and variability of response time during tasks. Research has demonstrated that performance on vigilance tasks tends to decline over time—a phenomenon known as the “vigilance decrement”. This decrement reflects cognitive fatigue and is exacerbated by factors such as normal aging or conditions like **ADHD** (Simon et al., 2023).

Initial theories suggested that sustained attention relied on a finite cognitive resource that depletes over

time. However, this has been challenged by paradoxical findings, such as easier tasks sometimes leading to greater vigilance decrement and the significant role of motivation in keeping dedicated focus (Esterman and Rothlein, 2019). In addition to these decrements, cognitive functions, including attention, oscillate at various temporal scales, from daily variations to fluctuations within milliseconds. Recent studies have shown that attentional performance oscillates between two states: moments of high accuracy and low response time variability (being “in the zone”) and moments of lower accuracy and higher response time variability (“out of the zone”). These fluctuations in response time variability are largely independent of vigilance decrements (Esterman and Rothlein, 2019).

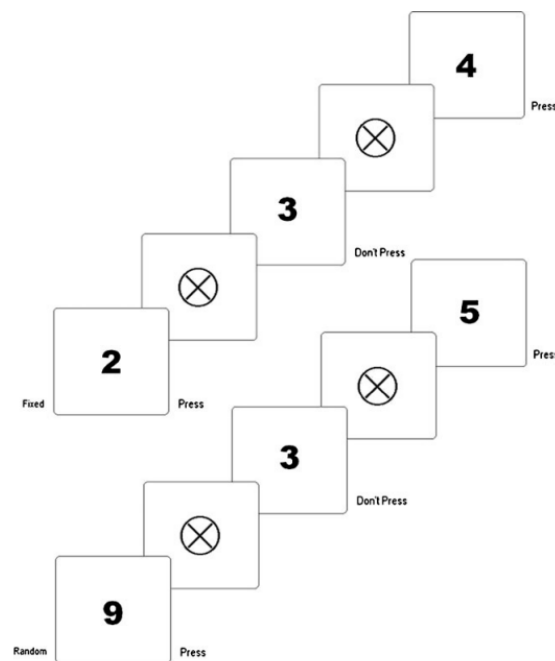


Figure 7: SART (from (Mcavinue et al., 2012a)).

The study by Mcavinue et al. (2012a) examined both sustained and selective attention, focusing on how these cognitive functions change across the lifespan. The section on selective attention will be addressed later, on its own section. The research aimed to explore the influence of aging on cognitive functions, one of them being sustained attention, which plays a central role in goal-directed behavior, influencing other cognitive abilities such as memory and reasoning. Sustained attention was assessed using the **Sustained Attention to Response Task (SART)**, in which participants were required to press a button for all numbers (1 to 9) except the number 3 (example on Figure 7, the top represents the fixed version in which numbers appear in a normal order and the bottom represents the random version in which numbers appear in any order). The task measured commission errors (incorrect presses), omission errors (failure to respond), and response time variability. This task placed a significant cognitive load on participants,

as it required fast, continuous responses, making it an ideal tool to measure sustained attention more rigorously than traditional vigilance tasks. The study found that sustained attention followed a U-shaped curve throughout life. Performance was weaker during childhood and adolescence, stabilized during young and middle adulthood, and deteriorated again in old age. This suggests that sustained attention not only declines with aging, but also demonstrates a distinct pattern of change over the life course. The sensitivity of the **SART** was crucial in detecting these declines, as it required continuous focus and rapid responses, which challenge the sustained attention system more intensely than other tasks. These findings highlight that while sustained attention is affected by aging, it is particularly vulnerable in early and late stages of life. This U-shaped pattern contrasts with other cognitive abilities that decline more steadily, emphasizing the unique trajectory of sustained attention as a cognitive function.

Another popular task to assess sustained attention is gradCPT. This task involves presenting 20 images, with 10% depicting mountains and 90% depicting city scenes. The images transition gradually, with each one remaining fully visible for 800 milliseconds. Participants are instructed to press a button for city images, while refraining from pressing for mountain images. There is an implicit response time limit, as each image transition occurs every 800 milliseconds. gradCPT is sensitive to various sustained attention metrics, including accuracy, reaction time, and the gradual decrement effect over time (Esterman et al., 2014).

The concepts of decrements and fluctuations, two broad categories of empirical findings related to sustained attention, have been outlined in paragraphs above. The focus now shifts to neurocognitive models that explain the core factors influencing sustained attention.

Starting with the **Arousal model**, which explains that our ability to sustain attention is directly influenced by physiological arousal levels. As previously mentioned, the locus coeruleus, is a brain region that releases the neurotransmitter norepinephrine, crucial in this process. When activity is too low, individuals' experience hypoarousal and become inattentive; when it is too high, hyperarousal leads to distraction. Optimal performance occurs at an intermediate arousal level. Studies using pupil size as an arousal indicator have shown that both mind-wandering and external distractions correlate with different arousal levels, impacting attentional performance (Esterman and Rothlein, 2019). Figure 8 represents a visual explanation of the process.

Moving on to the concept of **attentional allocation**. Beyond arousal, task performance is influenced by our ability to direct attention effectively, which involves both the intensity and the focus of attentional resources. "Mind wandering", for instance, describes instances when attention drifts to task-irrelevant thoughts, impairing performance. This drift may be intentional, where attention is consciously redirected

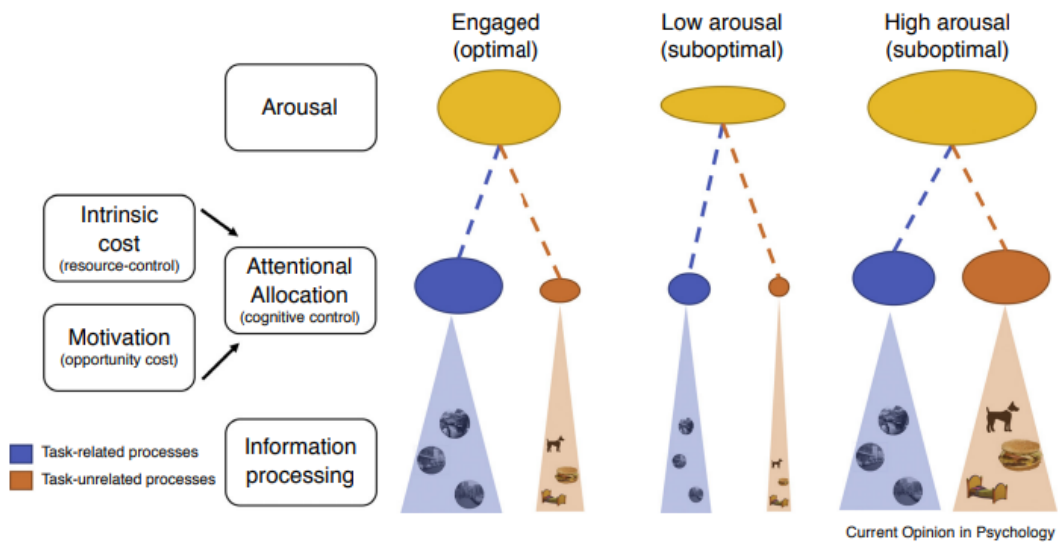


Figure 8: Explanation of the Arousal Model (from Esterman and Rothlein (2019))

from a task, or unintentional, occurring as the mind inadvertently shifts away. Research has found that task difficulty impacts this intentionality; as tasks become more challenging, intentional mind-wandering decreases, while unintentional drift becomes more frequent. Furthermore, motivation significantly influences attentional allocation; higher motivation levels typically reduce the likelihood of mind-wandering, enhancing task focus and performance (Esterman and Rothlein, 2019).

Then there's the **control model**, which explains that cognitive control comes at a "cost". The mind's default tendency is to wander, and sustaining focus requires cognitive resources that engage control networks, such as the **FPN**, while the **DMN** persistently exerts a pull toward mind wandering. For someone to maintain continuous attention, it requires a continuous override the natural inclination of the mind to wonder. Research indicates these two systems operate antagonistically: when one is active, the other generally decreases in activity. The model also proposes that attentional lapses correspond to heightened connectivity between the **DMN** and control regions, with lapses more frequent when this coupling is stronger. Additionally, momentary increases in **DMN** activity, paired with decreases in control network engagement, often lead to variability in focus, further linking the inverse activity patterns of these networks to sustained attention performance (Esterman and Rothlein, 2019).

The **opportunity cost model** explains that the difficulty in maintaining focus on a task may relate to "opportunity cost", or the subjective comparison between the value of paying attention to the task at hand versus engaging in other mental activities, such as mind wandering. If an alternative activity seems more valuable than the current task, sustaining attention feels more costly. Motivation can shift this balance; for instance, rewards can heighten one's capacity to maintain attention and reduce performance fluctua-

tions (Esterman and Rothlein, 2019). For example, Esterman et al. (2014) conducted experiments using rewards, finding that they improved overall accuracy and response stability but did not affect the vigilance decrement itself. This suggests a dissociation between mechanisms enhancing overall performance and those sustaining attention over time.

The previous models have all conceptualized attention as a resource that is in some way limited and thus must be allocated. However, what it means for attention to be allocated to a particular task versus task-unrelated thoughts is also of central importance. From an **information processing perspective**, attention enhances the brain's ability to process information efficiently. When someone is "in the zone", the brain communicates more effectively between visual and attentional control areas, allowing for streamlined processing. During moments of high motivation, communication strengthens between attentional networks and areas related to internal thoughts. Research indicates that focused states enable the brain to handle information with less cognitive effort, achieving greater efficiency. These sustained attentional periods are marked by optimized visual processing and resource use, minimizing unnecessary activation in control areas.

Selective attention

Selective attention is the ability to resist shifting attention from a selected target or goal (Carter et al., 2019; Araujo, 2022). It requires good levels of control and inhibition.

As discussed in the previous section, the work presented by Mcavinue et al. (2012a) also investigates selective attention, with the study addressing how aging impacts cognitive function.

Participants completed the Combi-TVA task (Mcavinue et al., 2012b), which measured selective attention by briefly presenting red target letters and blue distractor letters, requiring participants to identify the red letters. Various stimulus types were employed, including full and partial reports, as well as variations in exposure duration. The study's findings revealed a linear decline in both attentional capacity and selective attention with age. Notably, the greatest reductions with aging were observed in visual processing capacity and short-term visual memory capacity, both showing large effect sizes. These aspects peaked during adolescence or early adulthood, followed by a continuous linear decline throughout life. This supports previous research suggesting significant reductions in visual processing and short-term visual memory with aging. Overall, the study illustrates that aging affects various aspects of visual attention in different ways. While attentional capacity is particularly vulnerable to decline, selective attention is relatively more resistant in comparison, though they still experience notable reductions, especially in older age.

The Combi-TVA task, used to test selective attention in the study by Mcavinue et al. (2012a) was developed

based on the original work of Bundesen (1990) regarding the **Theory of Visual Attention (TVA)**. In this article, the author explains his theory that visual recognition and attentional selection are processes involving the perceptual categorization of elements in the visual field. When a categorization is stored in short-term memory, the element is selected and recognized as part of a specific category. The process of selection operates as a “race” between competing categorizations for entry into short-term memory (Figure 9). The speed of this race depends on three key factors: sensory evidence strength, perceptual decision bias, and relevance, which together determine the likelihood and timing of element recognition.

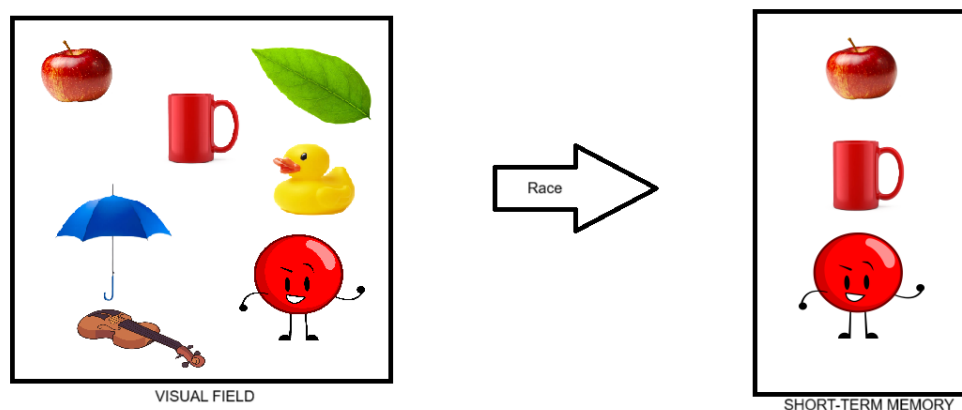


Figure 9: Theory of Visual Attention. The race model selects the categorizations to be attended as the first ones to arrive to the Short-Term Memory

TVA outlines two complementary mechanisms: filtering and pigeonholing. Filtering refers to the selection of specific elements in the visual field based on their features, such as prioritizing red objects when searching for something red. This process is governed by attentional weights, which are influenced by relevance values, allowing the system to process relevant elements faster than irrelevant ones. However, filtering does not alter how elements are classified; it only speeds up the processing of those deemed important. Pigeonholing, on the other hand, deals with classifying the selected elements into specific categories (e.g., letters, numbers). It is guided by perceptual bias, which increases the likelihood of categorizing elements into specific groups. This theory is able to explain various experimental findings related to visual attention, such as stimulus recognition, the interaction between targets and distractors, and the effects of the number of stimuli and practice. As such, it provides a solid foundation for understanding visual processing and attentional selection in recognition tasks (Bundesen, 1990).

Focused attention

Focused attention is the ability to single out one object in one's environment and respond to it while being able to *tune out* all the other elements around. Focused attention might be the most elementary subtype of the five, and its well-functioning is a crucial prerequisite for all other attentional subtypes (Carter et al., 2019; Araujo, 2022; CogniFit, 2023b; Bouchard et al., 2025). It might sound similar to selective attention, but while focused attention refers to the ability to initiate and maintain attention on a single stimulus in a basic and immediate way, selective attention involves maintaining that focus in the presence of distractions, requiring active filtering and more complex inhibitory control. Both involve concentrating on a specific target, but the former is about achieving focus, whereas the latter is about preserving it despite competing stimuli.

It's easier to pay attention to a stimulus or target activity if there are few environmental distractions, and it becomes more difficult to concentrate with more frequent or intense distractions, but people with good focused attention are capable of doing it. For example, an athlete being able to focus on the competition, while filtering out all the noise from the crowd. It's also very important in a school or university setting. Students must be able to correctly and efficiently pay attention to important stimuli in the classroom, which may be the teacher lecturing, for example. If the student is not able to keep their mental focus, they may see significant academic repercussions (Carter et al., 2019; Araujo, 2022; CogniFit, 2023b).

In addition to being vital for academic success and everyday efficiency, focused attention plays an essential role in overall cognitive functioning by enabling individuals to engage with the world in a targeted and meaningful way. Though minor lapses in focused attention are normal, significant impairments can severely impact a person's capacity to carry out even basic daily activities. Without sufficient focused attention, tasks that require sustained mental focus, such as driving, managing projects, or following instructions, can become challenging or even unsafe (CogniFit, 2023b).

Focused attention can be evaluated using eye-tracking technology to monitor gaze patterns during simple tasks. Participants may be required to follow specific moving objects on a screen, and the eye-tracker records eye movements to assess the accuracy of focus on the target's actual path. Additionally, participants might be asked to identify particular features (letters, shapes, or colors) within various orientations across multiple trials, which may include both relevant and irrelevant cues. Another common task involves scanning visual scenes, where eye movements are tracked to create a saliency map highlighting the most visually significant areas. Individual differences are very common, with attentional strategies and priorities that influence how and what people are attending to (Bouchard et al., 2025).

Assessing focused attention is therefore valuable in many settings. In academic contexts, evaluating a

child's focused attention can help educators identify those who may require more support to understand and retain information. Teachers can then tailor their approach, presenting information in a way that better suits the student's attentional capabilities. In clinical settings, an assessment of focused attention may reveal whether a patient can adequately respond to medical guidance or adapt to their environment, which is especially critical for treatment compliance and rehabilitation. In professional contexts, focused attention assessments can inform hiring processes for roles that demand precise attention to detail, such as quality control positions, drivers, or office roles that require careful management of tasks and information (CogniFit, 2023b).

In sum, focused attention is a fundamental cognitive skill with broad implications across various life domains, impacting personal, academic, and professional success by enabling individuals to concentrate on the most relevant information in any given context.

Alternating attention

Alternating attention is the ability to quickly shift focus from one stimulus to another, which requires a different sort of cognitive response. For example, someone reading a recipe, start cooking and occasionally stop cooking to check the recipe again. This type of attention displays mental flexibility to shift attentional focus from task to task and is also required in language processing, such as in bilingual individuals who shift their attention between speaking English and Spanish (Carter et al., 2019; Araujo, 2022; Bouchard et al., 2025).

This attention type really shows one's ability to react and deal with different stimulus, being able to detect which one to focus on. This can have impact in many situations. For instance, alternating attention is crucial in professional settings, such as when a nurse shifts focus between administering medication to a patient and monitoring vital signs on a nearby machine. Another common example can be found in social interactions, where individuals alternate their focus between listening to a conversation and observing non-verbal cues such as body language or facial expressions. These scenarios underscore the versatility of alternating attention in handling different situations where multiple sources of information compete for cognitive resources.

Many tasks evaluate alternating attention. A common measure is the Trail Making Test B. In this test, participants are shown 25 circles containing letters (A to L) and numbers (1–13). They are asked to connect the circles in ascending order, alternating between numbers and letters (1-A-2-B-3-C-4-D, etc.) without lifting their pen, as accurately and quickly as possible (Bouchard et al., 2025).

Another example is the Rule Shift Cards test, in which participants are presented with one black or red

playing card and must respond according to one of two rules (written below the card). The first rule is to say “No” for a black card and “Yes” for a red card. The second rule is to say “Yes” if the card is the same color as the previous one and “No” if it’s a different color (Bouchard et al., 2025).

Divided attention

Divided attention could be defined as our brain’s ability to attend to two different stimuli at the same time, and respond to the multiple demands of your surroundings, or in simpler terms, *multitasking* (Carter et al., 2019; Araujo, 2022). Alternating attention pretty much lays the groundwork so that one can multitask.

This type of attention is a crucial cognitive ability in daily life, enabling us to complete tasks quickly and efficiently by managing multiple perceptual, motor, or cognitive activities simultaneously. Effective performance on tasks requiring multiple simultaneous actions—such as driving, athletic performance, or academic tasks—relies directly on divided attention. In clinical contexts, understanding a patient’s divided attention capacity may inform recommendations, such as allowing additional time to process or gather information (CogniFit, 2023a).

During divided attention tasks, the **DLPFC**, particularly in the left hemisphere, plays a critical compensatory role. Studies show that the **middle-DLPFC** is recruited to manage multiple components of working memory, while the **posterior-DLPFC** modulates sensory cortex activation when sensory cortices struggle to sustain the required performance. Interestingly, individuals who achieved the best performance during divided attention tasks showed the greatest recruitment of their sensory cortices, indicating a trade-off between **DLPFC** activity and sensory region activation (Johnson and Zatorre, 2006).

The Simultaneity Test DIAT-SHIF, is a task used to evaluate divided attention, assessing cognitive flexibility and hand-eye coordination. In this test, the user must follow a moving ball with a pointer while simultaneously attending to words appearing at the center of the screen. When a word’s meaning matches its font color, the user is prompted to respond accordingly, requiring attention to both stimuli at once. This activity demands that users switch strategies, create responses in real-time, and manage both motor and visual skills simultaneously (CogniFit, 2023a).

Divided attention, is crucial in our daily life, but one’s should be careful when multitasking because there are certain behaviors that can hurt an individual’s attention in the short and long term. This will be detailed further in the next section.

Types Overview

The five types of attention allow an individual to focus on the tasks that impact is day to day life. They all interact with one another, being “called upon” when necessary. There are many instances where they are all involved.

When one’s driving, for example, all the types of attention are usually at play. Someone must focus on all the road-related stimuli (focused attention), pay attention for several minutes (sustained attention), resist being distracted by irrelevant stimuli like a text message on the cell phone (selective attention), change the attention from the mirrors, to the lanes beside us, and back to the lane (alternating attention), and finally, simultaneously perform actions such as turning the wheel, using the pedals, turning on your blinker, wipers, etc. (divided attention) (Bouchard et al., 2025).

These types of attention can also be viewed as part of a hierarchical structure, where they are organized by increasing complexity. At its foundation lies focused attention, the simplest form, which allows an individual to concentrate on a single stimulus. Building on this base, the hierarchy follows with sustained attention, selective attention, alternating attention, and finally, divided attention that represents the top and more complex type, requiring simultaneous engagement with multiple tasks (Figure 10) (Bouchard et al., 2025).

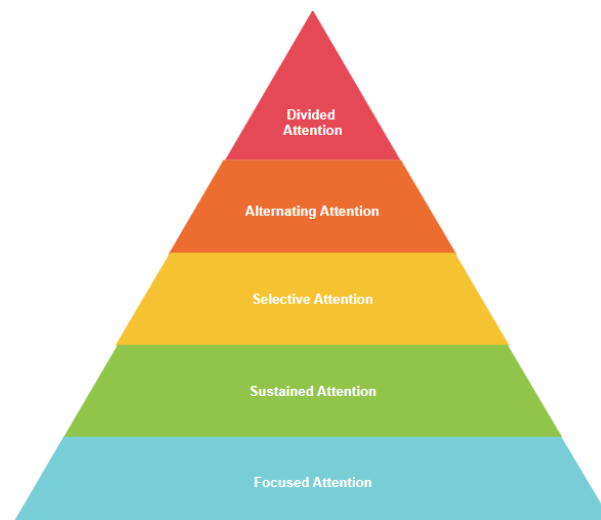


Figure 10: Types of Attention - Hierarchy of Complexity (idea from Bouchard et al. (2025))

This hierarchical perspective underlines how basic attentional processes serve as the foundation for more advanced and demanding cognitive functions. Everyday scenarios like driving exemplify not only the interplay but also the layered complexity of these attentional processes, demonstrating their indispensable role in effectively navigating modern life (Bouchard et al., 2025).

Figure 11¹ is an ontology that was developed to facilitate the visualization of the relationships between the five types of attention. This ontology visually represents the hierarchical organization of attentional processes, illustrating how each type builds upon the one beneath it, while also highlighting some key characteristics. It begins with focused attention, the most basic type, and its sensitivity to stimuli. Building upon this, sustained attention emerges, involving working memory to maintain focus over extended periods. Selective attention follows, being a type that needs the personal control and inhibition to resist changing focus from the intended target. Alternating attention adds the ability to shift flexibly between different tasks or stimuli, reflecting increased cognitive adaptability. At the highest level of complexity, divided attention enables multitasking by simultaneously engaging with multiple streams of information or actions.

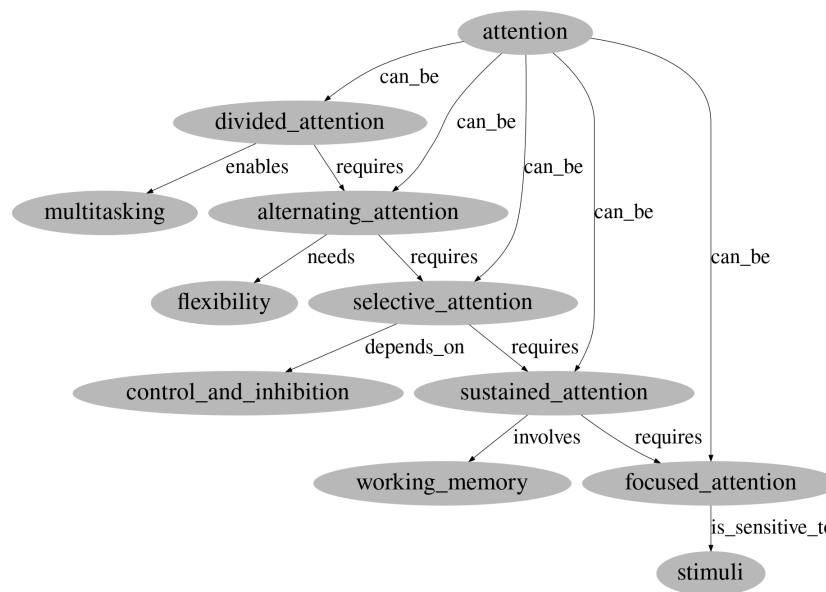


Figure 11: Ontology

In summary, attention is not linear and doesn't always involve the same reaction to stimuli. When people need to keep focus for a long period of time, sustained attention comes into to play. When distractions are present, selective attention helps one stay focused on a specific goal or task. When there's a need to respond to a single element while ignoring everything else, focused attention allows individuals' to tune out irrelevant details. When shifting between different tasks or stimuli, alternating attention ensures flexibility and adaptability. Finally, when managing multiple tasks or stimuli simultaneously, divided attention helps people respond effectively to overlapping demands. To perform at in the best way and make progress in academic or professional pursuits, maintaining the "health" of these various attention types is essential. Each plays a crucial role in ensuring focus, adaptability, and productivity in our complex, dynamic lives.

¹ Built using the platform available at <https://webontodl.ep1.di.uminho.pt/>

2.2.2 Attention Disturbances

The goal of this Master's project is to develop a platform aimed at helping individuals enhance their attention capabilities, with training. Many people across the world experience difficulties with attention and concentration, which, in most cases, are not linked to clinical disorders. The platform seeks to support these individuals by providing tools and techniques to improve their focus through non-clinical methods. Attention-related difficulties can manifest in various ways, such as being unable to remember recent events, struggling to sit still, or thinking clearly. Individuals might frequently lose things or fail to recall where they placed them, face indecision, or find it challenging to complete complex tasks. Additionally, a lack of focus, low physical or mental energy, and a tendency to make careless mistakes are common symptoms. Importantly, experiencing these issues does not necessarily indicate the presence of a clinical condition or disorder (Nall, 2023).

In many cases, lifestyle factors are a negative contributor to attention. Lack of sleep, hunger, insufficient exercise, fatigue, and stress are major factors that affect one's ability to concentrate (Nall, 2023).

However, gaining a deeper understanding of clinical conditions affecting attention is crucial and provides valuable insights. Unlike non-clinical attention challenges, these conditions cannot be resolved through training. This subsection will explore some of these disturbances, including **Attention Deficit Hyperactivity Disorder**, **Hemineglect Syndrome**, **Post-traumatic Stress Disorder** and **Schizophrenia**.

Attention Deficit Hyperactivity Disorder (ADHD) is one of the most prevalent neurodevelopmental disorders today, affecting over 5% of the global population. Despite being extensively researched, the diagnosis and management of **ADHD** remain challenging for clinicians due to its diverse manifestations and contextual variability (Kiliç et al., 2007; Drechsler et al., 2020; Kinman and Raypole, 2021; CDC, 2024).

ADHD significantly impacts core neuropsychological functions, affecting executive functioning. Deficits are consistently observed in response inhibition, sustained attention, working memory, and planning, with moderate effect sizes reported in research. Other affected domains include reaction time variability, selective attention, fluency, and general memory. Beyond cognitive impairments, individuals with **ADHD** often exhibit altered sensory processing, heightened emotional reactivity, and difficulty with social interactions. These deficits suggest not only impaired cognitive control but also disruptions in fundamental processing systems, which contribute to challenges in adapting to everyday demands and regulating behavior effectively, something that can be easier with help from the people around them (Kiliç et al., 2007; Drechsler et al., 2020; Ward et al., 2022).

ADHD is not strictly categorical but exists on a continuum. Many individuals exhibit subthreshold **ADHD**,

affecting nearly 10% of the population, with milder impairments that resemble **ADHD** but do not meet the full diagnosis. These individuals may exhibit better socioeconomic outcomes or display symptoms later in life (Drechsler et al., 2020).

ADHD frequently co-occurs with various neurodevelopmental and mental health disorders in childhood and adolescence. According to Drechsler et al. (2020) key comorbidities include, learning disorders, such as reading disorders (affecting 15–50% of individuals with **ADHD**) and dyscalculia (5–30%). Autism spectrum disorder is highly comorbid, observed in 70–85% of cases, reflecting substantial overlap in developmental challenges. Mood-related issues, including anxiety and mood disorders, affect 0–45% of individuals with **ADHD**. Furthermore, individuals with **ADHD** face heightened risks of substance misuse, including smoking, problematic media use, obesity, dysregulated eating patterns, enuresis (17%), and sleep disorders (25–70%).

Neuroimaging studies have revealed critical insights into the structural and functional differences associated with **ADHD**. Brain development in individuals with **ADHD** is characterized by delayed maturation, particularly in the frontal cortex. Structural abnormalities include reduced gray matter volume, cortical thinning, and compromised white matter integrity. Overall, **ADHD** brains show a 3–5% reduction in total brain size, with subcortical structures such as the amygdala, and cerebellum being disproportionately affected. In terms of functional connectivity, **ADHD** is associated with disrupted activity within the **DMN** and diminished interactions between the **DMN** and cognitive control networks, which contributes to attentional lapses. During task-based activities, individuals with **ADHD** exhibit hypoactivation in the **FPN**, which impacts their ability to perform executive tasks. Additionally, altered reward processing mechanisms, linked to dopamine deficiencies, reflect reduced sensitivity to rewards and impairments in learning and decision-making. Collectively, these findings demonstrate that **ADHD** affects the brain at multiple levels, providing a foundation for targeted interventions and further research (Drechsler et al., 2020).

Hemineglect syndrome, also known as unilateral or spatial neglect syndrome, is a behavioral disorder characterized by an inability to perceive, report, or orient to sensory stimuli on one side of space, typically the side opposite to a brain lesion. This syndrome commonly occurs following a stroke, particularly in the right hemisphere, which plays a critical role in attentional processes. The term hemineglect pretty much refers to a profound disruption in spatial awareness that can affect an individual's daily functioning and recovery trajectory (Ogden, 2012; Li and Malhotra, 2015).

The syndrome is most frequently observed after large strokes within the middle cerebral artery territory, being one of the reoccurring consequences. However, it can also arise from other pathological processes, including neurodegenerative diseases, trauma, or neoplasia. Patients with hemineglect may display an

array of symptoms, such as failing to respond to stimuli on one specific side, that being the contralesional side (opposite side of the injury), making it a primary disorder of attention rather than sensory function (Li and Malhotra, 2015).

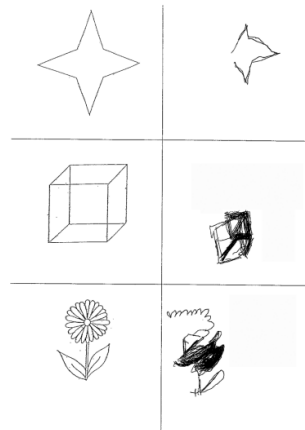


Figure 12: Copying task from a behavioural inattention test (from Li and Malhotra (2015))

The manifestations of hemineglect are diverse and can range from subtle to severe. Severely affected individuals may fail to fixate on people speaking to them, directing their gaze exclusively towards the side of the lesion. They may eat only from one side of their plate or neglect personal grooming on the contralesional side, a condition referred to as personal neglect. In reading, they may omit words or misinterpret text on one side of the page, a phenomenon known as neglect dyslexia. Additionally, some individuals exhibit motor neglect, avoiding use of their contralesional limb despite no physical weakness or sensory loss. Figure 12 shows how patients affected by this disease neglect one side, in this case only drawing the right-sided elements (Li and Malhotra, 2015).

Hemineglect syndrome most commonly arises from damage to the right cerebral hemisphere, frequently due to middle cerebral artery territory strokes. Although traditionally linked to parietal lobe lesions, these strokes typically affect multiple regions, including parietal, temporal, and frontal areas, leading to varying symptom combinations across patients. In some cases, neglect also follows subcortical strokes, which may induce hypoperfusion or cortical dysfunction in overlying areas. Other evidence also highlights the role of white matter tracts in mediating neglect. Damage to these tracts disrupts communication between cortical and subcortical regions essential for attentional control. Functionally, neglect involves an inter-hemispheric imbalance, where the relatively overactive left hemisphere after a right-hemisphere stroke biases attention and eye movements towards the right. This imbalance, combined with damage to right frontal and parietal regions critical for spatial functions, contributes to the hallmark spatial impairments of neglect. Additionally, these regions are implicated in non-spatial processes such as vigilance and sustained

task performance. Deficits in these domains often exacerbate neglect's severity and hinder recovery, especially in left-side neglect, where non-spatial impairments prolong the effects of spatial bias and recovery time (Li and Malhotra, 2015).

Post-traumatic Stress Disorder (PTSD) and **Schizophrenia** are complex mental health conditions that extend far beyond mere attention deficits. Both disorders are characterized by profound disruptions in cognition, emotion, and behavior, reflecting the severity of their impact on individuals' daily lives. However, despite their multifaceted nature, both **PTSD** and schizophrenia significantly influence attentional processes, underscoring the role of cognitive impairments in these conditions (Taylor-Desir, 2022; Ely et al., 2023; National Institute of Mental Health, 2024).

PTSD stems from traumatic experiences, such as assaults, natural disasters, or combat exposure. It is prevalent worldwide, with a trauma incidence of 28–85% and a resulting **PTSD** prevalence ranging from 6.8% to 9.2%. Key symptoms include intrusive memories, such as flashbacks and nightmares, as well as hyperarousal, severe anxiety, and avoidance of trauma-related stimuli. These symptoms lead to distress and functional impairments that significantly hinder the individual's ability to cope with daily challenges (Ely et al., 2023; Taylor-Desir, 2022). Schizophrenia, on the other hand, is a chronic and severe mental illness that profoundly affects thought, emotion, and behavior. Those diagnosed with schizophrenia often experience hallucinations, delusions, and disorganized thinking, creating a reality disconnect that can be devastating for both the individual and their support system. Schizophrenia typically emerges between the ages of 16 and 30, often following a first psychotic episode, and requires long-term management and care to address its wide-ranging impacts (National Institute of Mental Health, 2024).

In schizophrenia, attentional disturbances are extensively documented. Research identifies deficits across various dimensions, including sensory-motor gating, attentional set shifting, response inhibition, spatial cuing, and signal detection. One notable study demonstrated that individuals with schizophrenia who exhibited severe positive symptoms were unable to maintain effective discrimination between predictive and nonpredictive cues during testing, despite successfully doing so during training. This contrasts sharply with healthy control groups who retained this discrimination ability. Furthermore, the severity of positive symptoms in schizophrenia correlated with increased attention to nonpredictive cues during causal learning tasks. These findings suggest that impaired selective attention in schizophrenia leads to learning irrelevant causal associations, potentially underpinning the development of positive symptoms such as delusions. Such deficits show how attentional disruptions contribute to the broader cognitive and perceptual anomalies characteristic of the disorder (Morris et al., 2012).

Attention deficits in **PTSD** are similarly multifaceted, affecting various domains and contributing to the

disorder's symptomatic profile. Individuals with **PTSD** demonstrate poorer orienting attention, particularly in tasks requiring spatial cue utilization. These deficits suggest difficulties disengaging attention from non-emotional cues, which may exacerbate tendencies to interpret neutral environments as threatening, a hallmark of **PTSD**. Sustained attention is also significantly impaired in **PTSD**, as evidenced by increased variability in reaction times during tasks that demand prolonged focus. This variability may lead to practical challenges, such as maintaining consistent performance in professional or everyday settings. Additionally, broader attentional impairments are evident across different **PTSD** subtypes, further highlighting the pervasive nature of attention-related dysfunctions in this condition (Ely et al., 2023).

2.3 Attention in Today's World

The modern world has undergone significant changes over the last 15 years, profoundly altering how individuals process and allocate their attention. The rapid technological evolution, coupled with major global events such as the COVID-19 pandemic, has created an environment vastly different from that of previous generations. The human brain, shaped by gradual evolution, now faces unprecedented challenges in adapting to this rapidly changing landscape. This section explores these shifts, focusing on how contemporary factors have reshaped attention and the broader implications for cognitive function.

Burnout, prevalent among students and working adults, exemplifies the strain placed on the human brain and attention in today's demanding world. Workplace burnout, an occupation-related syndrome stemming from chronic stress, significantly impairs cognitive functions such as short-term memory and attentional focus, essential for effective job performance. Characterized by emotional exhaustion, cynicism, and feelings of inefficacy, burnout reflects a failure to manage stress through typical restorative measures like breaks and vacations. Among graduate students, burnout is especially pervasive due to the demanding nature of academic programs and professional obligations. Many doctoral candidates report high levels of exhaustion, negativity toward their work, and a diminished sense of accomplishment. These symptoms often correlate with reduced satisfaction, lower success rates, and intentions to leave academia. The prominence of exhaustion among students highlights its central role in the burnout phenomenon, underscoring the urgent need to address its root causes and mitigate its effects (American Psychological Association, 2023; Allen et al., 2021).

This section will delve deeper into two forces shaping attention in the modern era: the impact of the COVID-19 pandemic and the sometimes negative influence of contemporary technologies.

2.3.1 Technology and the Internet

In recent years, technology has become an integral part of our daily lives, offering countless benefits when used correctly. From enhancing productivity to providing innovative tools for learning and training, it has the potential to significantly improve how individuals' approach tasks and acquire new skills. When applied effectively, technology can support educational growth and personal development, making information more accessible and learning experiences more interactive and engaging (Araújo et al., 2024). However, alongside these advantages, the misuse of technology has also become a growing concern, especially as society has witnessed an unprecedented rise in its usage throughout the 21st century, that as caused many people to developed addictive and harmful behaviors (Méndez et al., 2024; Lodge and Harrison, 2019).

Despite its potential, many individuals find themselves using technology in ways that can be counterproductive, often leading to distraction, reduced focus, and poor time management. This excessive or misguided use can have far-reaching effects, particularly on attention span. The constant stream of information and notifications can fragment our focus, making it increasingly difficult to maintain attention on tasks that require deeper concentration.

This aforementioned expansion of technology, has brought smartphones and the internet into everyday life on an unprecedented scale. Historically, these technologies are extremely recent, yet they have quickly become integral to social interaction, education, and entertainment. This widespread availability has led to significant negative impacts, in terms of attention and cognitive control. **Internet Addiction (IA)** and **Smartphone Addiction (SPA)** are particularly prevalent among adolescents and young adults, leading to a host of psychosocial issues, including deficits in attention, increased anxiety, and even symptoms comparable to substance or gambling addiction, such as tolerance and withdrawal-like effects (Méndez et al., 2024; Loh and Kanai, 2016). These behavioral patterns highlight the need to understand the underlying neurological impacts of excessive internet and smartphone use, with particular attention to how these addictions disrupt cognitive control and attention regulation.

IA and **SPA** have been shown to impair several key brain functions tied to attention and executive control. Research has identified functional alterations in brain areas associated with reward processing and inhibitory control, including regions like the striatum, amygdala, **Anterior Cingulate Cortex (ACC)**, and insula. These changes can significantly disrupt the brain's ability to manage sustained attention and regulate impulsive behaviors, leading to compulsive technology use despite negative consequences. Specifically, neuroimaging studies reveal diminished activity in the **DLPFC** and frontoparietal circuits, which are critical for executive functions such as decision-making and impulse control. This reduction in

cognitive control explains why individuals with **IA** and **SPA** often struggle to manage their use of technology, even when it interferes with daily responsibilities. Other studies on these issues have highlighted the impact on the networks that constitute the triple-network model. Alterations in these networks hinder the ability to maintain sustained focus on tasks and to filter out irrelevant information. This negative impact on attentional capacity is exacerbated by repeated exposure to digital notifications and the instant gratification associated with online interactions. The weakened connectivity in regions responsible for sensory and executive functions implies a reduced ability to concentrate and sustain attention on non-digital activities, potentially harming cognitive development and learning (Áfra et al., 2024; Méndez et al., 2024).

One of the central issues in **IA** is a disruption in reward processing mechanisms. Those with **IA** and **SPA** exhibit heightened sensitivity to immediate rewards, but struggle to delay gratification for more significant long-term benefits. For example, in decision-making tasks like the Iowa Gambling Task, individuals with **IA** are more likely to opt for risky options that provide instant rewards despite potential long-term losses, demonstrating a focus on immediate gratification over careful evaluation of consequences (Loh and Kanai, 2016). These reward-processing deficits also correlate with reduced dopamine regulation, which further explains why excessive use continues despite negative impacts, as these individuals experience reinforced pleasure responses in response to digital stimuli (Méndez et al., 2024). However, the negative effects extend beyond cognitive control and reward processing. There is also an impact in language networks, visual processing, and executive functions. For instance, problematic smartphone and social media use has been associated with reduced connectivity in the language network, which could hinder language processing and verbal skills. This reduction is theorized to be a consequence of decreased face-to-face interaction and an over-reliance on technology-mediated communication, which often lacks the depth of traditional social interaction (Áfra et al., 2024).

As previously mentioned, problematic social media use has been associated with reduced connectivity in the language network. **Social Media Addiction (SMA)** is another common and growing issue in this day and age, that walks hand in hand with **IA** and **SPA** and has much more negative impacts than just worsening the language network.

SMA is a pervasive issue among young people, with around 97% of teens aged 13 to 17 actively using at least one social media platform. While these platforms, provide spaces for creative expression and connection, particularly for Gen Z, they also bring unique challenges to mental and emotional health. Specifically, social media use has been linked to psychological distress, reduced attention control, and a range of mental health symptoms. Many studies report that adolescents who spend more than two hours daily on social media are more likely to these issues. For individuals with lower attention control, social

media use may intensify the tendency to engage in prolonged and often passive browsing, which exposes them to more content that fuels social comparison and stress, including interactions like cyberbullying. In contrast, individuals with higher attention control appear more resilient, using social media with greater intentionality, which may help them avoid problematic engagement and focus more selectively on beneficial content (Mahalingham et al., 2022; Conte et al., 2024).

Similarly, the design of platforms like TikTok plays a significant role in fostering problematic social media use. Conte et al. (2024) discusses the concept of problematic TikTok and social media use, defined by excessive, unregulated engagement that can lead to symptoms like anxiety, depression, and memory impairment and attention issues. TikTok's short-form video format (that has shown up in other platforms like Instagram and YouTube), tailored by algorithms to individual preferences, encourages rapid, continuous scrolling, which can distort users' perception of time and lead to "flow states". This immersive experience is designed to capture users' attention for extended periods, making it harder for them to disengage.

Mindless and continuous scrolling, is a big problem of social media and impacts brain chemistry and attention. Each new post or notification triggers a small release of dopamine, creating a brief sense of reward that encourages users to keep scrolling in search of the next dopamine hit. While this can be pleasurable in the moment, it creates a paradox: the constant stimulation and anticipation for more dopamine reinforce the habit, even as it strains the brain's attention resources. This cycle of reward-driven scrolling not only diminishes immediate focus but may, over time, contribute to attention deficits by conditioning the brain to expect rapid, shifting stimuli rather than sustained concentration. Moreover, the repeated bursts of dopamine can be accompanied by increased stress, as much of the content encountered may be negative or emotionally charged. The brain becomes conditioned to a rhythm of quick, fluctuating emotions, which can interfere with attention control and the ability to concentrate on single tasks. In the long term, this pattern has been linked to reduced productivity and difficulties in maintaining focus, as it trains the brain to crave rapid, brief engagements rather than longer periods of sustained attention (Sauvé, 2023).

Individuals affected by these digital dependencies often find it difficult to disengage from their devices, leading to habitual patterns of checking notifications, switching between apps, and remaining digitally connected. This constant engagement fosters a fragmented attention span, where moments of deep, sustained focus become rare. As a result, the compulsive need to use digital devices leads into another issue, *multitasking*.

Multitasking was presented when describing divided attention, and it can be very beneficial in certain situations, demonstrating an individual's ability to manage multiple tasks simultaneously. When used

intentionally and effectively, multitasking can be advantageous in scenarios that require less focus on in-depth comprehension, allowing people to handle routine tasks efficiently. However, especially with more cognitively demanding activities, like learning or studying, the downsides of multitasking become clear. Often, ineffective multitasking involves a split between *on-task* activities (those directly related to the task at hand) and *off-task* distractions, like checking social media or responding to messages (Lodge and Harrison, 2019). For example, a student trying to study while watching a movie will likely be less focused on the material compared to someone dedicating their full attention to it.

The constant switching between these on-task and off-task activities can significantly reduce focus and impair performance. While on-task activities, such as taking notes or engaging directly with learning material, help enhance comprehension and retention, off-task distractions fragment attention, leading to poorer outcomes. This issue becomes particularly problematic in digital multitasking environments, where the presence of multiple devices or platforms creates a continuous stream of potential interruptions. In these cases, attention and memory are negatively impacted, as the brain struggles to maintain consistent focus when frequently shifting between tasks (Lodge and Harrison, 2019; Ophir et al., 2009).

Research shows that multitasking in digital contexts can be very damaging, as individuals habitually alternate between on-task and off-task activities, hurting their focus and cognitive performance. For example, students who engage in internet-based multitasking during class, such as browsing or messaging, demonstrate lower comprehension and memory retention compared to those who remain focused on traditional note-taking. This behavior not only affects the multitasker but also disrupts nearby peers, as the visibility of these distractions can diminish their concentration, illustrating a wider-reaching impact within educational settings. The frequency of switching between devices contributes to this fragmentation of attention, adding to the cognitive load as the brain continuously reorients between diverse stimuli (Loh and Kanai, 2016).

Individuals who engage extensively in media multitasking, switching between multiple media sources, tend to struggle with executive control, showing a reduced capacity to ignore irrelevant stimuli. Known as high-frequency media multitaskers, they adopt a more diffuse approach to information processing, which, while supporting the integration of multiple sensory inputs, diminishes concentration on primary tasks. This dispersed focus affects their ability to selectively attend to specific information, as evidenced in tests where high-frequency media multitaskers are less able to filter out distracting stimuli. While some studies argue that certain multitasking, such as video gaming, can enhance visual memory and neural efficiency, frequent multitasking with digital media has been linked to structural changes in the brain. These changes include reduced gray matter in the **ACC**, a region critical for cognitive control, suggesting potential cognitive deficits among habitual multitaskers (Loh and Kanai, 2016).

Another theory around new technology and how it's changing the way people live their daily lives is *screen inferiority*. Screen inferiority is a term that refers to the tendency for digital reading to be less effective than reading on paper, especially concerning comprehension and focus. Studies in this area reveal that while reading speed remains relatively constant across formats, readers often retain and understand material less effectively on screens than on paper, partly due to differences in cognitive load and metacognitive strategies required by each medium (Lodge and Harrison, 2019; Kong et al., 2018; Ronconi et al., 2022). With the increasing prevalence of digital materials in education and daily life, the dynamics of reading are changing. Many educational institutions now depend heavily on digital learning systems, such as Blackboard, which is utilized in thousands of schools and universities worldwide, leading students to interact with digital texts more frequently than with print. This shift affects not only reading habits but also comprehension strategies, as the approach to understanding information on screens differs significantly from that on paper (Kong et al., 2018). While digital texts are integrated across all educational levels, research has largely concentrated on university students and young children, leaving a gap in understanding its effects on adolescents, who are particularly influenced by digital interactions in both their studies and leisure activities. For this age group, who often experience lower motivation toward reading, the medium of text presentation may affect comprehension levels significantly, with digital reading sometimes perceived as easier or more enjoyable but potentially less effective for deep comprehension (Ronconi et al., 2022). Research consistently shows that reading on paper tends to support deeper comprehension, particularly when identifying central ideas in a text. Digital reading, by contrast, can foster a more superficial approach, as theorized in the *shallowing hypothesis*, which suggests that screens may encourage a less engaged reading style. This may be partly due to additional cognitive load from adjusting to screen glare, page scrolling, and digital interfaces, which can interfere with concentration and processing. Additionally, studies show that readers are less likely to notice their own distractions while reading on screens, leading to overconfidence in their comprehension accuracy. This effect is seen particularly among adolescent readers, who may not yet have developed the skills to synthesize and prioritize information on screens as effectively as adults. The self-assessment of understanding is often less accurate in digital contexts, contributing to the overestimation of comprehension (Ronconi et al., 2022). In terms of reading speed, studies show minimal differences between digital and paper formats, implying that speed is less affected by the medium. However, reduced effort and focus during digital reading may cause more rapid but less thorough reading, particularly among younger readers, whose comprehension may suffer as a result. Despite the inevitability of digital reading's continued rise, these findings underscore the importance of teaching readers, especially students, strategies to mitigate screen-related distractions and enhance metacognitive

skills for more effective digital comprehension (Kong et al., 2018).

2.3.2 The impact of Covid-19

In 2020, the world changed in an unforeseen way with the Covid-19 pandemic, which forced everyone to stay at home and adapt their daily lives. Not everyone caught Covid, but its effects were felt universally, leading to significant changes. When it comes to attention issues with people who caught Covid-19, a lot of studies have come forward in the last few years exploring these impacts.

The COVID-19 pandemic has not only impacted global health through acute illness but has also left a significant mark on survivors' cognitive functions. Among the post-COVID-19 complications, cognitive deficits (often termed "brain fog") stand out, manifesting as slowed thinking, forgetfulness, and difficulties in concentration and attention. These impairments have been documented extensively, with approximately 20% of survivors showing objective cognitive deficits, particularly in memory and attention. These issues affect daily functioning, work productivity, and overall quality of life (Velichkovsky et al., 2023; Arbula et al., 2024).

One key aspect of post-COVID-19 cognitive dysfunction is the disruption of short-term memory and attention. Survivors frequently exhibit poorer performance in verbal and visual short-term memory tasks, as well as difficulties in selective attention, as evidenced by their lower scores in tasks such as the Flanker and Visual Arrays tests. These deficits suggest that the cognitive disruptions observed in post-COVID patients can not only subjective complaints but also measurable and noticeable (Velichkovsky et al., 2023). The deficits are hypothesized to stem from neuroinflammatory processes, structural changes in gray and white matter, and potential damage to the hippocampus, all of which are compounded by high levels of stress and anxiety commonly reported in this population (Arbula et al., 2024). However, while standardized neuropsychological tests indicate mild objective deficits, there is a clear discrepancy between subjective complaints and objective findings. This gap highlights the complex interplay between the psychological and neurological sequelae of COVID-19. Arbula et al. (2024) report that survivors often perceive their cognitive difficulties as more severe than what is detected in clinical evaluations, pointing to a potential amplification of symptoms by underlying psychological distress.

Psychological factors such as depression, anxiety, and **PTSD** are significant contributors to post-COVID cognitive impairments. Elevated stress levels and associated increases in cortisol have been linked to declines in memory and attention, further exacerbating cognitive difficulties. Moreover, 61% of post-COVID patients report failures in prospective memory (essential for planning and remembering future tasks) underscoring the depth of cognitive disruption. This pattern aligns with broader functional cognitive disorders,

including fatigue, mental lapses, and difficulties with word retrieval, all of which are consistent with the experience of “brain fog” (Velichkovsky et al., 2023; Arbula et al., 2024).

But the people who caught COVID-19 are not the only ones that were affected by it. The whole world felt its effects in one way or another. This is especially true for children and young people, who experienced two to three years of historically unprecedented disruption to their formative years. The pandemic reshaped their education, social development, and daily routines in ways that had no modern parallel. While the full extent of these impacts may only become clear in the years to come, early studies and observations reveal concerning trends.

One of the most prominent effects of the pandemic on children and adolescents was the disruption to their attention spans and self-regulation. The closure of schools, interruption of daily routines, and emotional stress led to increased impulsivity, reduced persistence in tasks, and difficulty concentrating. Studies found that children from families experiencing significant pandemic-related disruptions, such as financial instability or bereavement, exhibited more pronounced difficulties in self-regulation. Parental anxiety further exacerbated these challenges, with children of anxious caregivers struggling more to concentrate and manage their impulses. Surprisingly, children with pre-existing **ADHD** diagnoses demonstrated relative stability in comparison with children without **ADHD**, likely due to their ongoing access to structured support systems, such as therapy or medication (Shuai et al., 2021; Raghunathan et al., 2021).

The shift to online learning posed additional challenges, particularly for younger children who struggled with short attention spans and for older students who lost motivation and interest in academics. Families faced obstacles such as poor internet connectivity, lack of proper learning devices, and limited capacity to help children navigate the new educational platforms. Many parents described remote learning as frustrating, with some noting that their children became emotionally distressed, disengaged, and academically stagnant. The lack of face-to-face interaction with teachers and peers further alienated students, contributing to feelings of isolation and, in severe cases, depression, causing a very damaging effect on one’s ability to maintain focus (Rapaport and Polikoff, 2024).

The return to in-person schooling brought its own set of challenges. Children who had spent months in isolation found it difficult to adapt to structured environments and rebuild social relationships. Teachers and caregivers observed reduced motivation and attention, along with heightened emotional struggles in some students. For children who had experienced severe mental health issues during the pandemic, this adjustment period was particularly taxing. Schools and families reported an urgent need for targeted interventions to support these students in regaining their pre-pandemic levels of cognitive and emotional functioning (Raghunathan et al., 2021; Rapaport and Polikoff, 2024).

An issue that can also be traced back to the pandemic is the rise in **SMA**. The lack of physical interaction created a reliance on social media, fulfilling a need for belonging but often resulting in compulsive behaviors. Studies indicated that this overuse was driven by the frustration of social needs rather than their satisfaction, leading to increased feelings of loneliness and depression. While social media helped mitigate immediate social isolation, its addictive nature disrupted natural attention cycles and exacerbated emotional vulnerabilities (Cheng and Lau, 2022).

2.4 Attention Training

As previously mentioned, the primary goal of this Master's project is to develop a web platform for **Attention Training (AT)**. **AT** has gained widespread acceptance as an effective tool and has seen substantial growth over the past few decades. Its benefits extend to a broad audience, supporting individuals who struggle with attention-related challenges as well as those looking to enhance their focus and mental strength.

In this section, the evolution of **AT** is explored, delving into its historical roots and development over time. Additionally, various methods used to train attention are examined.

2.4.1 History

The concept of **AT** has roots that trace back more than a century, marking its significance in psychological research and practice. Among the pioneers of this field, Yuzero Motora (Figure 13) stands out as one of the earliest to develop structured methods for training attention, particularly in children with focus-related difficulties. Born in 1858, his groundbreaking work positioned him not only as a trailblazer in **AT** worldwide but also as the founder of modern psychology in Japan (Takeda T, 2015; Sato et al., 2016).

Motora's academic journey was deeply influenced by his mentor, Granville Stanley Hall, a pivotal figure in American psychology and a student of Wilhelm Wundt. Hall's focus on child development and learning processes profoundly shaped Motora's interest in cognitive and behavioral improvement among children. Upon returning to Japan in 1888 after studying under Hall, Motora established the first psychology laboratory in the country and began his exploration of perception, attention, and educational psychology. This foundational work set the stage for his later focus on **AT** (Takeda T, 2015).

His initial efforts in **AT** began in the early 20th century within the context of Japanese public schools. Recognizing the difficulties faced by children who displayed attention deficits, Motora developed innovative experiments to assess and improve their focus and cognitive performance. His approach involved



Figure 13: Yuzero Motora (from Wikipedia)

non-pharmacological interventions, reflecting a progressive understanding of cognitive challenges that preceded the modern diagnosis of **ADHD** (Takeda T, 2015).

Central to Motora's method was his theory that attention could not be trained directly but rather indirectly, through sensory exercises targeting vision and hearing. For instance, one exercise involved a machine that challenged participants to identify specific colors amid distractions, while another required children to listen to and distinguish sounds produced by various bells. These exercises aimed to enhance what he termed the "rule of thought" (organizing ideas) and the "rule of effort" (sustaining focus amidst distractions), which he believed were critical to managing attention effectively (Takeda T, 2015).

Motora was notably ahead of his time in addressing attention deficits, a domain that was largely underexplored in his era. His differentiation between concentration and the ability to shift focus anticipated later cognitive theories. While other historical figures laid the theoretical groundwork for understanding attention, Motora uniquely emphasized practical training and measurable improvement. As such, the influence of Motora's work is still discernible today. While his specific methods were not widely adopted after his death in 1912, his emphasis on structured **AT** informed later psychological and educational practices. His early adoption of empirical testing and his integration of psychology into education highlight his lasting impact (Takeda T, 2015; Sato et al., 2016).

2.4.2 AT Approaches

The platform to be developed aims to train attention by implementing practices designed to strengthen cognitive control and improve self-regulation. At its core, **AT** involves repetitive exercises that engage executive control mechanisms, such as conflict resolution and working memory tasks. These practices seek to

foster the development of brain networks responsible for executive functions and task performance (Tang and Posner, 2009).

Various methods have been shown to enhance attention and self-regulation. Tang and Posner (2009) broadly categorizes them, into two groups: **Attention Training (AT)** and **Attention State Training (AST)**.

In this Master's project, these methods will be divided in three groups:

- **Attention State Training:** Doesn't use any type of technology, focus on body-mind balance and restoration.
- **Simple Attention Training:** Repetitive cognitive practice to improve specific tasks using simple technology, for example computer programs.
- **Advanced Attention Training:** Repetitive cognitive practice to improve specific tasks using complex methods, for example **Virtual Reality**.

This three-part division was chosen because the two types of **AT** differ significantly in terms of test complexity and the technologies involved. This distinction reflects not only the varying levels of technological integration but also the increasing sophistication required to deliver advanced training methods. Separating these categories allows for a clearer justification and analysis of the unique demands and potential benefits of each approach, enabling a more targeted discussion of their respective impacts.

Attention State Training

Attention State Training (AST) emphasizes achieving changes in mental and physical states through holistic practices without targeting specific cognitive tasks. Drawing on traditions such as mindfulness, meditation, and exposure to natural environments, **AST** aims to foster body-mind balance and restoration. Its approach typically involves two phases: an initial stage requiring effortful control and discipline, followed by a more advanced stage of effortless practice. **AST** not only enhances attention but also produces transferable benefits across emotional regulation, stress management, and social behaviors. By engaging the autonomic nervous system, **AST** helps create a relaxed yet alert mental state, complementing the cognitive focus developed through **AT**. This type of training doesn't really look to foster the development of brain networks like the other two types seek to do (Tang and Posner, 2009; Posner et al., 2015).

It can be said that **AST** involves training brain states to cultivate conditions conducive to attention and self-regulation. Practices such as exposure to nature, aerobic exercise, and meditation have shown significant benefits for cognitive function. Spending time in natural environments, for instance, has been linked to

enhanced attentional capacity, while aerobic exercise broadly improves cognitive functions, including attention. Meditation, as a core component of **AST**, directly counters mind wandering by fostering sustained focus. It achieves this through two approaches, concentrating on specific content, such as breathing, or maintaining a present-centered awareness without attachment to thoughts. These methods can result in improved executive attention and mood, characterized by heightened positive affect and reduced negative states (Tang and Posner, 2009; Posner et al., 2015).

An example of **AST** in practice is the mindfulness training program tried by Mrazek et al. (2013). His study emphasizes focused-attention meditation, teaching participants to anchor their attention on sensory experiences, particularly breathing, while effectively managing distractions. Techniques such as distinguishing thoughts, reframing distractions, and counting exhalations were employed alongside sensory mindfulness exercises and daily 10-minute meditation sessions outside of the classroom.

The study revealed that a two-week mindfulness training program led to significant improvements in working memory capacity and reading comprehensions. These cognitive enhancements were attributed to reduced mind wandering, facilitated by sustained attentional focus during mindfulness practices. Notably, the observed reduction in mind wandering is linked to decreased activity in the **DMN**, associated with task-irrelevant thoughts. By decreasing **DMN** activation, mindfulness enhances focus and prevents task-relevant information from being displaced by distractions. This decreased activity in **DMN** doesn't mean that the neural network is being trained (because **AST** doesn't cause that), but it shows that the program helps people getting out of their natural mind wandering state. Unlike **AT**, this approach demonstrates improvements in attention and memory without direct task-specific practice, suggesting the broader applicability of **AST** techniques (Mrazek et al., 2013).

Another study by Haukaas et al. (2018) also touched on a mindfulness intervention, this time in a three-week three-session trial intervention. The intervention included sessions that introduced mindfulness concepts and guided breathing exercises in the first week, self-compassion and loving-kindness meditation in the second, and strategies to integrate self-compassion into daily life during the final session. The study claims and defends that mindfulness practices enhance attention and promote emotional regulation and that the results were maintained at the 6-month follow-up. It also noted that reductions in anxiety and depression symptoms were observed, as well as positive developments in attention flexibility.

In conclusion, in an increasingly technology-driven world, where remote work and constant connectivity dominate, practices aligned with **AST** offer simple yet effective ways to support mental and physical well-being. Methods such as spending time in nature or engaging in guided mindfulness exercises require minimal effort and resources (taking often only a small investment of time) and can have a profound

impact on cognitive and emotional health. With this **AST** can be a versatile and complementary approach to both simple and advanced **AT**, as training brain states can also improve overall attention, memory, and emotional resilience. It also can address broader challenges like anxiety and stress, making it a valuable companion to cognitive training methods. Techniques such as mindfulness, aerobic exercise, and exposure to natural environments not only foster attention but also improve executive functioning and mood. Ultimately, the simplicity and accessibility of **AST** practices really show their potential to make a significant difference in individuals' lives.

Simple Attention Training

Attention Training (AT) is a structured approach designed to enhance cognitive abilities through the repetitive practice of tasks that engage executive control mechanisms. These tasks often involve managing conflict, maintaining focus, and utilizing working memory, all of which require sustained attention and deliberate effort. The primary aim of **AT** is to train and strengthen the executive attention networks, non-autonomic control systems responsible for higher-order cognitive functions. While the exercises can lead to mental fatigue due to their intensity, the benefits extend beyond attention, with measurable improvements observed in other cognitive domains such as memory and problem-solving (Tang and Posner, 2009; Posner et al., 2015).

Since **AT** is a subset of cognitive training, it often overlaps with techniques that enhance broader cognitive abilities. By developing attention, many of these methods simultaneously improve other type of cognitive skills. Whether through digital exercises or simpler, analog tools, training the brain in general helps it grow healthier and more versatile.

Simple **AT** focuses on practical, accessible interventions. These methods range from computer-based exercises designed to provide targeted cognitive challenges to non-digital interventions that utilize straightforward materials like headphones or even pen and paper.

In the aforementioned study from Haukaas et al. (2018), there was mention of the work done in it around mindfulness. However, it wasn't the only thing analyzed in that article, as it also included another different intervention centered around **Attention Training Technique (ATT)**.

The **Attention Training Technique** by Wells (1990) is one component of metacognitive therapy (Wells, 2009), a form of cognitive-behavioral therapy. **ATT** utilizes a structured 12-minute auditory listening exercise, divided into three distinct phases: **Selective Attention** (5 minutes), where participants focus on specific auditory stimuli while ignoring others; **Attention Switching** (5 minutes), which involves rapidly shifting attention between different auditory inputs; and **Divided Attention** (2 minutes), during which

participants simultaneously attend to multiple stimuli. Through these practices, ATT fosters improved attentional skills in many ways.

By improving the specific types of attention targeted in the training, **ATT** also generates broader benefits for individuals. It helps people rediscover control over their attention, challenge false beliefs about attentional limitations, and cultivate the ability to move attention flexibly across different stimuli and contexts.

The effectiveness of **ATT** extends beyond attentional improvement, as evidenced in some studies. In the work of Haukaas et al. (2018), participants engaged in **ATT** exercises and discussions centered on self-focused versus external attention. The intervention yielded significant reductions in anxiety and depression symptoms, while enhancing self-compassion, and attention flexibility. Attention flexibility emerged as a crucial mechanism, suggesting its potential as a target for optimizing treatment strategies for emotional disorders.

Similarly, in the work of Fergus and Limbers (2019), **ATT** was examined as an intervention for test anxiety in adolescents. The study revealed notable reductions in test anxiety for students with strong metacognitive beliefs about worry. These reductions persisted at a follow-up, highlighting **ATT**'s potential for sustained effects. The study also emphasized that the effectiveness of **ATT** is contingent on baseline metacognitive profiles, as students with weaker positive metacognitive beliefs experienced limited benefits. This suggests that targeted screening for individuals most likely to benefit could optimize outcomes in educational settings.

By addressing worry directly and disengaging individuals from maladaptive cognitive patterns, **ATT** not only improves attention processes, but also mitigates anxiety, contributing to better emotional and cognitive self-regulation. As an auditory-based form of simple **Attention Training**, **ATT** demonstrates significant positive outcomes in both clinical and educational contexts.

AT and cognitive training can have many positive impacts in different people from all ages, from young children (for example Kirk et al. (2019) and Tamm et al. (2013)) to older adults (for example Klimova (2016) and Lampit et al. (2014)). In the case of young children, getting in touch with computerized methods and exercises not only to train attention, but also learning can help them get more in touch with new technologies and help them develop tools for the future such as computational thinking (Araújo et al., 2024).

AT and cognitive training encompass structured interventions designed to enhance attentional control and other cognitive abilities through focused and repetitive tasks. These interventions aim to strengthen brain networks involved in executive functioning, such as working memory, multitasking, and attentional modulation, thereby fostering neuroplasticity and creating transferable cognitive improvements. Many studies illustrate the varied approaches and outcomes of **AT** highlighting its ability to improve reaction

times, attentional flexibility, and multitasking performance. These findings show the relevance of **AT** in diverse contexts, including aging populations, educational environments, and scenarios requiring rapid cognitive adaptation (Aben et al., 2019; Zendel et al., 2016; Baniqued et al., 2014).

It is also important to note that these diverse contexts also require well-thought-out and calculated exercises and conditions, that suit the participants and their improvement. It has been shown for example that home-based training for example usually causes worse results and fewer improvements when compared to training in controlled conditions (Lampit et al., 2014).

The training tasks employed in the work of Aben et al. (2019) were designed to evaluate the potential for attentional modulation and its generalization across varying contexts, with the help of E-Prime. The primary training task utilized an arrow-priming paradigm, structured to challenge and enhance cognitive control mechanisms. During the trials, participants were required to respond to the direction of a target arrow, with congruent trials (where all arrows pointed in the same direction) and incongruent trials (where the target arrow's direction conflicted with flanking arrows). Following the training phase, a direct transfer task was introduced to measure how well participants had internalized the attentional control skills developed during the primary task. This task mirrored the arrow-priming paradigm but excluded the inducing trials, focusing exclusively on the core elements of responding to the direction of the target arrow. In another task, numerical stimuli replaced the arrow stimuli. Participants were tasked with responding based on whether a number, ranging from 1 to 9 (excluding 5), was greater or less than 5. By shifting the focus to numerical comparison, this phase assessed whether the attentional skills honed during the primary training could transfer to related but distinct tasks. The last task, utilized the **AX Continuous Performance Task (AX-CPT)** to probe the extent to which training could generalize to entirely novel scenarios. The **AX-CPT** involved four cue-probe combinations, AX, AY, BX, and BY, with AX trials representing the majority (70%) and requiring a target button response. In the remaining trials, AY, BX, and BY, participants had to press the non-target button. BY trials, involving entirely invalid cue-probe pairs. This task demanded sustained attention and dynamic adjustment of responses, offering insights into the adaptability of attentional control mechanisms trained in earlier phases. All these tasks from Aben et al. (2019) can be seen in Figure 14.

a) is the target arrow training task, b) the direct transfer task, c) the numerical stimuli task, d) the **AX-CPT**

The study by Zendel et al. (2016) used dual-task paradigms to train attentional allocation and multitasking abilities. Participants engaged in an alphanumeric equation task and a visual detection task, with experimental conditions that included single-task, fixed divided attention, and variable divided attention exercises. Variable training required alternating attention allocation between tasks (80%-20% or 50%-50%), with performance feedback provided to optimize cognitive deployment. The dual-task condition mim-

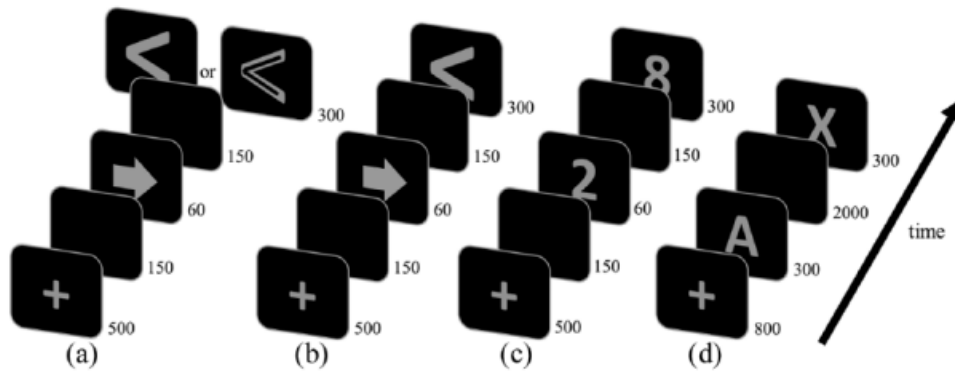


Figure 14: The four tasks from Aben et al. (2019)

icked real-world multitasking scenarios, and participants' performance across different attention allocation schemes demonstrated how adaptive strategies could enhance multitasking efficiency.

Baniqued et al. (2014) examined a suite of brain-training games targeting reasoning, working memory, and attentional control. Tasks included the N-back, attentional blink, and task-switching exercises. For instance, task-switching required participants to alternate between judging numbers as odd/even or greater/less than five, based on background color. Another example is the trail making task, in which participants first connected numbers distributed across a sheet of paper by drawing a line between numbers in ascending order. They then connected numbers and letters in alternating and ascending order on a second sheet. Exercises analyzed many capabilities, for example, response accuracy and processing speed under varied conditions. This approach allowed researchers to assess whether such gamified interventions could yield transferable cognitive benefits beyond the trained tasks.

The findings from these studies collectively demonstrate that **AT** significantly enhances various cognitive functions. Aben et al. (2019) reported persistent improvements in reaction times and attentional modulation across tasks. This shows the durability of training effects and their applicability to diverse cognitive contexts. Similarly, Zendel et al. (2016) revealed substantial gains in multitasking accuracy and adaptability among older adults, particularly in the variable **AT** group, highlighting the potential of targeted training to mitigate age-related cognitive decline. Baniqued et al. (2014) demonstrated that a brain game intervention could enhance divided attention and task-switching abilities, particularly for individuals with lower baseline cognitive abilities. Together, these studies validate the efficacy of **AT** in enhancing various types of attention, attentional control and many more attentional and cognitive functions.

Baniqued et al. (2014) touched on the use of video games for **AT**. The reality is that he is not the only one who defends that video games can bring attention improvements, either with the intention of training attention or as an unintended benefit (Kirk et al., 2019; Mishra et al., 2012; O'Brien et al., 2013; Wang

and Huang, 2013; Vergara and Gallardo, 2023; Moradi et al., 2024). This growing body of research highlights how the interactive and immersive qualities of video games create unique opportunities to enhance attentional capacities across diverse populations.

Video games present unique advantages as tools for **AT**, offering personalized and engaging environments that can sometimes surpass traditional methods. Through their interactive elements and the integration of multimedia, video games simulate diverse cognitive challenges, enabling repeated practice of attention-focused operations. These iterative and dynamic practices are inherently motivating and highly adaptable, allowing individuals to experience scenarios that would be difficult or impossible to replicate in real life. As such, they represent a promising approach to designing **AT** programs that merge entertainment with measurable cognitive benefits (Wang and Huang, 2013).

It as already been mentioned that **AT** can have positive impacts in people of all different ages.

Interactive video games hold significant potential for helping children develop various cognitive functions from an early age. Beyond enhancing attention and reaction times, these games can also foster problem-solving abilities, critical thinking, and decision-making skills. Such games provide an engaging platform for children to practice and refine cognitive operations in an iterative and stimulating manner. Research delves into this topic not only in children without diagnoses but also in those with specific conditions, such as ADHD, showing improvements across a wide range of cognitive domains (Kirk et al., 2019; Vergara and Gallardo, 2023; Moradi et al., 2024; Wang and Huang, 2013).

For example, Kirk et al. (2019) uses a program called the Tali Train that involves a gamified **Attention Training** regimen tailored for children. Over five weeks, participants engaged in 25 daily sessions of 20 minutes each, focusing on selective, sustained, and executive attention processes. The program used an adaptive difficulty system to align challenges with the child's performance, integrating seamlessly into school schedules via touch-screen tablets. Similarly, another study Vergara and Gallardo (2023) introduced a series of interactive activities targeting cognitive skills. These included tasks like stimulus selection, identifying hidden figures, solving mazes, and memorizing drawings, which encouraged attention and problem-solving in an entertaining format. Examples from both studies can be seen in Figure 15. While these are just two examples, they highlight the vast potential for creating diverse game-based tools aimed at enhancing children's cognitive abilities through playful, targeted training.

AT using video games for young children has demonstrated a wide range of benefits. Studies have shown significant improvements in both short-term and long-term perception, working memory, and various types of attention, such as selective and sustained attention. These enhancements have contributed to better academic performance and classroom engagement, as evidenced in some cases by a decrease in com-



Figure 15: Two examples of games as exercises (left from Kirk et al. (2019) and right from Vergara and Gallardo (2023))

mission errors. Additionally, reductions in inattentive and hyperactive behaviors have been reported, highlighting the effectiveness of these interventions in managing symptoms commonly observed in conditions like ADHD. This shows that **AT** programs for children not only bolster cognitive performance but also promote behavioral regulation in young learners (Kirk et al., 2019; Vergara and Gallardo, 2023; Moradi et al., 2024).

Action video games have consistently demonstrated their potential to enhance attention and working memory in young adults. These games uniquely challenge players to process complex, fast-paced environments, fostering the development of several types of attention, including spatial, temporal, object-based, and sustained attention. For instance, action game players often excel at tracking multiple moving objects amidst distractions, showcasing superior object-based attention. These players also exhibit faster response times without compromising accuracy in tasks requiring sustained attention and impulsivity control, such as the Test of Variables of Attention, which measures the ability to respond to rare targets while resisting frequent non-targets (Mishra et al., 2012).

These findings demonstrate that action video games are a compelling example of how games can naturally enhance attention stimuli without being explicitly designed as training tools. This suggests that incorporating attention-focused games into structured training programs could provide a valuable approach for conducting **AT**. By leveraging the engaging and dynamic nature of action games, such interventions could amplify the benefits of **AT**, offering a practical and enjoyable method to foster cognitive improvements in young adults (Mishra et al., 2012).

Video game-based approaches have also proven effective for **AT** in older adults. For instance, adaptations derived from speed-of-processing training have shown significant improvements in visual attention, perception, processing speed, and memory (O'Brien et al., 2013). This system includes adaptive exer-

cises designed specifically for seniors, creating a game-like environment to make cognitive training both accessible and engaging. Such interventions highlight the potential of video games not only to slow the natural decline of cognitive functions associated with aging but also to help maintain and even enhance key mental faculties.

When discussing the **Attention Training Technique**, it was highlighted as a powerful tool for improving cognitive performance, and its applications have extended to anxiety-related disorders. Evidence suggests that other forms of simple **AT** can be adapted to address challenges like social anxiety, providing targeted support for individuals in high-stress situations. For example, Heeren et al. (2011) demonstrated that **AT** significantly benefits individuals with social phobia, particularly in public speaking scenarios. The study revealed that **AT** reduced anxiety during speech tasks by enabling participants to disengage more effectively from social threat cues, offering a practical approach to mitigating symptoms associated with social anxiety, with the help of better selective attention control.

It was previously mentioned that **AT** and **AST** can be mixed and tried together and the study by Mrazek et al. (2022) tries to somewhat that. He does this by integrating mindfulness-based **AT** into educational settings, showcasing its potential to improve focus and emotional regulation among high school students. Conducted over 22 days, the intervention employed digital platforms to deliver four structured lessons complemented by daily exercises that aligned with students' preferences, such as listening to tailored music genres. While teachers facilitated the program, students independently engaged with the content, and their progress was monitored via pre-test and post-test surveys. Key skills taught included anchoring attention, maintaining focus, and releasing distractions, all of which were reinforced through daily practice. The study also measured multiple outcomes, such as classroom focus, mind-wandering, emotional regulation, and students' beliefs about their ability to improve attention. Self-reported benefits and lesson completion rates further evaluated the program's feasibility and impact, demonstrating its accessibility even under remote learning conditions.

The results underscored the intervention's effectiveness in enhancing critical attentional and emotional competencies. Participants reported a significant reduction in mind-wandering during both daily life and classroom activities, along with noticeable improvements in self-perceived classroom focus and the adoption of growth mindsets about attention. Moreover, the program positively impacted students' emotional regulation by improving their ability to cognitively reappraise stressors, reflecting the dual benefit of **AT** in fostering both cognitive engagement and emotional resilience. These findings highlight the potential for scalable, digitally delivered mindfulness-based **AT** programs to address the growing attentional and emotional demands faced by adolescents, offering practical tools to support their academic and personal

development (Mrazek et al., 2022).

In summary, Simple **Attention Training** is highly adaptable, enabling the design of exercises suited to a wide range of contexts and individual needs. Its flexibility allows it to be used effectively across all age groups, from young children to older adults. Depending on the exercises applied, it can lead to significant improvements in various types of attention and working memory, demonstrating its broad cognitive benefits. Moreover, its potential extends beyond cognition, offering support in addressing challenges like anxiety. This versatile and ever-evolving field continues to grow, solidifying its place as an excellent tool for helping people across diverse circumstances.

Advanced Attention Training

The objectives of advanced **Attention Training** align closely with those of its simpler counterpart, aiming to enhance cognitive abilities through the repetitive practice of tasks that engage executive control mechanisms. However, what sets advanced **AT** apart is its integration of cutting-edge technologies, enabling more complex exercises and detailed analyses.

This section delves into two innovative approaches to **AT**, one using **Virtual Reality (VR)**, and the other utilizing **Eye Tracking** technology. Additionally, it explores studies employing advanced tools like **Functional magnetic resonance imaging (fMRI)** to analyze the effects of **AT** on the brain and neural networks. The aim for these technologies is not only enhance the training experience but also provide a deeper understanding of how **AT** impacts cognitive and neural processes.

VR has emerged as a groundbreaking tool in **AT** and cognitive rehabilitation. By simulating real-world environments, **VR** offers an immersive, three-dimensional platform that transcends traditional approaches like paper-and-pencil tests or standard computer-based tasks. Through head-mounted displays and other technologies, **VR** delivers precise visual and auditory stimuli, enabling ecologically valid settings that closely simulate real-life scenarios. This realistic approach not only improves attention during standard tests but also shows significant potential for broader cognitive rehabilitation efforts (Kim and Lee, 2022; Johansen et al., 2024; Lorentz et al., 2023).

Furthermore, **VR** can enhance user engagement through gamification, incorporating elements like goals, rewards, and adaptive difficulty levels. These features can increase motivation, creating a highly interactive environment conducive to effective **AT**. Repetitive tasks within **VR** environments, such as head movements in response to visual stimuli, can induce behavioral plasticity and improve attentional orientation. This highlights **VR** as a particularly valuable tool for addressing attention impairments and supporting recovery from cognitive deficits (Kim and Lee, 2022; Johansen et al., 2024; Lorentz et al., 2023).

Virtual Reality training also facilitates the delivery of performance feedback, which can significantly enhance attentional capacities. According to Kim and Lee (2022), participants who received immediate feedback during **VR** training demonstrated substantial improvements in overt attention, particularly in their ability to disengage from irrelevant stimuli. This feedback mechanism allowed individuals to monitor and adjust their responses in real-time, fostering greater self-regulation and attentional control. Furthermore, the study highlights the potential of **VR** based feedback interventions in addressing specific cognitive challenges, such as improving attentional efficiency in individuals with conditions like Sluggish Cognitive Tempo. An example of how feedback may be seen in this type of intervention is on Figure 16 where green is positive feedback and red is negative.



Figure 16: **VR AT** with the aid of feedback (from Kim and Lee (2022))

All in all, **VR** offers a sophisticated and engaging method for **AT**, leveraging immersive environments to simulate real-world tasks and improve attentional processes. Its ability to generalize treatment effects beyond laboratory settings, combined with the adaptability of its gamified elements, underscores its potential as a transformative tool in cognitive rehabilitation and attentional enhancement (Kim and Lee, 2022; Johansen et al., 2024; Lorentz et al., 2023)

While **VR** is a very valuable and promising way of **AT**, its application is sometimes limited by hardware requirements and the need for physical mobility. Eye tracking provides a more accessible alternative, delivering significant improvements in attention and cognitive performance by leveraging eye movement data to guide targeted interventions (Moon et al., 2022; Wang et al., 2024).

Eye tracking technology monitors eye movements such as fixations and smooth pursuits, providing real-time, millisecond-level feedback. This precision allows for the development of exercises that directly address deficits in attention, memory, and processing speed. Moreover, eye tracking is highly adaptable, offering tailored exercises that respond to individual attention habits. Its reliance solely on eye movements makes it particularly beneficial for individuals with motor or speech impairments, ensuring greater inclusivity and accessibility. Additionally, the interactive and adaptive nature of eye tracking tasks enhances

user engagement, driving more effective results (Moon et al., 2022; Wang et al., 2024).

Beyond attention improvements, eye tracking-based training holds promise for related cognitive domains, such as memory and executive function. Research has shown its potential in disturbances and conditions like schizophrenia and **ADHD**, improving cognitive processing speed, visual perception, and memory. Emerging evidence also suggests potential benefits for stroke patients, highlighting its utility in post-stroke cognitive impairment rehabilitation (Moon et al., 2022; Wang et al., 2024).

Eye tracking is not limited to **AT**. It has also been studied in contexts like anxiety management, with applications such as Eye Movement Desensitization and Reprocessing therapy. This therapeutic approach integrates bilateral eye movements to process traumatic memories, leveraging the smooth pursuit eye movement system. The technique demonstrates how eye tracking can extend beyond attention to address emotional and psychological challenges, further broadening its applications (Kapoula et al., 2010).

In summary, eye tracking represents a versatile and precise tool for enhancing attention and cognitive functions. Its adaptability and accessibility make it suitable for diverse populations, including individuals with physical impairments. By providing real-time feedback and personalized exercises, eye tracking not only improves attention but also holds promise in related areas like memory and anxiety management.

Advanced **AT** also included neuroimaging and electrophysiological techniques to assess the direct impacts of training on neural activity. These methods provide invaluable insights into the brain's plasticity and the specific mechanisms underlying cognitive work and improvement. The approaches **Functional magnetic resonance imaging (fMRI)**, **Electroencephalography (EEG)** and **Event-related potentials (ERPs)** will be analyzed in this section and how they offer good information of how **AT** influences brain activity and cognitive performance (O'Brien et al., 2013; Lorentz et al., 2023; Zendel et al., 2016).

fMRI enables the observation of brain activity by detecting changes in blood oxygenation levels, particularly during engagement with high-stimulus tasks. An example comes from a study using **VR** in an MRI scanner, where participants performed a visual attention task (Lorentz et al., 2023). The results revealed increased activation in clusters associated with attention, such as the visuomotor circuits essential for action preparation. Additionally, **fMRI** studies indicate that high-engagement stimuli often reduce activation in the **DLPFC** while increasing activation in the insula. These patterns suggest that stereoscopic **VR** and other immersive technologies can effectively stimulate neural regions tied to attentional control.

EEG is a non-invasive method for detecting brain states and capturing large-scale electrical signals from the nervous system and **ERPs** refer to neurophysiological signals that allow for the analysis of the timing of various cognitive processes. Among the most studied **ERPs** components in **EEG** is the N200, associated with attentional control, inhibitory mechanisms, and conflict monitoring. Studies have shown that the

N200 amplitude improves post-**Attention Training**, reflecting enhanced task performance and neural plasticity. For example, research demonstrates that even in older populations, **AT** can enhance the brain's capacity to form new neural connections (Zendel et al., 2016). Other **ERPs**, such as the N2pc (measuring visual attention allocation) and P3b (associated with attentional processing capacity), further highlight **EEG's** versatility in assessing attention and cognitive function (O'Brien et al., 2013).

These tools offer a window into the brain's adaptability, revealing improvements in cognitive performance and giving researchers a better understanding of the underlying neural mechanisms. By utilizing these technologies, researchers and clinicians can refine interventions, ensuring targeted, effective, and scalable solutions for cognitive rehabilitation. As this field continues to grow, these advanced tools can play an essential role in expanding our understanding of **Attention Training's** transformative potential.

Summary

This section explored **Attention Training** by examining its diverse methodologies, showcasing the vast potential and adaptability of this field. Through the investigation of various approaches, it becomes clear how expansive and impactful **AT** can be for cognitive development and well-being.

The first approach, **Attention State Training (AST)**, focuses on achieving changes in mental and physical states through holistic practices like mindfulness, meditation, and exposure to natural environments. Unlike other types of training, **AST** emphasizes body-mind balance and relaxation rather than directly targeting cognitive networks. It offers benefits that extend beyond attention, including emotional regulation and stress management, providing a complementary strategy to the cognitive focus of other **AT** methods.

Simple Attention Training (AT), on the other hand, takes a more structured and task-focused approach. It aims to improve cognitive abilities through repetitive tasks that engage executive control mechanisms. This type of training includes practical, accessible exercises, ranging from computer-based challenges to analog tools like pen-and-paper tasks or audio-based interventions. Video games also play a role in this domain, offering engaging and effective methods for cognitive enhancement. Simple **AT's** focus on accessibility and broad applicability makes it a versatile option for diverse age groups and contexts.

Finally, **Advanced Attention Training (AT)** builds on the principles of its simpler counterpart while leveraging advanced technologies such as **VR** and eye tracking. These tools enable the creation of immersive, interactive, and highly personalized training environments. Advanced **AT** also integrates neuroimaging techniques like **fMRI** and electrophysiological methods, such as **EEG** and **ERPs**, to analyze the neural impacts of training. These approaches not only enhance the training experience but also provide insights into the brain's plasticity and cognitive mechanisms, allowing for a deeper understanding of how **AT** affects

neural processes.

As this Master's project focuses on developing a web-based application, the selected approach for implementation is Simple **AT**. This decision is based on its practicality, accessibility, and effectiveness in delivering targeted attention exercises. The next section, which is the proposal, will delve into the specifics of this implementation.

Chapter 3

BEAM: Proposal

In this Master's project, a web-based platform will be developed to provide an accessible and efficient tool for **Attention Training** and diagnostic evaluation. Its name will be **Brain Engagement and Attention Management (BEAM)** and it will integrate structured exercises designed to enhance attention and cognitive abilities, accompanied by diagnostic tools to assess user performance. Additionally, the platform will be tested to evaluate the effectiveness of the implemented techniques and their impact on attention. In this chapter, **BEAM** system will be designed. The entities present in the domain are identified and a data model is proposed. To identify everything that is required to **BEAM**, the most important use-cases are presented and then the requirements are listed. The system architecture is sketch through a block diagram. At last the technological decisions to implement **BEAM** are discussed. These components will serve as the foundation for building a versatile and effective solution, capable of advancing **AT** and providing valuable diagnostic insights.

3.1 Data Model

To gain a better understanding of the problem at hand, as well as the entities involved in the system, the domain model shown in Figure 17 was developed. This is the initial-proposed model and during development there might be some changes.

The data model designed in represents the key entities in the system and their relationships. At the core, the User entity includes attributes like name, email, password, and type, distinguishing between Patients and Therapists. A Therapist can care of multiple Patients, while each Patient has test results and is associated with multiple Notes, written by the Therapist. Patients perform Exercises, which store details such as name, duration, type, and results. These results contain metrics like score, reaction time, and accuracy. Exercises can be classified as either Diagnostic Tests (with an attempt number) or Training sessions (with difficulty level and session number). This structure ensures that therapists can track patient

progress efficiently while maintaining a structured record of tests and training sessions.

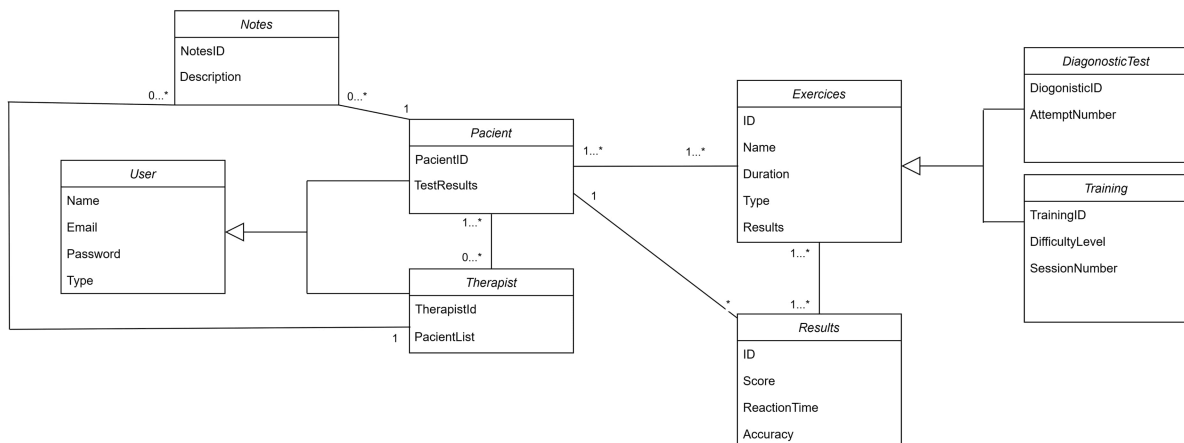


Figure 17: BEAM Data Model

To better understand the system's context, a *Use Case* diagram was developed (Figure 18). This diagram outlines some of the main functionalities of the system and the actors involved. It also identifies which functionalities each actor in the system will have access to.

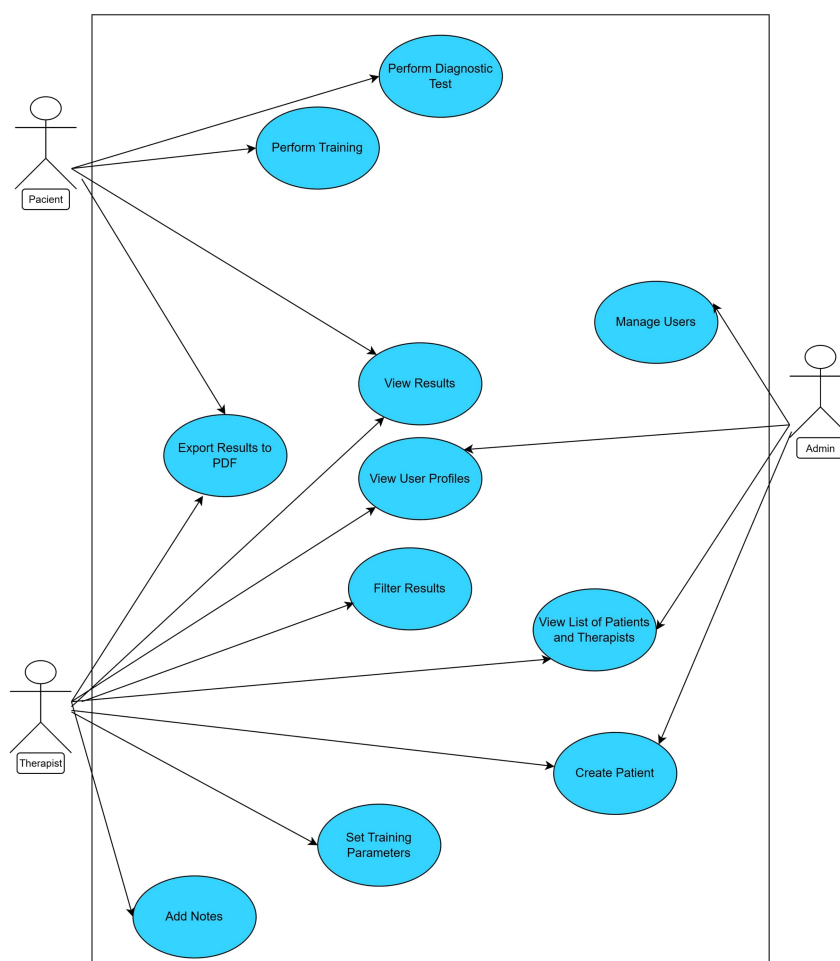


Figure 18: Use Cases Diagram

As depicted in the diagram shown in Figure 18, the system supports three types of actors: the Admin, the Therapist, and the Patient.

The **Patient** is the user in the platform who will perform the diagnostic tests and training exercises. Their access within the system is limited to functionalities directly related to these tasks, ensuring a straightforward and focused interaction.

The **Therapist**, on the other hand, has extensive privileges within the system. Therapists can manage a list of patients, review diagnostic and training results, add new patients, and configure training exercises. Their role is crucial for tailoring the system's functionalities to the needs of each patient.

The **Admin** is responsible for the overall configuration and maintenance of the platform. In the final implementation of the platform, it is likely that one or more therapists will assume the role of admin.

Below is a tabular specification of each *use case* considered, providing clarity and structure for the implementation of the system's functionalities.

Each table provides a detailed specification of a use case, including at least one associated actor. These

tables outline the normal scenario for each use case, describing the expected sequence of interactions between the actor and the system. Additionally, they may include alternative scenarios or failure cases, ensuring a comprehensive representation of the system's behavior in different situations in reaction to the user inputs and actions. Tables 1 to 11 describe formally the scenarios for each use case sketched in the diagram of Figure 18

Each use case description follows a standardized structure, listing the actors, preconditions, and postconditions, followed by a sequential table that details the interaction flow between the actor and the system, and any alternative scenarios that may occur.

Perform Diagnostic Test

Use case	1	
Primary Actor	Patient	
Secondary Actor	-	
Preconditions	The Patient is authenticated on the platform	
Postconditions	The test results are stored in the system	
Normal Scenario	Actor Input	System Response
	1 - The Patient navigates to the tasks page 3 - The Patient selects a diagnostic test 5 - The Patient performs the diagnostic test 7 - The Patient is redirected back to the tasks page	2 - The System displays the list of available tasks 4 - The System loads the selected test 6 - The System records the test results
Alternative 1: [Exit the task page] (Step 3)	3.1 - The Patient exits the task page	3.2 - The System sends the Patient to the new page

Table 1: Specification of the *use case* "Perform Diagnostic Test"

Perform Training

Use case	2	
Primary Actor	Patient	
Secondary Actor	-	
Preconditions	The Patient is authenticated on the platform	
Postconditions	The training results are stored in the system	
Normal Scenario	Actor Input	System Response
	1 - The Patient navigates to the tasks page 3 - The Patient selects a training exercise 5 - The Patient performs the training exercise 7 - The Patient is redirected back to the tasks page	2 - The System displays the list of available tasks 4 - The System loads the selected training exercise 6 - The System records the training results
Alternative 1: [Exit the task page] (Step 3)	3.1 - The Patient exits the task page	3.2 - The System sends the Patient to the new page

Table 2: Specification of the *use case* "Perform Training"

View Results

Use case	3	
Primary Actor	Patient, Therapist	
Secondary Actor	-	
Preconditions	The Patient or Therapist is authenticated on the platform	
Postconditions	The requested results are displayed to the user	
Normal Scenario	Actor Input	System Response
	1 - The Therapist navigates to the Patient's profile 3 - The Therapist selects the "View Results" button	2 - The System displays the list of results for the Patient 4 - The System displays the results
Alternative 1: [View Results in the Task Page] (Step 1)	1 - The Patient or Therapist navigates to the specific task page 3 - The Patient or Therapist selects the "View Results" button	2 - The System displays the exercise details and its related results 4 - The System displays the results for the selected exercise
Exception 1: [No results available] (Step 6)		6.1 - The System informs the user that no results match the filter criteria

Table 3: Specification of the *use case* "View Results"

Export Results to PDF

Use case	4	
Primary Actor	Patient, Therapist	
Secondary Actor	-	
Preconditions	The Patient or Therapist is authenticated on the platform	
Postconditions	The results are exported as a PDF file	
Normal Scenario	Actor Input	System Response
	1 - The Patient or Therapist navigates to the results page 3 - The Patient or Therapist selects the "Export to PDF" button	2 - The System displays the list of results 4 - The System generates a PDF file containing the selected results 5 - The PDF file is downloaded to the user's device
Exception 1: [No results available to export] (Step 2)		2.1 - The System informs the user that no results are available for export

Table 4: Specification of the *use case* "Export Results to PDF"

Filter Results

Use case	5	
Primary Actor	Therapist	
Secondary Actor	-	
Preconditions	The Therapist is authenticated on the platform	
Postconditions	The filtered results are displayed to the Therapist	
Normal Scenario	Actor Input	System Response
	1 - The Therapist navigates to the results page 3 - The Therapist selects a filter option (e.g., exercise type, date) 5 - The Therapist observes the filtered list	2 - The System displays the list of results 4 - The System applies the filter and updates the displayed results
Exception 1: [No results available] (Step 4)		4.1 - The System informs the Therapist that no results match the filter criteria

Table 5: Specification of the *use case* "Filter Results"

View User Profiles

Use case	6	
Primary Actor	Therapist, Admin	
Secondary Actor	-	
Preconditions	The Therapist or Admin is authenticated on the platform	
Postconditions	The selected user's profile is displayed	
Normal Scenario	Actor Input	System Response
	1 - The Therapist or Admin navigates to the dashboard 3 - The Therapist or Admin observes the list and searches for the user 4 - The Therapist or Admin selects a user from the list	2 - The System displays the list of users 5 - The System loads and displays the profile of the selected user
Exception 1: [User profile not found] (Step 4)		4.1 - The System informs the user that the selected profile is not available

Table 6: Specification of the *use case* "View User Profiles"

View List of Patients and Therapists

Use case	7	
Primary Actor	Therapist, Admin	
Secondary Actor	-	
Preconditions	The Therapist or Admin is authenticated on the platform	
Postconditions	The list of users is displayed	
Normal Scenario	Actor Input	System Response
	1 - The Therapist or Admin navigates to the dashboard 3 - The Therapist or Admin scrolls through the list	2 - The System displays the list of Patients and Therapists
Alternative 1: [Search for a specific user] (Step 3)	3.1 - The Therapist or Admin enters a search term in the search bar	3.2 - The System filters the list based on the entered search term
Exception 1: [No users found] (Step 2)		2.1 - The System informs the Therapist or Admin that no users are available

Table 7: Specification of the *use case* “View List of Patients and Therapists”

Manage Users

Use case	8	
Primary Actor	Admin	
Secondary Actor	-	
Preconditions	The Admin is authenticated on the platform	
Postconditions	The user information is updated or the user is deleted	
Normal Scenario	Actor Input	System Response
	1 - The Admin navigates to the user management page	2 - The System displays the list of users
	3 - The Admin selects a user to manage	4 - The System loads the user's information
	5 - The Admin updates the user's details and submits the changes	6 - The System saves the updated information and confirms the update
Alternative 1: [Delete a User] (Step 3)	3.1 - The Admin selects the delete option for a user 3.3 - The Admin confirms by entering their password	3.2 - The System prompts a confirmation screen 3.4 - The System deletes the user and confirms the action
Exception 1: [User not found] (Step 2)		2.1 - The System informs the Admin that the selected user does not exist

Table 8: Specification of the *use case* "Manage Users"

Create Patient

Use case	9	
Primary Actor	Therapist, Admin	
Secondary Actor	-	
Preconditions	The User is authenticated on the platform	
Postconditions	The new patient is added to the system	
Normal Scenario	Actor Input	System Response
	1 - The Therapist or Admin navigates to the "Create Patient" page	2 - The System displays the patient creation form
	3 - The Therapist or Admin fills in the form with the patient's name, email, and password	4 - The Therapist or Admin selects the Therapist to associate with the patient
	5 - The Therapist or Admin submits the form	6 - The System validates the data and creates the new patient
Exception 1: [Patient already exists] (Step 6)		6.1 - The System informs the Therapist that a user with the same email already exists
Exception 2: [Invalid form data] (Step 6)		6.1 - The System informs the Therapist of the specific errors in the form

Table 9: Specification of the *use case* "Create Patient"

Set Training Parameters

Use case	10	
Primary Actor	Therapist	
Secondary Actor	-	
Preconditions	The Therapist is authenticated on the platform	
Postconditions	Training parameters are configured for a patient or updated globally	
Normal Scenario	Actor Input	System Response
	1 - The Therapist navigates to the task management page 3 - The Therapist selects a specific task to assign to a patient 5 - The Therapist sets the difficulty level and toggles feedback options	2 - The System displays the list of tasks 4 - The System loads the task's configuration options 6 - The Therapist assigns the task to the desired patient 7 - The System confirms the parameters and saves the assignment
Alternative 1: [Edit Task Configuration] (Step 3)	3.1 - The Therapist selects the option to edit the task itself 3.3 - The Therapist modifies the task parameters, such as images or specific scenarios	3.2 - The System loads the task editor 3.4 - The Therapist saves the updated task configuration and names this version 3.5 - The System confirms the updates as a new way of doing this task and makes them available for future assignments

Table 10: Specification of the *use case* "Set Training Parameters"

Add Notes

Use case	11	
Primary Actor	Therapist	
Secondary Actor	-	
Preconditions	The Therapist is authenticated on the platform	
Postconditions	A new note is added to the patient's profile	
Normal Scenario	Actor Input	System Response
	1 - The Therapist navigates to the patient's profile 3 - The Therapist selects the Notes section 5 - The Therapist writes the note in the provided field 6 - The Therapist submits the note	2 - The System displays the patient's details 4 - The System displays a form for adding a new note 7 - The System saves the note and displays it in the notes list
Exception 1: [Empty Note Field] (Step 6)		6.1 - The System prevents submission and informs the Therapist to fill in the note

Table 11: Specification of the *use case* "Add Notes"

3.2 Requirements

Defining system requirements is a critical step in the development of any software application, as it ensures that all functionalities align with user needs and project goals. Analyzing carefully the use cases and their scenarios presented in the previous section for the **BEAM** platform it is possible to identify the following key specifications.

Functional Requirements

- The system must support different user types.

- The system must include a diagnostic test.
- The system must offer a training exercise.
- The system must store the results of diagnostic tests.
- The system must store the training results of users.
- The system must allow users to log in.
- The system must allow users to log out.
- The system must support different difficulty levels in training exercises.
- The system must store overall performance metrics for future analysis.
- The system must allow the registration of diagnostic test results.
- The system must allow the registration of training results.
- The system must enable filtering and sorting of results by date or type of exercise.
- The system must allow the addition of notes or observations to results.
- The system must allow the history of results to be viewed.
- The system must allow the assignment of tests.
- The system must support exporting results in PDF format for sharing or offline analysis.
- The system must allow the customization of training exercises based on difficulty levels.
- The system must allow the creation of different versions of exercises to provide new challenges.
- The system must provide feedback to users.
- The system must allow control over the provision of feedback during exercises.
- The system must record the duration of each exercise completed.
- The system must allow the registration of reaction times during exercises.
- The system must allow the registration of accuracy during exercises.
- The system must include explanations for how each exercise works.
- The system must provide comparative metrics to assess performance.

Non-Functional Requirements

Usability

- Therapists and Admins must be able to create Patients.
- Admins must be able to delete and manage Users.
- The system must include explanations for how each exercise works to guide users effectively.
- The system must record Patients' reaction times and accuracy during exercises.
- The system must provide comparative metrics to enable performance assessment against general averages.
- Therapists must be able to access diagnostic test results and training results of their patients.
- Therapists must be able to filter and sort results by date or type of exercise.
- Therapists must be able to add notes or observations to patient results.
- Patients must be able to view their history of results and see tests assigned to them by their Therapists.

Security

- The system must ensure data security through encryption of personal information and results.
- The system must include role-based access control to ensure appropriate permissions.

Maintenance and Support

- The system must allow easy addition of new features without major disruptions to existing functionalities.
- The system must include error logging and reporting to facilitate debugging.

Operational

- The system must work as a web application.

Cultural and Political

- The system must be in Portuguese.

3.3 Design

This section outlines the platform's design, focusing on the various pages and components it will encompass. This exercise is crucial to understanding the web flow and serves as a guide for structuring the application during the implementation phase.

Each part of the system will be examined individually, accompanied by brief descriptions to clarify their purpose and functionality. Finally, the complete system layout will be presented for an overarching view. It is important to note that the designs provided are simple mock-ups intended solely as a guiding reference. In Figure 19, the initial mock-ups of the platform are presented, showcasing the primary pages available before a user logs in. These include the home page, which provides general information about the platform, the registration page (restricted to therapists), and the login page for all users. After logging in, users are directed to the welcome page, where they can access personalized options based on their roles. Additionally, the mock-up includes a header design visible throughout the application, which facilitates navigation to key sections of the platform.

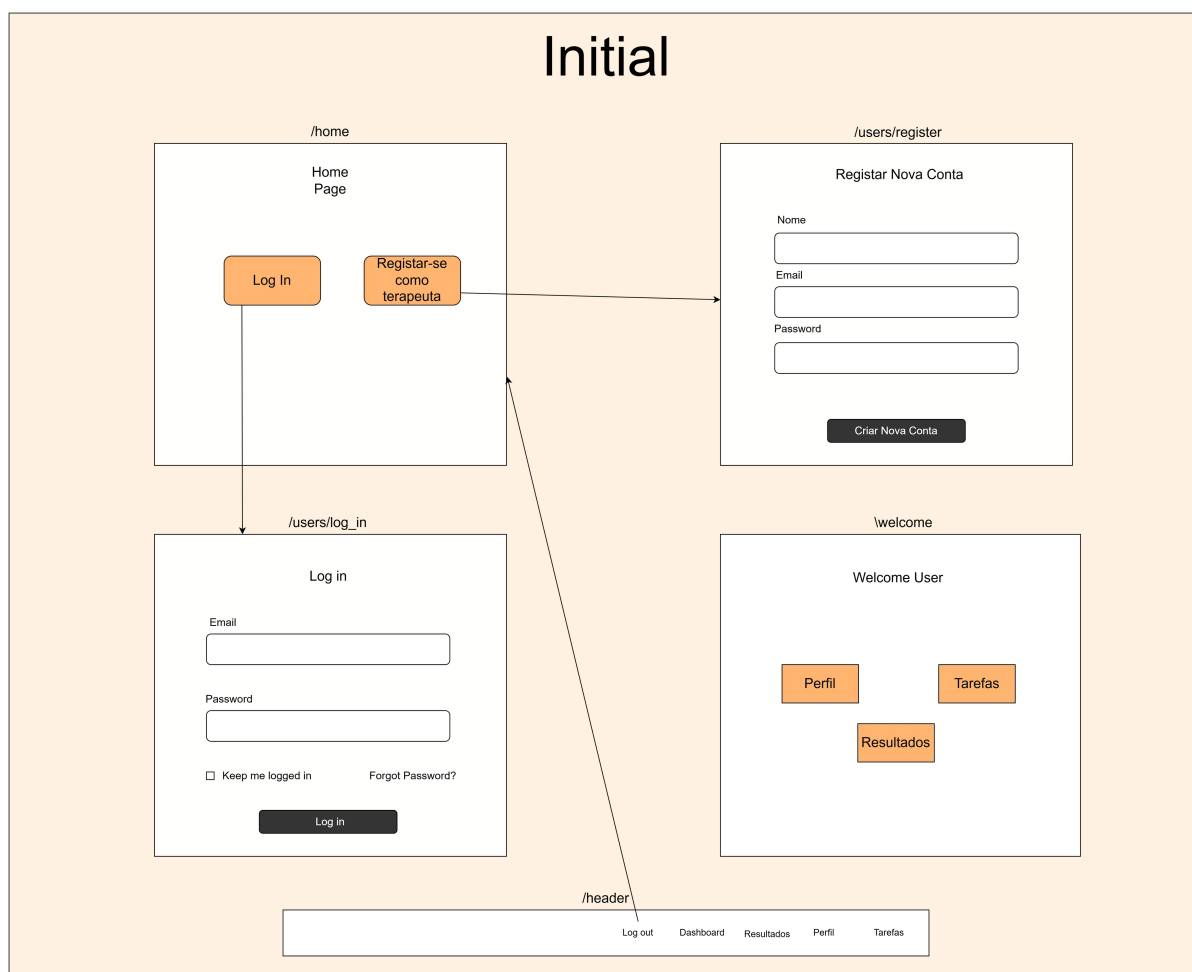


Figure 19: Initial Mockup

In Figure 20, the dashboard mock-up is presented, designed exclusively for therapists and admins, to provide an overview of platform activity. The dashboard displays the total count of therapists and patients registered on the platform, as well as separate lists for each group. These lists include a search function, allowing users to quickly find specific individuals by name. This centralized screen ensures that users can stay updated and manage platform interactions effectively.

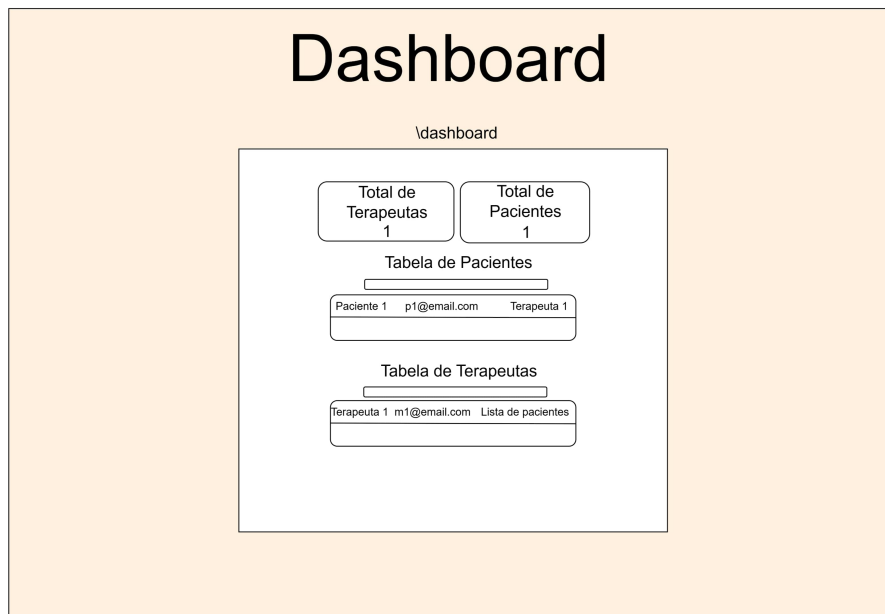


Figure 20: Dashboard Mockup

In Figure 21, the mock-ups for the tasks section are illustrated. The initial tasks page features a comprehensive table containing all tasks available within the application. Selecting a task redirects users to a dedicated page where a detailed explanation of the task's functionality is provided. Another screen showcases the exercise template editor, enabling therapists to adjust and customize tasks extensively to meet diverse needs. Additional mock-ups represent individual task screens, allowing users to interact with specific exercises tailored for training or diagnostics.

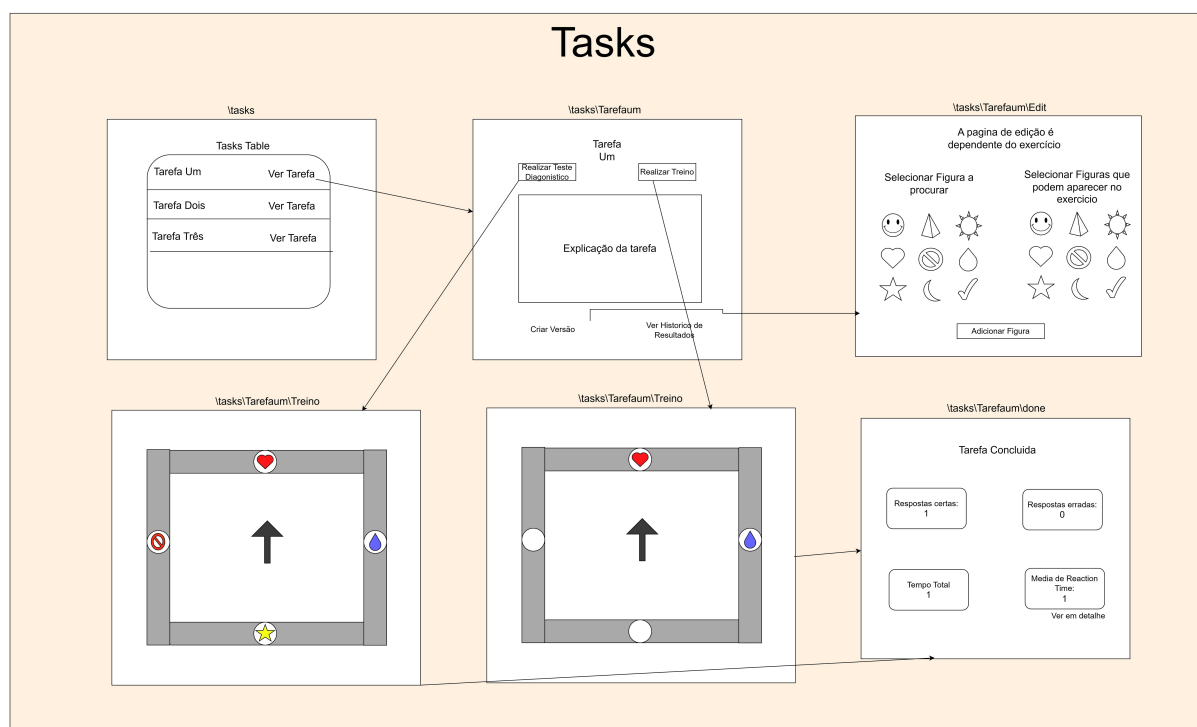


Figure 21: Tasks Mockup

The task in the mock-up is titled “Searching for the Answer” and can be seen in Figure 22. This task operates on a straightforward principle: within a set time limit, various figures appear in designated circles, with one being the figure to look out for. The patient’s objective is to quickly press the arrow key corresponding to the direction of the figure’s location. The task aims to train reaction speed and directional awareness. This is one of the tasks that will be implemented during the development of **BEAM**.

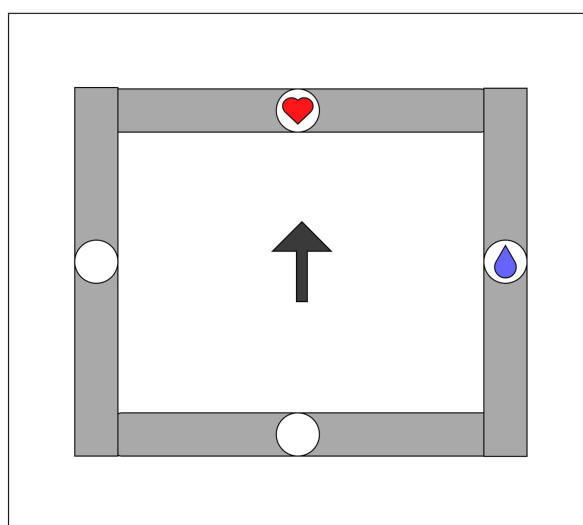


Figure 22: Searching for the Answer Mockup

The results section, in Figure 23 begins with a landing page that provides access to three distinct result views: by exercise, by user, or general platform results. Each of the subsequent screens represents one of these views, offering tailored data based on the selection. The mock-ups presented are examples, with the final pages containing more detailed information. A specific example shown includes a screen featuring a graph, demonstrating what might appear when users delve into more detailed data. In the displayed mock-up, the graph highlights the reaction time results for a patient, complemented by a table presenting all individual reaction time outcomes. Additionally, the graph provides the average reaction time for the patient across various tasks, offering a comprehensive perspective of performance.

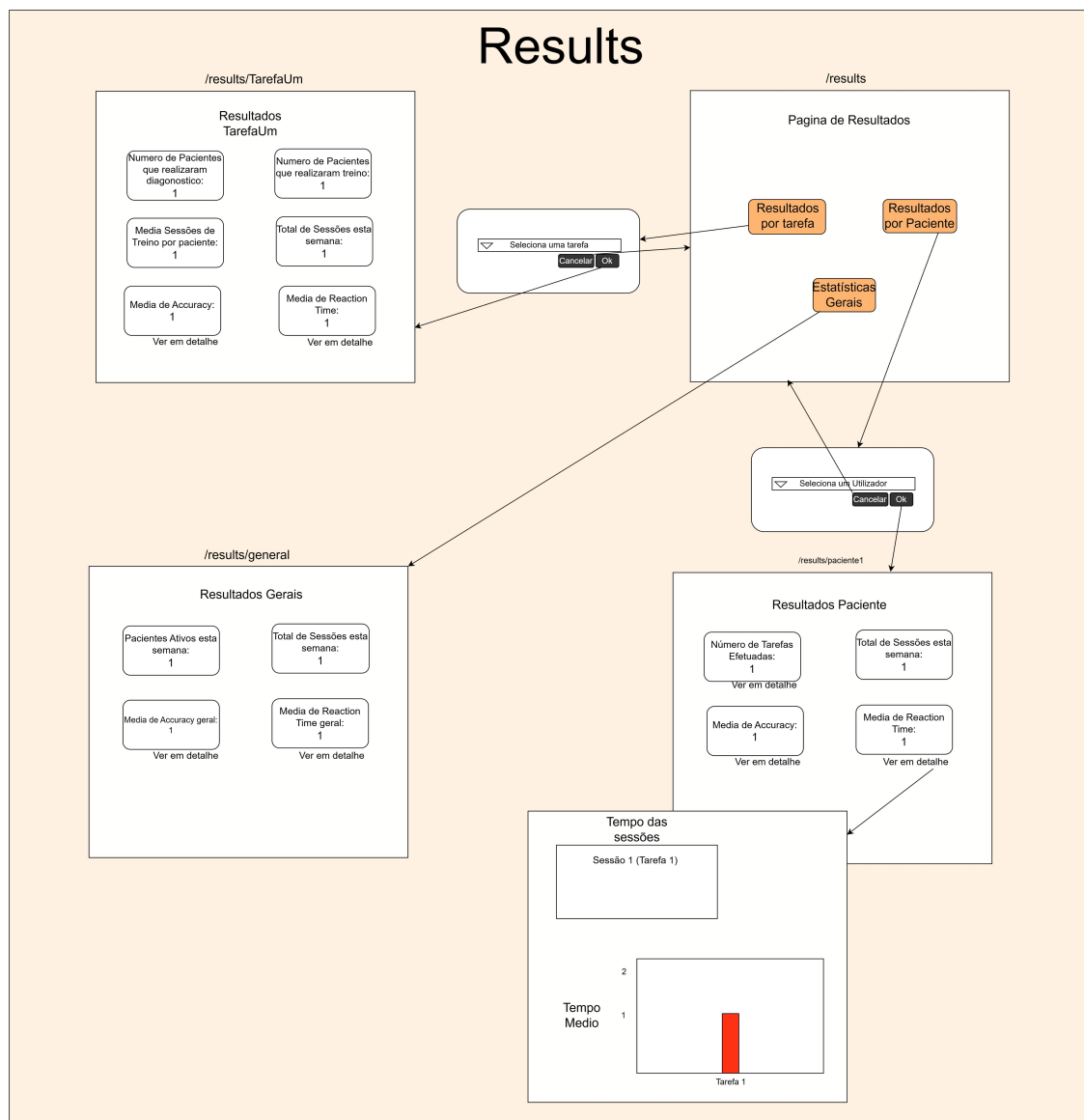


Figure 23: Results Mockup

The profile section (Figure 24) showcases the functionality for therapists to manage their own profiles. In

this section, they can choose to edit their information, leading them to the respective menu. From their profile page, therapists can access their list of patients, where they can select a specific patient's profile to view. Within a patient's profile, therapists have the option to view the patient's results, which redirects them to the results section previously described. Alternatively, therapists can choose to access the notes page for a patient. This page enables therapists to add notes about any relevant observations or updates related to their patient. The notes feature is designed to help therapists stay informed and maintain comprehensive records for effective patient management. Through the patient's profile, therapists can also recommend an exercise for them to do.

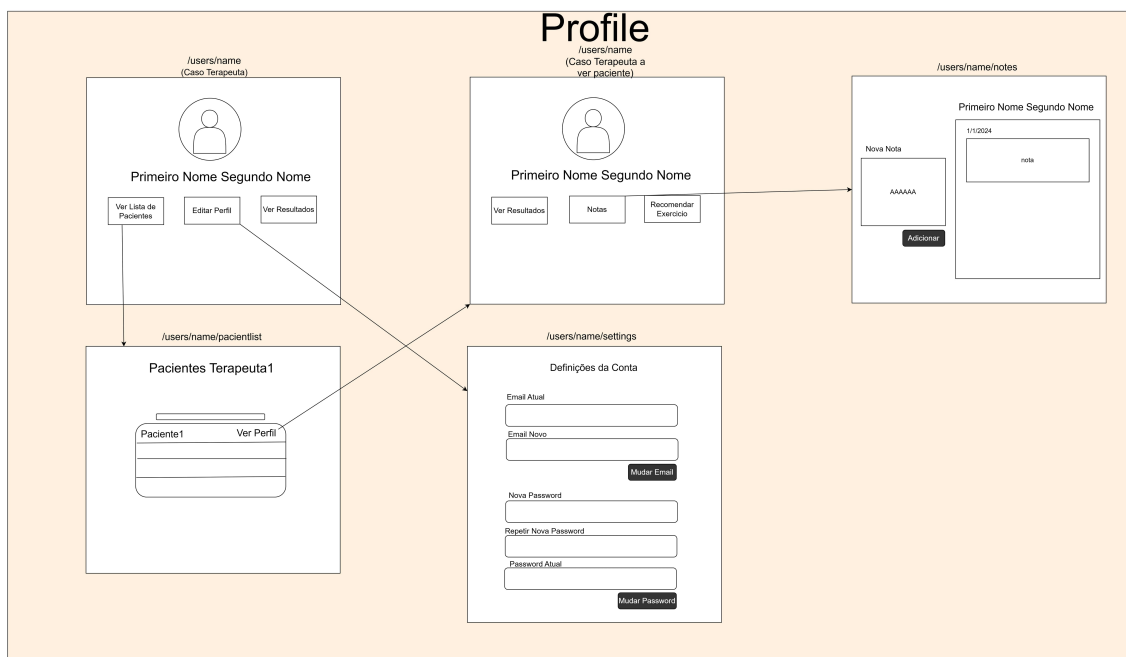


Figure 24: Profile Mockup

The completed mockup can be seen in Figure 25 representing the different sections and which relations with each other.

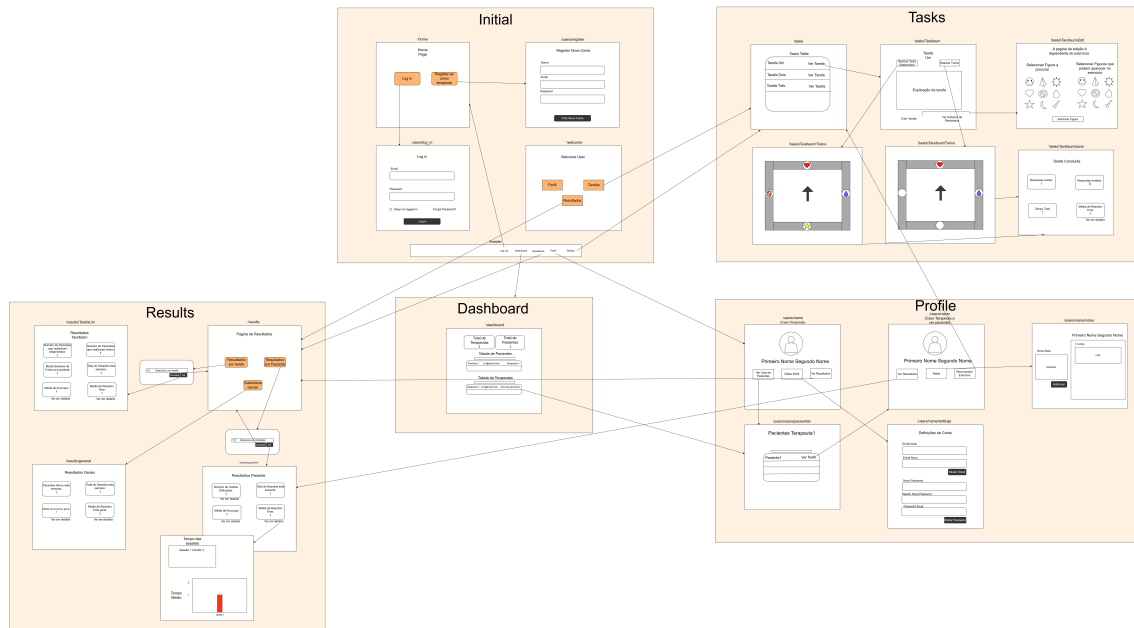


Figure 25: Full Mockup

3.4 Architecture

The system architecture for the proposed **AT** web application **BEAM** is structured into three main components: two frontend interfaces (therapist side and patient side) and the backend processes that connect and manage data for these interfaces. It can be seen in Figure 26.

The therapist side of the application is designed to enable therapists to manage their patients and monitor their progress effectively. This frontend allows the therapist to work on the system through the **Therapist Interface**, which serves as the primary interaction point for the therapist. Through this interface, therapists can access the following modules: the **Patient Manager Module**, that allows therapists to manage patient information, and the **Exercise Definition Module** that provides tools for therapists to define and customize exercises tailored to the specific needs of each patient and the **Results Module** that enables therapists to view detailed results of patient performance on exercises, facilitating analysis and decision-making.

The patient side of the application is designed to provide an engaging and intuitive experience for patients as they interact with the exercises. This frontend allows the therapist to work on the system through of the Patient Interface, which enables patients to interact with the following modules: **Exercise Solving Module** that provides the exercises and the tools required for the patient to complete their assigned training and improve his skills and the **Results Module** that displays exercise outcomes to patients,

helping them visualize their performance and helping them track their progress over time.

The backend is responsible for managing and processing all data within the system, acting as the intermediary between the therapist and patient sides to ensure data integrity and seamless communication. Central to this are the **Data Managers** on both sides, which manage and help the correct handling of the information. Key components within these managers include the Results Updater, which transfers updated patient results to the therapist, and the Recommendations Updater, which propagates exercise recommendations from the therapist to the patient.

Additionally, the **Patient Processes** handle core operations like verifying patient responses through the **Answer Verifier** and generating detailed performance data via the **Exercise Results Generator**, ensuring the accuracy of results provided to both parties. At the center of the backend is the **Database**, which securely stores all patient progress, therapist input, exercise configurations and much more information, serving as the backbone of the system.

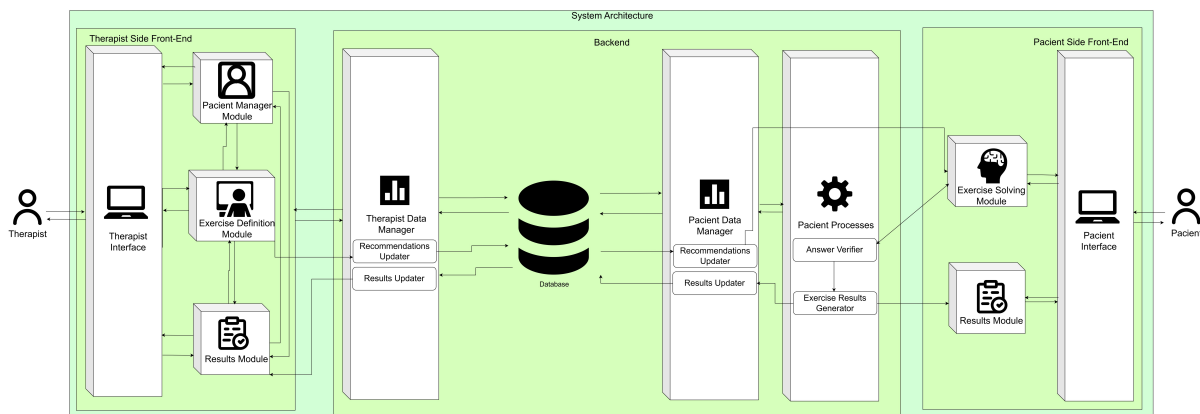


Figure 26: System Architecture

3.5 Technology

The **BEAM** platform will be built using the **Phoenix Framework** (Figure 27), a robust tool specifically designed for developing web applications. Phoenix is powered by Elixir, a functional programming language running on the Erlang VM, known for its scalability, performance, and fault-tolerance. This framework was chosen for its ability to simplify the development process and provide a flexible environment suitable for crafting sophisticated web applications.



Figure 27: Phoenix Framework

For the design and styling of the platform, **Tailwind CSS** will be used. Tailwind CSS is a utility-first CSS framework that simplifies the creation of visually appealing and responsive user interfaces. Its versatility and ease of use will ensure the platform's design aligns seamlessly with user experience requirements.

The platform's database be built and managed using **PostgreSQL**, a powerful and open-source relational database system. Known for its reliability and advanced feature set, PostgreSQL is ideal for managing the platform's data requirements.

GitHub will be used to manage version control and serve as the central repository for the project's codebase. It facilitates collaborative development, providing tools for tracking changes, managing contributions, and ensuring the integrity of the platform's code throughout its lifecycle.

This combination of technologies ensures that **BEAM** will be developed on a solid foundation, enabling efficient implementation, scalability, and maintainability.

Chapter 4

BEAM: Development

This chapter provides a comprehensive description of how the **BEAM** platform was built from a technical standpoint. It documents the development process, including the choice of technologies, the backend and frontend structure, the organization of code into modules and contexts, as well as the deployment strategy. The goal is to explain how each part of the system contributes to the platform's overall functionality, maintainability, and long-term scalability.

4.1 Initial Setup and Framework Configuration

As discussed in the previous chapter, the development of the **BEAM** platform relies on the Phoenix Framework and its underlying technologies. The initial setup of the project was significantly streamlined thanks to the tools and generators provided by Phoenix. In particular, the *mix phx.gen.auth Accounts User users* command proved especially valuable in establishing a foundational authentication system. This feature allowed the quick implementation of essential functionalities such as user registration, login, and access control, saving substantial development time and reducing the likelihood of implementation errors in these critical areas.

Another key aspect of the development process was the use of Phoenix LiveView. By enabling the creation of interactive, real-time user interfaces without relying heavily on JavaScript, LiveView greatly simplified the construction of dynamic features within the application. This allowed for a smoother development workflow and a more cohesive integration between frontend and backend logic.

The combination of Phoenix's code generators, LiveView capabilities, and Elixir's expressive syntax provided a robust and developer-friendly environment that supported rapid prototyping and efficient iteration. These tools not only facilitated the early stages of project setup but also contributed to maintaining consistency and clarity throughout the application's construction.

The core logic of the **BEAM** platform resides within the `lib/beam` directory, which serves as the foundation

of the backend implementation. This directory encapsulates the domain logic and database interactions through well-organized modules and contexts, ensuring a clear separation of concerns and promoting maintainability.

4.2 Backend Structure and Domain Separation

The backend is divided into two main contexts: *Accounts* and *Exercices*. This architectural separation reflects the two primary domains of the application: user management and exercise/task management. The *Accounts* context is responsible for handling user-related functionality, including authentication, registration, profile updates, and role-specific behavior. Within this context, schemas such as *User*, *Therapist*, and *Patient* are defined, mapping directly to corresponding database tables. These schemas include data validation logic, changesets for data transformations, and utility functions such as password hashing and user role verification. For instance, the *User* module manages sensitive user data and provides mechanisms for authentication, password management, and profile customization.

As hinted above, the *Accounts* context introduces two role-specific schemas: *Therapist* and *Patient*. These modules establish clear domain boundaries by associating each user with a particular role through foreign keys and unique identifiers. Therapists are linked to a *user_id* and assigned a unique *therapist_id*, while patients are associated not only with a *user_id* and *patient_id* but also with a therapist via a *therapist_id* relationship. This structure models real-world clinical hierarchies, where patients are assigned to therapists and their interactions are managed accordingly.

Another important entity within the *Accounts* context is the *Note* schema, which allows therapists to create written observations or comments linked to specific patients. Each note is associated with both a therapist and a patient, forming a triadic relationship between different user roles and their ongoing interactions. This functionality enhances the platform's clinical utility by providing a space for tracking qualitative data alongside quantitative performance metrics.

All these domain models are orchestrated through the *accounts.ex* module, which serves as the main interface for interacting with user data, performing role-based logic, and managing associations between therapists, patients, and notes. This module includes functions for user verification, role assignment during registration, profile updates, age calculation, and more. By abstracting database access and business logic into a centralized layer, the application maintains consistency, reduces duplication, and simplifies integration with the LiveView interface and controller logic.

On the other hand, the *Exercices* context focuses on the interactive tasks and activities users engage

with. This part of the system includes not only the exercises themselves, but also functionality related to performance tracking and personalized support. For instance, modules are responsible for generating result entries based on user interaction, calculating statistics such as accuracy and reaction time, and allowing therapists to recommend specific tasks to their patients. These capabilities support a more adaptive training experience tailored to individual user needs.

Each interactive task is encapsulated within its own module under the *tasks* subdirectory. This directory holds the logic responsible for dynamically generating exercises with randomized elements, such as shapes, colors, or numbers, based on the selected difficulty. Modules also include validation functions to assess user input. Some tasks feature added complexity, such as overlapping items or movement, which are implemented to increase difficulty progressively. This modular approach promotes flexibility and allows for seamless addition of new tasks over time.

Additionally, the platform supports the creation of customized task configurations, enabling therapists to adjust specific parameters of an exercise to better suit the cognitive profile or therapeutic needs of a patient. This is achieved through a dedicated behavior module called *Configurable*, which defines the structure for editable tasks, including the default values, editable fields, and validation logic. These configurations are stored in the *ExerciseConfiguration* schema, where each entry includes the task it applies to, the configuration data, and the therapist who created it. Access to these configurations is controlled via the *ExerciseConfigurationAccess* schema, allowing therapists to selectively assign them to patients. This mechanism offers a practical way to adapt exercises, not only by difficulty, but by concrete task behavior, enhancing personalization and engagement, particularly in clinical contexts.

In structural terms, the Exercises context is anchored by the *Task* schema, which defines the core metadata for each cognitive activity available in the platform. Tasks include attributes such as name, description, image, and tags, serving both as identifiers for the exercises and as references for linking results and recommendations.

Performance data is stored in the *Result* schema, where key metrics like correct, incorrect, and omitted answers, as well as accuracy and reaction time, are saved per user and task. To distinguish the context of each interaction, the platform also defines *Training* and *Test* schemas, which track repeated attempts and associate them with their corresponding results.

The *Recommendation* schema supports therapist-to-patient task suggestions. Each recommendation links a therapist, patient, and task.

All functionality within this context is coordinated through the *exercices.ex* module. This module acts as the central access point for retrieving tasks, saving results, managing recommendations, and linking user

interactions to their corresponding entries in the database. By consolidating these operations in one place, the module ensures consistency across the system and simplifies the development of LiveView modules and controllers that depend on exercise-related data.

Together, these contexts enable a clean and scalable structure for the backend, making it easier to implement new features, manage data integrity, and maintain clear boundaries between different areas of the application's domain.

4.3 Frontend Structure and LiveView-Based Interface

The *beam_web* directory serves as the interface layer of the application, handling how data and functionality from the backend are rendered and delivered to users. This directory is structured according to Phoenix conventions, separating reusable components, layout templates, and controllers that respond to HTTP requests. At the core of this interface structure lies the router, defined in *router.ex*, which plays a crucial role in connecting URLs to the appropriate controllers and LiveView modules. It ensures that user requests are correctly routed to their corresponding components, providing the foundation for navigation across the entire platform.

Although the application heavily relies on Phoenix LiveView, there are still some modules implemented through the standard controller and template-based system. These are primarily structured around HTML templates with the *.heex* extension and corresponding controller modules. Many of these files were generated during the initial scaffolding of the Phoenix project and later adapted to fit the specific requirements of the platform.

For example, *home.html.heex* and its corresponding controller handle the rendering of the homepage, while layout templates such as *app.html.heex* and *root.html.heex* define the overall structure and wrapper for the user interface. These layouts ensure consistency across different pages by providing shared components like headers, footers, and navigation bars.

A particularly relevant file in this structure is *core_components.ex*, which defines reusable UI elements and helpers. This module, also generated by Phoenix by default, allowed to create standardized and declarative components that can be used across the application to ensure consistent styling and behavior. It plays a crucial role in reducing duplication and improving maintainability, especially in larger projects where interface elements are reused frequently.

Controllers under *controllers/* are responsible for handling classic HTTP requests and passing data to templates. These include modules such as *PageController* and *TaskController*, which map incoming routes

to actions and render appropriate views. In practice, only a small portion of the application logic remains in these traditional controller-based modules, as most of the interactive functionality is implemented using LiveView.

While the application includes some controller-based views, the majority of the user interface is implemented using Phoenix LiveView, which plays a central role in delivering a dynamic and responsive experience. LiveView allows developers to build real-time, stateful interfaces without writing custom JavaScript, streamlining the development process and improving maintainability.

LiveView modules are located within the *lib/beam_web/live* directory. Each file represents a specific view or feature of the platform and encapsulates both rendering and event-handling logic in a single, cohesive unit. This approach promotes modularity and clarity, as all the behavior and structure related to a particular page are maintained in one place. For example, files such as *user_registration_live.ex*, *user_login_live.ex*, and *user_confirmation_instructions_live.ex* handle authentication flows. These modules were generated and scaffolded using built-in Phoenix commands, then adapted to the platform's specific needs. As previously mentioned, this greatly accelerated development and ensured a solid foundation for security-related features.

A key part of the exercise flow is managed by two specific LiveView modules: *dynamic_task_live.ex* and *task_explanation_live.ex*. The *DynamicTaskLive* module acts as a dispatcher that receives a task identifier and redirects the user to the correct task module based on its type. This dynamic routing system allows for a scalable way to handle multiple exercises while maintaining clean and organized code. On the other hand, *TaskExplanationLive* is responsible for presenting information about a given exercise, such as its description, difficulty options, and training or test modes, before the user begins the task. This view also supports additional interaction for therapists, enabling them to recommend exercises directly to patients. In addition to recommending existing tasks, therapists can also create and manage personalized task configurations directly through the interface. The *ConfigEditLive* module enables this by dynamically generating input forms that correspond to the specific configuration schema defined by each task. Therapists can name the configuration, adjust relevant parameters (such as time limits or number of rounds), and decide whether the configuration should be visible to other therapists. This feature is particularly useful for tailoring exercises to specific patient needs in a structured and reusable way.

To further support the sharing and reuse of these configurations, the *EditionsReserveLive* module provides an interface where therapists can view all previously created configurations. From this page, they can assign configurations to specific patients for permanent access, recommend them for immediate use, or simply review the parameter details. This structured approach to configuration management improves

usability, encourages personalization, and strengthens the therapist's ability to deliver targeted cognitive training.

Several other LiveView modules contribute to enhancing the overall user experience. These include features such as user profile management (*user_profile_live.ex*, *user_settings_live.ex*) and note-taking functionality (*user_notes_live.ex*), for example. A reusable help component was also implemented to provide contextual guidance across various pages, helping users understand how to interact with the platform effectively. LiveView provides a seamless and fluid experience that adapts well to the needs of both therapists and patients using the platform.

To better organize the core interactive and analytical functionalities of the platform, the LiveView layer is further divided into two key subdirectories: tasks and results. This separation ensures a clear distinction between the modules responsible for executing exercises and those that handle the display and analysis of results, aligning with the two fundamental user flows in the application—performing cognitive tasks and reviewing performance.

The results subdirectory includes all views related to performance tracking. The *results_end_live.ex* module is responsible for showing a summary of results immediately after a user completes a task, providing metrics such as accuracy, number of correct or omitted answers, and average reaction time. This module is the primary point of feedback for patients. In addition, patients can access the *results_per_user_live.ex* view, which displays a personal history of task results over time.

Therapists, on the other hand, have access to more comprehensive data visualization tools. The main entry point is *results_main_live.ex*, a central hub where therapists can choose between different modes of analysis. From there, they can access modules such as the already mentioned *results_per_user_live.ex* and also *results_general_stats_live.ex* for platform-wide performance trends and *results_per_exercise_live.ex* to inspect how users perform on specific tasks. There is also *results_per_category_live.ex* which allows therapists to see all the results from a specific category (for example, focused attention or working memory). This layered structure allows therapists to gain both macro and micro-level insights into user progress and task effectiveness.

To render the visual statistics and graphs present in these result views, the platform integrates the *Chart.js* library on the frontend. Through custom hooks, multiple charts, such as bar graphs for accuracy and reaction time, as well as pie charts for patient distributions, are dynamically generated based on live data. This allows for clear and interactive visualizations that complement the numerical reports, enhancing the therapist's ability to interpret patient performance at a glance.

Each LiveView module within the tasks directory is dedicated to managing the full interactive flow of a

specific cognitive exercise. These modules are designed to encapsulate both the visual presentation and the behavioral logic of each task, ensuring a smooth and consistent user experience. They are tightly integrated with their respective backend modules in *lib/beam/exercices/tasks*, allowing for clean coordination between data generation, user input handling, and result evaluation. Despite variations in purpose and mechanics, all tasks follow a common interaction pattern: initialization, active rounds of user engagement, feedback, and automatic result persistence. This unified structure simplifies development, promotes consistency across tasks, and establishes a scalable framework for introducing new exercises in the future. Each task and its specific functionality will be explored in detail in later sections.

Each task LiveView operates independently but follows a shared structure that includes preparation phases, interaction rounds, real-time feedback, and automatic persistence of performance metrics. This consistent structure simplifies the addition of new exercises in the future and ensures a uniform experience across different types of cognitive activities.

4.4 Deployment

The deployment of the **BEAM** platform was designed to ensure stability, maintainability, and ease of access for its target users. The application is publicly accessible via the address `beam.ep1.di.uminho.pt`, which is hosted on a CentOS 7 Linux server maintained by the EPL at the Department of Informatics of the University of Minho.

To transfer the application code to the server, the **GitHub repository** was used. The codebase is updated directly from this repository during each deployment cycle, enabling streamlined version control and simplifying the update process.

The running system is orchestrated using two Docker containers. One container is responsible for running the main Phoenix application, while the second hosts the PostgreSQL database used for persistent storage. This containerized architecture ensures strong isolation between the application logic and the database layer, improving fault tolerance and simplifying upgrades or maintenance.

Unlike more complex orchestration setups that rely on tools such as *docker-compose*, the **BEAM** platform opts for a single, well-structured *Dockerfile* to build and run the Phoenix application. This Dockerfile was generated using Phoenix's built-in deployment commands and then manually refined to suit the specific needs of the platform and hosting environment.

Phoenix plays a helpful role in this process by providing a suite of deployment tools and configuration templates. These include a base Dockerfile, asset bundling strategies, and runtime configuration hooks

that significantly reduce the overhead typically associated with deploying web applications. This built-in support allowed for a smoother transition from development to production and enabled the creation of a robust deployment pipeline with minimal manual intervention.

Through this deployment approach, the **BEAM** platform remains consistently accessible, easy to update, and securely hosted, aligning with the broader goals of reliability and user accessibility.

4.5 Updated Domain Model

To illustrate the progress and structural evolution of the platform throughout its development, an updated domain model is presented in Figure 28. This model reflects key architectural additions and refinements that differentiate it from the initial version shown in the project proposal (Figure 17).

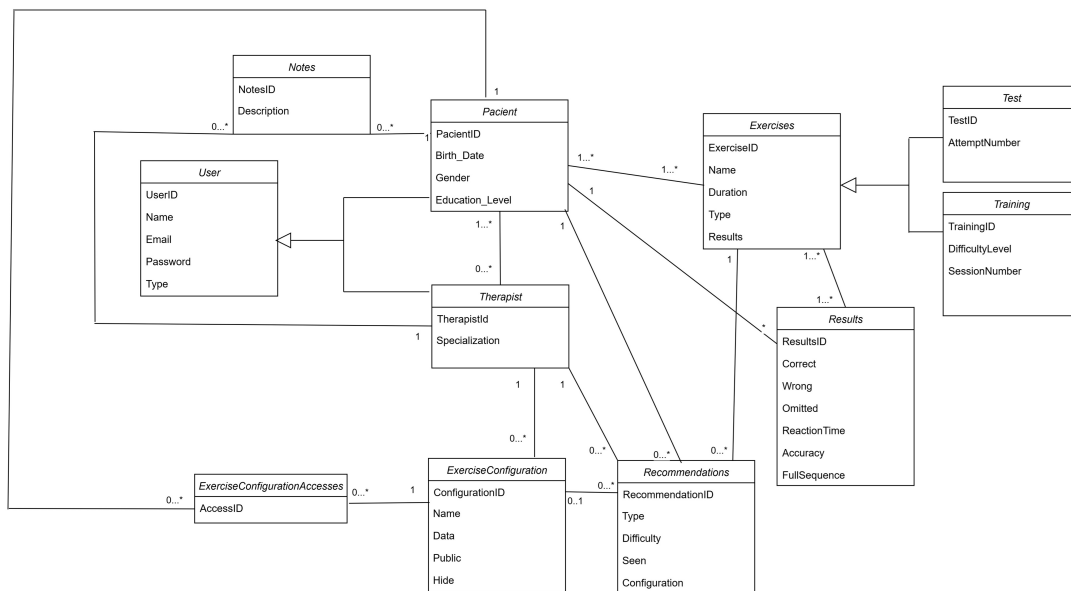


Figure 28: Updated Domain Model

Compared to the initial model, this updated version introduces several new entities and extends existing ones to support new features implemented during development. One of the most notable additions is the **ExerciseConfiguration** entity, which enables therapists to create and manage personalized configurations for cognitive tasks. These configurations allow adjustments to parameters such as time limits, difficulty, or sequence length, offering greater flexibility and personalization in cognitive training.

The **ExerciseConfigurationAccess** relation (represented implicitly in the diagram) manages access control by associating configurations with specific patients, meaning that therapists can associate parents with the configurations they can practice at will.

Another important addition is the **Recommendations** entity. This structure allows therapists to suggest specific exercises or configurations to individual patients, either for training or testing purposes. Each recommendation includes metadata such as the difficulty level, whether the suggestion has been seen, and, optionally, a direct link to a custom configuration. This mechanism enhances the therapeutic workflow by enabling proactive guidance and tailored interventions.

Furthermore, the existing entities were extended to include additional fields. For example, the **Results** entity now stores information such as *FullSequence*, which is relevant for certain tasks that require recalling or reproducing entire sequences of elements.

The diagram also emphasizes the bidirectional relationships between therapists, patients, and their corresponding data, such as notes, configurations, and results. This reflects the platform's core goal of enabling structured and traceable interactions between healthcare professionals and their patients.

Chapter 5

BEAM: The System at a Glance

This chapter introduces the main areas of the **BEAM** platform from a user perspective. It describes the different parts of the user interface and outlines what each part allows users to do, whether they are patients engaging in cognitive training or therapists monitoring and guiding their progress. The goal is to provide a clear and structured guide of the platform's features and how they are accessed and used in practice.

5.1 Overview

The **BEAM** platform is organized into several distinct pieces, each designed to support a specific aspect of the user experience. From the moment users log in, they are guided through a structured interface that adapts to their role, whether as a patient performing exercises or as a therapist managing patient activity. This section offers a general overview of the main areas of the application. It briefly touches on key elements such as the task library, where patients can find and complete exercises, the results section, which presents detailed feedback and performance history and many other features and pages the application offers.

One important usability feature present throughout the platform is the help component, which is accessible on multiple pages. This element offers brief, but relevant guidance tailored to the user's role, presenting different information depending on whether the user is a therapist or a patient. Its presence ensures that users can quickly access instructions, clarifications or even learn features they weren't aware of.

Access to the **BEAM** platform begins with a welcoming landing page (Figure 29) that clearly communicates the core purpose of the project: to train focus and improve attention through cognitive exercises. This initial screen offers two main options one allowing either account creation for new professionals and the other access for existing users of any type.



Figure 29: Opening Page

If a user wishes to register as a healthcare professional, they are directed to the registration form (Figure 30, left), where they must provide basic information such as name, email, password, and area of specialization. Once registered, the user can access the platform through the login page (Figure 30, right) by entering the credentials previously defined. Patients, on the other hand, must be created within the platform by their therapist, in a different menu shown later in this chapter. However, the login procedure for patients follows the same process once their accounts are set up.

Registrar uma conta de Terapeuta

Já está registado? [Inicie sessão na sua conta agora.](#)

Nome

Email

Palavra-passe

Especialidade

Terapeuta

[Criar conta](#)

Iniciar sessão na conta

Não tem uma conta? [Registe-se para criar uma agora.](#)

Email

Palavra-passe

☐ Manter-me autenticado

[Iniciar sessão →](#)

Figure 30: Therapist Registration Page (left) and Login Page (right)

Upon logging in or signing in, users are directed to the welcome page (Figure 31). This screen acts as a central hub and provides quick access to the three core areas of the platform: viewing tasks, checking results, and looking at the user profile. The interface is intentionally minimalistic and user-friendly, with a clean layout and clearly labeled buttons.

Bem-vindo, José Barbosa!

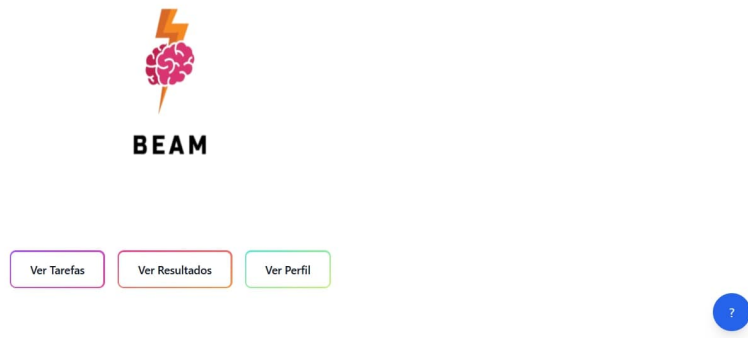


Figure 31: Welcome Page

At the top of every page within the platform, a consistent header is displayed, adapting its content to the user's role. For therapists, the header (Figure 32) includes navigation links to the profile page, task list, dashboard, results, and a logout option. This allows quick and structured access to all key areas of the platform. For patients, the header includes the links to only the profile and therapist sections and the ability to log out, as well as a notification icon. This bell icon provides a direct link to recommendations issued by their assigned therapist, encouraging engagement with suggested training.



Figure 32: Therapist Navigation Header

The next key section of the platform is the task selection page (Figure 33), which serves as the central interface for browsing all available cognitive training exercises. Each row in the table corresponds to a distinct task, displaying a small visual preview, its title, and the cognitive domains it targets. The tags for trained capacities provide a quick overview of which cognitive functions are stimulated by each task, such as attention, memory, or reaction speed. This helps both patients and therapists select exercises that best fit specific training objectives.

By clicking on the name of a task, users are redirected to the corresponding task explanation page, where the selected exercise can be initiated. These tasks are described in further detail in the Tasks section of this chapter. In addition, the interface provides two action buttons for therapists next to each task: the pencil icon allows therapists to create custom configurations for that specific exercise, while the notepad icon gives access to previously saved configurations. This structure enables therapists to easily personalize

tasks according to individual patient needs while maintaining a clear and intuitive interface for navigation.

Treinos Cognitivos e de Atenção

Jogo	Nome	Capacidades Treinadas
1	Matemática	Atenção Focada, Memória de Trabalho, Atenção Sustentada, Velocidade de Reação
2	Procurar uma resposta	Atenção Seletiva, Atenção Sustentada, Velocidade de Reação, Atenção Visual
3	Menor que 5	Atenção Focada, Velocidade de Reação, Inibição de Resposta, Atenção Sustentada
4	Sequência Inversa	Memória de Trabalho, Atenção Sustentada, Atenção Focada, Manipulação Cognitiva
5	Código de Símbolos	Memória de Trabalho, Atenção Sustentada, Velocidade de Reação, Atenção Visual

Figure 33: Tasks Page

When users select a task from the main list, they are directed to the task explanation page (Figure 34). This screen provides a brief description of the selected exercise, explaining its purpose and mechanics in clear terms. On the therapist version of this page, additional options are available: besides the ability to return to the task list, therapists can recommend the task to a patient or try it themselves for testing purposes. For patients, the same page presents two options: starting the task as a training session or as a test. This distinction allows both user roles to easily navigate towards their intended usage of the task.

Matemática

Voltar

Recomendar Tarefa

Experimental Tarefa

Esta tarefa testa a sua capacidade de calcular rapidamente. Irá aparecer no ecrã uma operação matemática, cuja complexidade depende do nível de dificuldade que o utilizador terá alguns segundos para resolver, seleccionando a opção correta. Pode ser jogada com rato e em ambiente táctil.

Matemática

Voltar

Começar Treino

Começar Teste

Esta tarefa testa a sua capacidade de calcular rapidamente. Irá aparecer no ecrã uma operação matemática, cuja complexidade depende do nível de dificuldade que o utilizador terá alguns segundos para resolver, seleccionando a opção correta. Pode ser jogada com rato e em ambiente táctil.

Figure 34: Task Explanation Pages for Therapist and Patient

For therapists aiming to personalize the training experience, each task includes a dedicated configuration page (Figure 35). Here, therapists can create new custom configurations by adjusting task-specific parameters. For example, in the case of the *Matemática* task, parameters such as equation display time,

response time, maximum number values, and type of operations can be freely edited. This flexibility allows therapists to tailor task difficulty and cognitive demands based on each patient’s individual needs.

Once a configuration is saved, it appears on the configuration management page, where all previously created configurations for that task are listed (also shown in Figure 35). Therapists can recommend any saved configuration directly to a specific patient using the *Recomendar* button, or grant permanent access through the *Pacientes* option, which allows selected patients to freely access that configuration whenever they choose to train. Additionally, therapists can review the details of each configuration to recall its settings, or permanently delete configurations that are no longer relevant.

This system of configurable tasks ensures that **BEAM** offers both standardized training protocols and highly adaptable exercises. Therapists are empowered to fine-tune each task to optimize cognitive stimulation for different patient profiles, while patients benefit from a flexible training experience that matches their abilities and progress over time.

Editar tarefa: Matemática

Nome da Configuração

Tempo de Exibição da Equação (ms)

2000

Tempo para Responder (ms)

4000

Valor Máximo nas Operações

100

Tipo de Operações

Arbitrárias

Guardar Configuração

Configurações Guardadas

Nome	Ações
Contas de Somar Básicas	Recomendar Pacientes Ver detalhes

Figure 35: Task Configuration Editing and Management

The platform also provides a dedicated profile page for both therapists and patients, offering quick access to personal information and role-specific actions. The therapist profile page (Figure 36) presents basic user details, such as the therapist’s name and email address, followed by three main options: viewing their assigned patients, editing their personal profile information, and accessing overall results. This clear structure allows therapists to efficiently manage their patients and monitor performance data directly from their own profile dashboard.

When therapists access a specific patient’s profile (also shown in Figure 36), they are presented with a detailed summary of the patient’s information. This includes the patient’s name, email, age, gender, and educational level, alongside two main action buttons: one to view the patient’s task results and another to access patient notes. Importantly, when patients themselves view their own profile, they see a nearly identical interface, except for the absence of the *View Patients* option, which is exclusive to therapists.

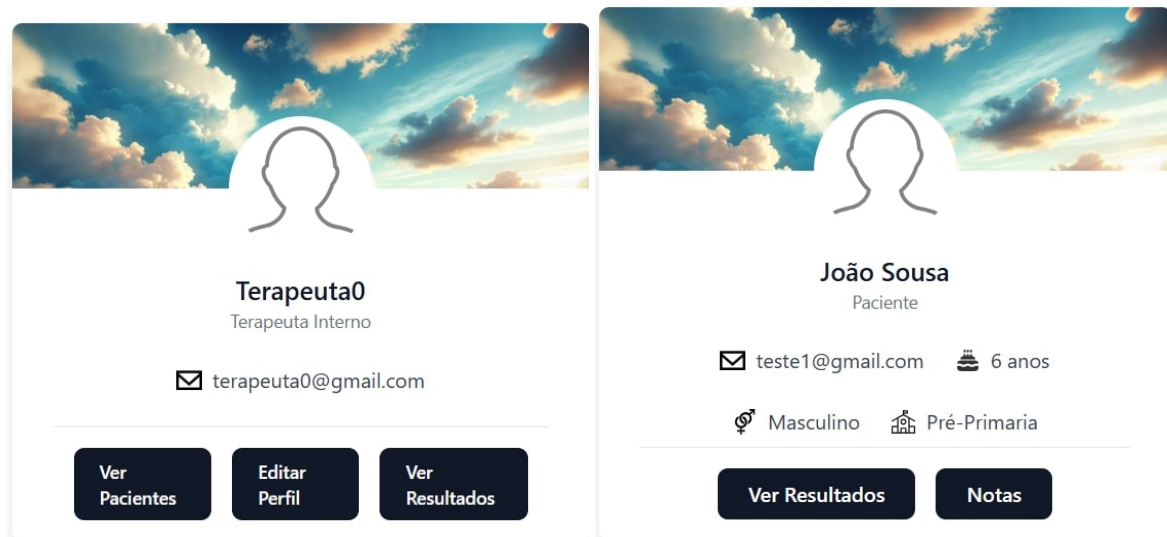


Figure 36: Therapist and Patient Profile Pages

By clicking on the *Notas* button within the patient profile, therapists are directed to the note management page (Figure 37). This page enables therapists to create and store personalized notes for each patient, providing a space for clinical observations, session comments, or progress updates. Therapists only have access to the notes of their own assigned patients, ensuring privacy and maintaining appropriate professional boundaries within the multi-user system.

This integrated notes system allows therapists and professionals to complement the quantitative performance data with qualitative observations, offering a more complete picture of each patient's progress. These notes can be deleted as needed, helping therapists to maintain organized and up-to-date records throughout the intervention process.



Figure 37: Patient Notes Page

Therapists have access to a dedicated dashboard (Figure 38), which serves as the central workspace for managing both patients and other professionals within the platform. The top section of the dashboard provides a quick overview of the system, displaying the total number of registered patients and therapists. Below, therapists can access and manage their assigned patients. A filtering option labeled *Apenas meus pacientes* allows therapists to restrict the view to only display the patients under their direct care. Additionally, a search bar is available to quickly locate specific patients based on their name. They can also access the screen to create new patients from here. Further down the page, therapists can view a full list of all professionals registered on the platform. This secondary table provides an overview of other therapists present in the system, facilitating professional coordination if necessary.

Bem-vindo, Terapeuta Interno!

Total de Pacientes
5

Total de Terapeutas
5

Tabela de Pacientes
[Novo Paciente](#)
☒ Apenas meus pacientes

Q

Nome do Paciente	Email do Paciente	Terapeuta Associado
rodolfo	rodolfo@gmail.com	Terapeuta0
rodolfa	rodolfa@gmail.com	Terapeuta0

Tabela de Profissionais
Q

?

Figure 38: Dashboard Page

When accessing the “Create Patient” — *Criar Paciente* — screen (Figure 39), therapists are presented with a straightforward form to input the new patient’s essential information. The required fields include name, email, password, date of birth, gender, and education level, all necessary to personalize the experience and allow meaningful performance comparisons. This registration process ensures that therapists can quickly onboard new patients that from that point forward can access the platform from anywhere.

Criar Paciente

Preencha os detalhes abaixo para registrar um novo paciente.

Nome

Email

Palavra-passe

Data de Nascimento

Género

Masculino

Escolaridade

Pré-Primaria

Figure 39: Create Patient Page

Therapists can also navigate to the results section (Figure 40), which provides access to different views for analyzing user performance. The interface presents four main options: general statistics, results by user, results by task, and results by category. These options allow therapists to explore both high-level trends and detailed individual data, making it easier to track patient progress, identify patterns, and evaluate the effectiveness of training programs across the platform.

Resultados

Ver Estatísticas Gerais

Ver Resultados por Utilizador

Ver Resultados por Exercício

Ver Resultados por Categoria

Figure 40: Results Menu

The *General Statistics* page (Figure 41) offers a high-level overview of activity on the platform, providing information into the general users of the app or just the patients under the therapist accessing the page. It presents basic metrics such as the number of exercises completed and active users during the current week, as well as the average age of all patients on the platform and of those under the therapist's care.

Below this summary, four buttons give access to key data visualizations: general accuracy, general reaction time, accuracy for the therapist's patients, and reaction time for the therapist's patients. These options help therapists gain quick insights into how their users are performing compared to the overall population. Clicking on one of the graph buttons opens a dynamic chart (Figure 41, right) that visualizes performance metrics per task. In the case of the reaction time chart shown, therapists can compare average response times across different exercises, with filters for age, gender, and educational level. This enables a more refined analysis of trends and outliers, allowing therapists to better tailor their interventions. The visualization tools on this page play a central role in supporting data-driven therapeutic decisions, combining accessibility with meaningful insights.

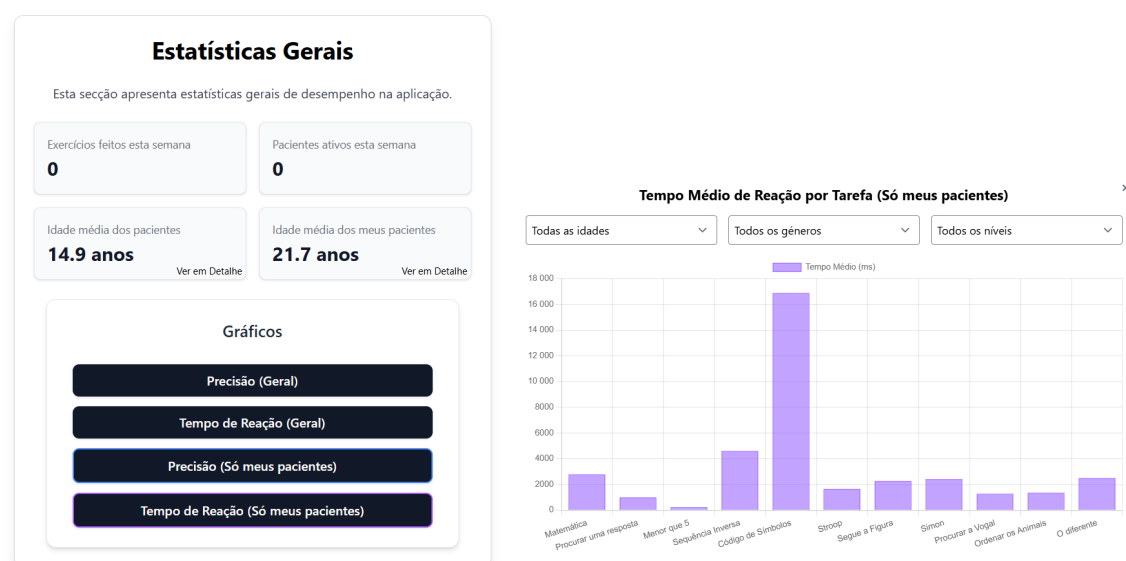


Figure 41: General Statistics Page (left) and Reaction Time Graph (right)

The remaining results pages, by user, by task, and by category, share a common structures. Each of these pages includes a comprehensive table listing key metrics such as task type, number of correct and incorrect answers, omissions, accuracy, average reaction time, and the date of completion. Therapists can filter these records according to the selected task or if the task has training or a test, allowing them to focus on specific areas of performance. This structure ensures consistency across different views while supporting flexible exploration of the data, whether the focus is on an individual user, a specific cognitive task, or broader task categories.

In the case of the *Results by User* page (Figure 42), a performance graph is also displayed above the table. This visual representation summarizes metrics such as average accuracy across tasks and supports side-by-side comparisons when needed. For example, therapists can observe which tasks present greater difficulty for a given patient and track performance trends over time.

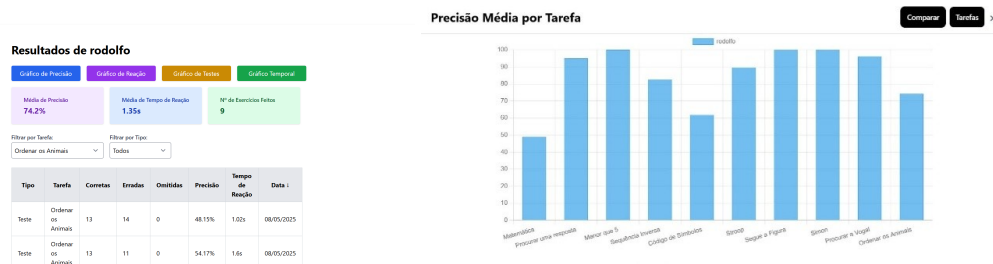


Figure 42: User Results Table (left) and User Results Graph (right)

5.2 Tasks

At the heart of the **BEAM** platform lies its most essential component: the set of interactive tasks and exercises through which patients can actively train their cognitive abilities. These tasks are designed to provide engaging, structured activities that stimulate core mental functions such as attention, working memory, response control, and cognitive flexibility. By combining clinical insights with interactive technology, **BEAM** offers users a dynamic environment where they can strengthen these skills through repeated practice and feedback.

The exercises serve not only as tools for cognitive training but also as opportunities for assessment and progress tracking. Each task is carefully crafted to challenge specific domains of cognition, adapting in complexity across different levels of difficulty. This ensures that users are continuously engaged and appropriately challenged according to their individual capabilities.

In this section, each of the tasks available in the **BEAM** platform will be briefly introduced and explained. The following descriptions aim to highlight the purpose, structure, and cognitive focus of each exercise, offering an overview of the system's training potential and its relevance in supporting cognitive development and rehabilitation.

5.2.1 Math Operation

The first of these exercises is *Math Operation* task — *Matemática* — (Figure 43). As discussed by Ardila (2024), engaging in mental math tasks strengthens the brain's ability to store and manipulate information while maintaining focus under pressure. Based on this principle, the task was created to stimulate executive functions such as sustained attention and working memory.

In each round of the task, the user is shown a simple arithmetic operation and must choose the correct result from a selection of possible answers. The operations range in complexity depending on the selected

Resolve a Operação Matemática

70 - 54

Resolve a Operação Matemática

16

13

8

11

Figure 43: Math Operation Task

difficulty level, with harder versions introducing more challenging calculations. Time is limited, pushing the user to focus, reason quickly, and respond without hesitation. This time constraint is not just a challenge—it is a deliberate part of the exercise's cognitive training value, forcing the user to remain alert and engaged. Through this interaction, the task helps develop the ability to concentrate for extended periods and process information rapidly, skills that are essential for daily functioning. The session is composed of many quick operations, offering repeated practice that reinforces cognitive endurance and strengthens the connection between mental effort and accuracy. The simplicity of the exercise makes it accessible, while the pace and structure ensure it remains mentally demanding.

5.2.2 Searching for an Answer

The *Searching for an Answer* task — *Procurar uma Resposta* — (Figure 44) was inspired by a minigame from the *Super Mario Party* series, adapted and redesigned to serve a cognitive training purpose. While not directly based on a specific scientific protocol, its conception was guided by findings in the literature, including the work of Mishra et al. (2012), which highlights how videogames can positively impact attention and memory. This task leverages those benefits in a controlled and structured environment.

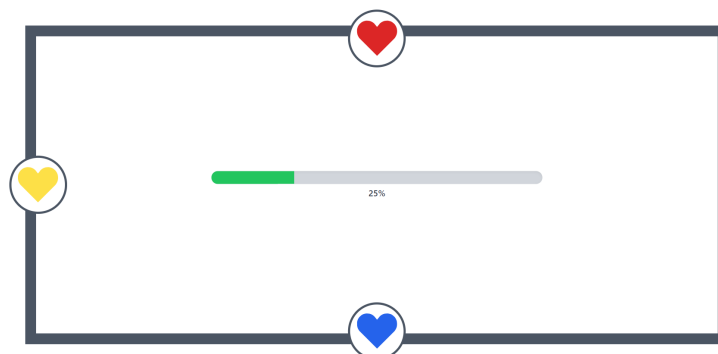


Figure 44: Searching for an Answer Task

In this activity, the user is presented with a target figure, defined by a specific combination of shape and

color, that must be located among distractors. Across each round, the target appears in one of four possible directions (top, bottom, left, or right), and the user must quickly press the corresponding arrow key to indicate its location. Distractor figures are also positioned in those directions to challenge visual discrimination and increase cognitive load.

This exercise touches on several interconnected cognitive functions: selective attention, to distinguish the target from similar distractors; sustained attention, to maintain performance across multiple rounds; visual attention, to scan the layout efficiently; spatial processing, to match target location with the correct key; and reaction speed, to respond within a limited timeframe. By integrating elements of gamified interaction with targeted mental challenges, *Procurar uma Resposta* offers a compelling way to train attention and perceptual agility.

5.2.3 Less Than Five

The *Less Than Five* task — *Menor que 5* — (Figure 45) was adapted from an exercise described in Aben et al. (2019), which was previously discussed in the State of the Art. In the original study, numerical stimuli were used to assess attentional control and response inhibition by asking participants to react only when a number was either greater or less than a specific value. This particular example stood out for its simplicity and cognitive relevance, making it a strong candidate for adaptation within the platform.

1

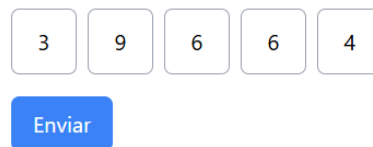
Figure 45: Less Than Five Task

In this version, a number from 1 to 9 (excluding 5) appears on the screen, and users are instructed to press the spacebar only when the number is less than 5. The task requires participants to maintain continuous focus, quickly evaluate each stimulus, and inhibit their response when the condition is not met. Its apparent simplicity masks the high demand for attentional control and fast decision-making.

The task effectively trains several key cognitive abilities: focused attention, to monitor and evaluate each number as it appears; reaction speed, by encouraging rapid and accurate motor responses; response inhibition, to suppress incorrect impulses; and sustained attention, as users must remain engaged and alert over multiple trials without losing focus.

5.2.4 Reverse Sequence

The *Reverse Sequence* task — *Sequência Inversa* — (Figure 46) was inspired by a clinical exercise proposed in the work of Lima (2015). In this task, users are asked to memorize a sequence of digits and then reproduce it in reverse order. The challenge lies not only in retaining the original sequence, but also in mentally manipulating it before submitting the final response.



The image shows a digital interface for the Reverse Sequence task. It consists of five white rectangular boxes with rounded corners, each containing a digit. The digits are 3, 9, 6, 6, and 4, arranged horizontally from left to right. Below these boxes is a single blue rectangular button with rounded corners and the word 'Enviar' in white text.

Figure 46: Reverse Sequence Task

The difficulty of the task is modulated by the length of the sequence, which increases based on the selected difficulty level. Throughout each round, the sequence is displayed briefly, requiring users to maintain concentration and encode the information efficiently. Once the input phase begins, digits must be entered without the possibility of correction, encouraging deliberate attention and self-monitoring.

This exercise actively trains several core cognitive abilities: working memory, by requiring the temporary storage and transformation of information; sustained attention, to remain focused throughout the task; focused attention, to capture and recall each digit accurately; and cognitive manipulation, to reverse the sequence internally.

5.2.5 Code of Symbols

The *Code of Symbols* task — *Código de Símbolos* — (Figure 47) was also inspired by a clinical exercise described in the work of Lima (2015). In this activity, users are shown a reference code that links a set of colored symbols to specific numbers. After memorizing these associations, they begin the task by clicking a start button, which triggers a timed challenge where symbols appear in random positions within a grid. The user's goal is to recall the correct number associated with each symbol and input it accordingly. The structure of the grid and the number of unique symbol-number pairs vary with difficulty: the easy level uses 4 unique symbols in a 4x4 grid, the medium level uses 6 symbols in a 5x5 grid, and the hard level uses 8 symbols in a 6x6 grid. Once the task begins, the user has 45 seconds to complete as much of the grid as possible, reinforcing the need for both speed and precision.

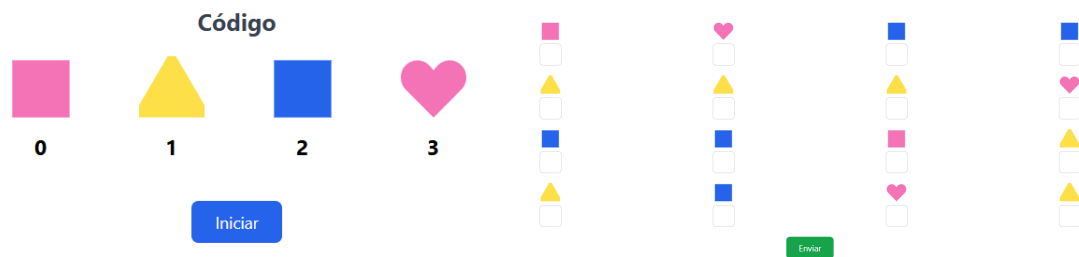


Figure 47: Code of Symbols Task

This task engages multiple cognitive processes. It strengthens working memory through the memorization and retrieval of symbol-number associations, supports sustained attention over the duration of the timed session, exercises visual attention as users scan and interpret the randomized layout, and enhances reaction speed by encouraging quick and accurate responses under time constraints.

5.2.6 Stroop

The *Stroop* task (Figure 48) is based on the well-known cognitive interference test introduced in psychological literature, and it follows the principles discussed in Scarpina and Tagini (2017). This classic task is designed to challenge the brain's ability to manage conflicting information. In each round, users are presented with a word that denotes a color but is displayed in a different font color. After the brief presentation, users are prompted to identify either the word itself or the color it was shown in, depending on the instruction.

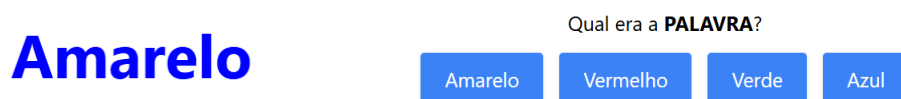


Figure 48: Stroop Task

The difficulty level affects the duration of the word's display, with shorter exposure times at higher levels. This increases the cognitive demand placed on the user, requiring quick perception, decision-making, and inhibition of automatic responses. For example, identifying the font color rather than reading the word directly involves suppressing a dominant reading response, a key aspect of inhibitory control.

The Stroop task exercises a range of executive functions. It promotes selective attention by requiring users to focus on one specific attribute while ignoring another; trains response inhibition, as automatic reading

must be suppressed when focusing on color; enhances reaction speed by encouraging fast and accurate replies; and supports cognitive flexibility, as users must adapt to changing instructions across different trials.

5.2.7 Follow the Figure

The *Follow the Figure* task — *Segue a Figura* — (Figure 49) was inspired by a bonus minigame from the *Super Mario Bros* series called *Wanted*, in which players must find a specific character hidden among others. This task adapts that core idea into a cognitively demanding activity by replacing characters with colored geometric shapes and embedding the concept into a focused cognitive training structure. Its development was also informed by research such as Baniqued et al. (2014), which emphasizes how well-designed videogames and videogame-like tasks can enhance attention and memory performance.

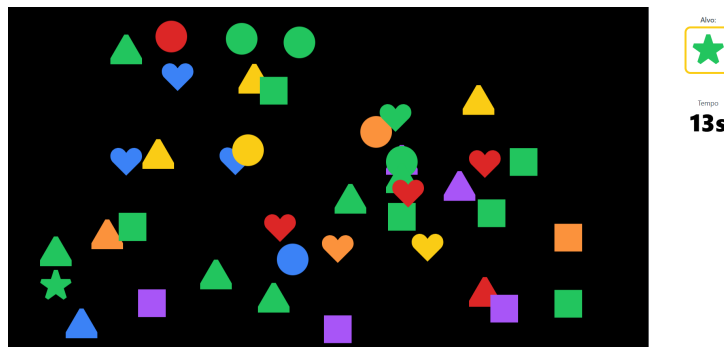


Figure 49: Follow the Figure Task

In each round, a target shape with a unique color and form is shown to the user. The screen is then filled with numerous distractor figures, shapes that vary in color and type, randomly positioned on the screen. The user must visually scan the environment and click on the correct figure as quickly as possible. The task progresses through 20 rounds or ends when the available time expires, maintaining a continuous cognitive demand throughout the session.

This exercise is particularly effective for training selective attention, as users must filter relevant from irrelevant stimuli under time pressure. It also promotes sustained and visual attention, requiring consistent engagement and active visual searching. Lastly, by prompting quick and accurate responses in a cluttered space, it reinforces reaction speed—making *Segue a Figura* is a tool for strengthening core attentional processes.

5.2.8 Simon

The *Simon* task (Figure 50) is inspired by the well-known electronic memory game and is supported by findings in the literature such as those presented in Raja et al. (2023), which emphasize its effectiveness in training attention and memory-related functions. This activity presents users with a growing sequence of colored buttons. The objective is to reproduce the sequence in the correct order, relying on short-term memory and perceptual recall.

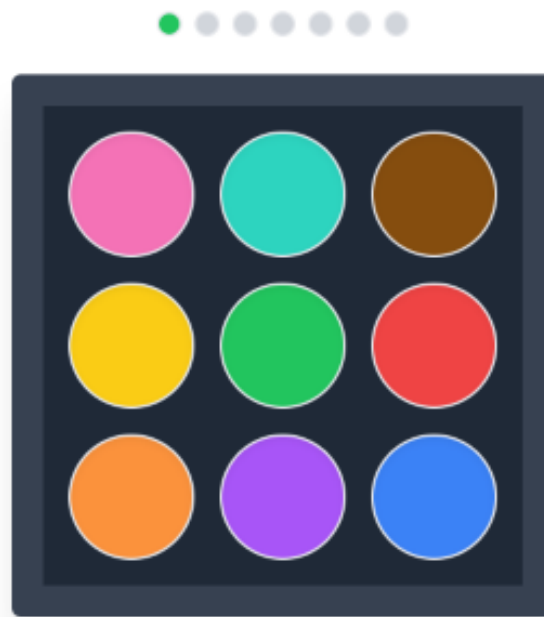


Figure 50: Simon Task

The difficulty level defines the number of colored buttons available (4 for easy, 6 for medium, and 9 for hard) directly influencing the visual complexity and cognitive load. Each correct response extends the sequence by one element, up to a maximum of seven repetitions (of course therapist can create a configuration where this number is different). The task is time-limited to two minutes, and any mistake causes the current sequence to reset, while also registering an error.

This exercise engages several key cognitive domains. It enhances working memory by requiring the user to store and reproduce dynamic sequences, fosters sustained attention over the course of multiple rounds, and promotes visual attention and reaction speed under time constraints. By blending simple mechanics with escalating difficulty, *Simon* provides an effective and engaging way to strengthen core executive skills.

5.2.9 Searching for a Vowel

The *Searching for a Vowel* task — *Procurar a Vogal* — (Figure 51) was developed as a direct follow-up to the Searching for an Answer exercise. While the underlying concept and layout remain essentially the same (locating a target among several distractors) the key difference lies in the interaction method and the nature of the stimuli. This variation was introduced after internal discussions highlighted the potential value of adding a visuomotor coordination component to the activity.

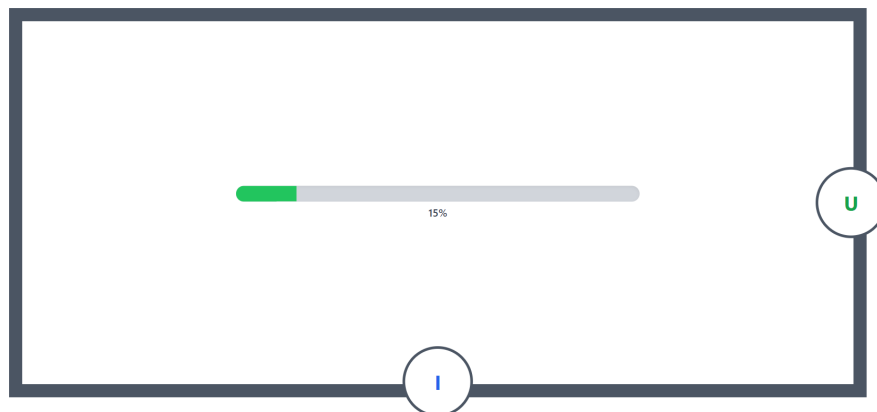


Figure 51: Searching for a Vowel Task

Instead of using arrow keys to indicate the position of the target, users must now identify and click directly on the correct vowel using the mouse. This shift in input modality adds a layer of motor planning and visuomotor integration to the task, engaging users not only cognitively but also physically through coordinated hand-eye response. Visually, the setup remains similar, with vowels replacing the colored figures and distractors occupying analogous positions across the screen.

This task continues to train selective and sustained attention, visual scanning, spatial discrimination, and reaction speed. However, by incorporating direct manual interaction, it also reinforces visuomotor coordination, an important skill in both academic and daily functional contexts. *Procurar a Vogal* thus expands the original exercise's scope, offering a complementary approach to training cognitive and perceptual-motor abilities.

5.2.10 Order Animals

The *Order Animals* task — *Ordenar os Animais* — (Figure 52) was developed with the goal of combining memory and attention training with a more playful and visually appealing format, particularly suited for younger users and was inspired by a visit to the *Centro Terapêutico ConSigo de Braga*. To increase

engagement and intuitiveness, the task incorporates drag-and-drop functionality, allowing users to interact directly with familiar animal figures.



Figure 52: Order Animals Task

In this exercise, a series of animals is presented one at a time across several rounds. After the presentation phase, the user is challenged to recall and reconstruct the original sequence by dragging the animals into the correct order. The number of animals shown per round depends on the selected difficulty level, with higher levels increasing the memory load.

This task promotes the development of several essential cognitive skills: focused attention, to observe and register each animal accurately; sustained attention, to remain engaged throughout the rounds; visual attention, to differentiate and locate the animals during the recall phase; and working memory, to retain and mentally organize the sequence.

5.2.11 Odd One Out

The “Odd One Out” task — *O diferente* — (Figure 53) was inspired by a visit to the *Centro NeuroSensorial de Braga*. In this task, users are presented with a grid containing a series of visually similar characters with only one character differing subtly from the rest. The user’s objective is to identify and click on this different character in the middle of the pack.

E	E	E	E	E	E
E	E	E	E	E	E
E	E	E	E	E	E
F	E	E	E	E	E
E	E	E	E	E	E
E	E	E	E	E	E

[Desisto / Seguir em frente](#)

Figure 53: Odd One Out Task

The difficulty of the task is dynamically adjusted based on the user's performance: correctly identifying the odd character may result in an increased grid size in the following round, while incorrect or missed attempts can reduce the complexity. This adaptiveness helps keep the challenge balanced and engaging. The task is designed to train several cognitive abilities, including visual attention, focused attention, visual discrimination, and selective attention, as users must carefully scan and compare multiple stimuli to detect small differences under time pressure. Its simplicity and flexibility make it suitable for a wide range of users, from children to adults.

5.3 Summary

This chapter provided a guided walkthrough of the **BEAM** platform, detailing its structure, user roles, and core functionalities. From registration to patient management, from performing tasks to analyzing results, the system offers a cohesive and role-aware interface designed to support both cognitive training and clinical oversight.

The task system forms the heart of the platform, offering a diverse set of scientifically grounded cognitive exercises targeting key domains such as attention, memory, inhibition, and flexibility. Each task is designed not only to challenge but also to adapt, through difficulty levels and customizable parameters, allowing therapists to fine-tune sessions according to individual needs. This balance between structure and adaptability ensures that **BEAM** remains relevant across different therapeutic contexts and user profiles. Complementing the exercises is a robust results system that supports performance monitoring from multi-

ple angles: per user, per task, per cognitive category, or general statistics. Visual graphs, sortable tables, and personalized notes allow therapists to combine data-driven insights with clinical judgment, helping to shape effective and responsive intervention strategies.

In summary, the **BEAM** platform combines cognitive studies, usability, and personalization in a comprehensive digital tool. By offering a structured yet flexible training environment, supported by clear analytics and therapist-led customization, it empowers users to engage in meaningful cognitive development while enabling professionals to guide and assess progress with precision.

Chapter 6

Testing and Evaluation

This chapter presents the strategies used to test and evaluate the **BEAM** platform. It outlines the different phases of validation, therapist feedback, and patient trials. The focus is on assessing both technical stability and user experience, with the goal of ensuring that the platform is functional, intuitive, and well-adapted to the needs of its target users. By detailing this process, the chapter highlights the steps taken to validate the platform's effectiveness and reliability in real-world settings.

As previously mentioned, the development of the platform benefited from continuous collaboration with therapists from different clinical institutions. Throughout the process, feedback was collected not only on the design and usability of the interface but also on the therapeutic relevance of the implemented features and exercises. This close involvement helped refine the platform to better suit clinical workflows and cognitive goals. Fortunately, two partner institutions, *Centro Terapêutico Consigo* and *Centro Neurosensorial*, agreed to conduct initial testing with their own patients. These trials involved real interactions with the platform's exercises.

Thanks to the support of these two institutions and additional individuals who were contacted throughout the project, it was possible to conduct a substantial number of real interactions with the platform. At the time of writing, over 200 exercises have been completed by users, providing valuable data on performance and engagement. Participants spanned a wide range of age groups — from as young as 6 years old to older adults aged 70 and above — although the majority of completed exercises were performed by children, adolescents, and young adults. This diversity proved especially valuable, as **AT** is beneficial across all age groups, and having access to such a broad range of participants allowed us to have more diverse data. To better understand how the platform performs across different age groups, three separate charts were analyzed. Figure 54 displays the average accuracy per task for three distinct age ranges: users aged 60 and above (left), users aged 7 to 12 (center), and users of all age groups combined (right).

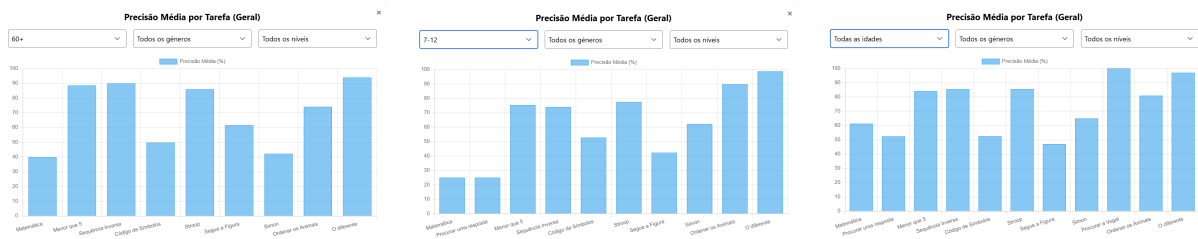


Figure 54: Average Task Accuracy by Age Group: 60+ (left), 7–12 (center), and All Ages (right).

From the data shown in Figure 54, accuracy levels in the general population are generally higher than those seen in younger children or older adults. This finding aligns with existing literature, which highlights that children are still developing key cognitive abilities, while older adults may face age-related cognitive decline, both of which can impact attention and memory.

At the time of writing, all three groups had completed the majority of tasks, with the only exception being the 60+ group, who did not attempt the *Procurar a Vogal* task. Despite this, some clear trends emerge: for example, the task *O Diferente* consistently achieved high accuracy across all three groups, suggesting it is widely accessible. Conversely, tasks like *Matemática* and *Procurar uma Resposta* showed lower average accuracy, particularly among the 7–12 age group, potentially due to their higher complexity and required cognitive effort.

The *Sequência Inversa* and *Stroop* tasks were relatively successful among older users, both achieving over 85% accuracy. The *Simon* task, however, showed significantly lower performance in this group, possibly reflecting the difficulty of sequential memory under time constraints. Meanwhile, in the 7–12 age range, tasks like *Ordenar os Animais* and *O Diferente* performed well, while *Matemática* and *Segue a Figura* presented greater challenges.

This diverse set of results helps reinforce the importance of a flexible and age-inclusive approach to cognitive training. The differences in accuracy across tasks and demographics demonstrate the platform's potential to reach and adapt to users with varying cognitive profiles and developmental stages.

In addition to the analysis by age group and task, it is also relevant to examine user performance by cognitive category. Among the attention-related categories, Focused Attention currently presents the highest average accuracy at 84.9%, indicating that users generally perform well when the task requires concentration on a single stimulus. Selective Attention follows with an accuracy of 75.6%, showing slightly more difficulty but still a decent result. Sustained Attention and Visual Attention present more challenges for users, with average accuracies of 67.4% and 67.9%, respectively. These lower values may reflect the greater cognitive demands of maintaining attention over time or processing multiple visual stimuli, aligning with what is often reported in the literature regarding attention fatigue and visual overload.

While the previous analysis summarized overall performance across cognitive categories, a more detailed examination of individual tasks can provide deeper insights into user interaction and task-specific challenges. Analyzing all 11 cognitive exercises implemented in the platform would be too extensive for this section; therefore, four tasks, *Segue a Figura*, *Código de Símbolos*, *Stroop*, and *O Diferente*, were selected among the most frequently executed on the platform. These exercises have become therapist favorites due to their relevance to core cognitive skills such as attention, visual scanning, and inhibition. Interestingly, some of them are also among the most challenging for patients, offering valuable insights into user performance and task design.



Figure 55: Results for the Follow the Figure task

In the Follow the Figure task (Figure 55), the average accuracy is relatively low at 47.1%, making it the most challenging exercise on the platform in terms of correctness. With a mean reaction time of 3.91 seconds, user interaction is decently paced, yet the true complexity lies in the task's core mechanics. Users must visually scan a grid filled with similar shapes to identify the exact target figure, demanding sharp visual attention, rapid recognition, and quick motor responses, all within a short time window that can even decrease with wrong answers adding to the pressure. Despite its difficulty, it has proven highly engaging, especially among younger users, who often attempt it multiple times — not necessarily to achieve perfection, but to improve, beat their previous results, or simply enjoy the challenge. This replayability factor makes it both a diagnostic and motivational tool. As such, Follow the Figure exemplifies how cognitively demanding tasks, when presented in an intuitive and engaging format, can foster intrinsic motivation and

sustained participation, even in users who struggle with initial performance.

Resultados da Tarefa: Código de Símbolos



Figure 56: Results for the Code of Symbols task

The Code of Symbols task (Figure 56) also displays a relatively low average accuracy of 52.4%, placing it among the most challenging exercises for users. While the mean reaction time is strikingly high at 41.12 seconds, this is partly due to the task's structure: the timer only stops once the user has completed all symbol pairs and submitted the response. However the task itself demands considerable cognitive effort. Users must memorize a sequence of symbol–number associations and then accurately apply that knowledge under pressure. This dual challenge, of memory encoding and timed recall, makes the task cognitively demanding, explaining both the elevated reaction times and the lower overall performance. Additionally, during testing sessions, it was observed that many patients often chose to repeat this task immediately after their first attempt, aiming to either improve their score or maintain their previous performance. This behavior suggests that, despite its difficulty, the exercise remains highly engaging and motivating, a quality that is especially valuable in sustained cognitive training.

Resultados da Tarefa: Stroop



Figure 57: Results for the Stroop task

In contrast with the previous two tasks, the Stroop task (Figure 57) stands out from the other two with a considerably higher average accuracy of 85.6% and a much faster mean reaction time of 2.15 seconds. This exercise is a clear fan favorite among therapists, likely due to its status as a well-established classic in cognitive science. Its popularity may stem from its strong scientific foundations and intuitive design, making it a reliable choice for training and evaluating inhibitory control. The task effectively holds users' attention from beginning to end, combining simplicity in format with cognitive depth. Despite targeting a complex function, the ability to suppress automatic responses, users tend to perform with high precision and consistency, which only improves with continued training. This promising outcome suggests that the Stroop task strikes an effective balance between challenge and accessibility, offering both engaging interaction and meaningful performance insights. Its proven relevance in clinical and research contexts further reinforces its value as a core component of the platform's training suite.

Resultados da Tarefa: O diferente



Figure 58: Results for the Odd-One-Out task

The Odd-One-Out task (Figure 58) is notable for its exceptionally high average accuracy of 97.1%, placing it among the best-performing tasks on the platform in terms of correctness. Its consistent success rate has made it a fan favorite, both for its simplicity and for the clarity it offers in assessing attention and visual discrimination. However, unlike other tasks, the key differentiator in evaluating user progress here lies not in accuracy, but in reaction time. Because users have unlimited time to identify the item that differs from the others, performance differences often manifest in the speed of execution rather than in correctness. This is evident in the results table: although all users achieved perfect accuracy, their reaction times vary considerably. This task highlights an important insight: precision alone does not always tell the full story of user performance. In tasks where a performance ceiling is reached in accuracy, reaction time becomes a crucial metric for assessing cognitive development. Furthermore, it was observed during testing that many users chose to repeat this task voluntarily, aiming to improve their speed and track their own progress. This self-motivated repetition, paired with the engaging nature of the activity, reinforces its value as a useful and enjoyable cognitive challenge.

Overall, the data from these four tasks demonstrate a balance between cognitive challenge and usability. While two of them are among the hardest for patients in terms of accuracy, they also provide therapists with valuable data for assessing and training attention-related skills. These results reinforce the need for diverse task formats that can both challenge users and inform individualized intervention strategies.

To further illustrate the analytical capabilities of the platform, a detailed case study of an individual patient

is presented. This analysis showcases how therapists can gain insights by examining specific patterns in both accuracy and reaction time, filtered by age group, task, or difficulty level.

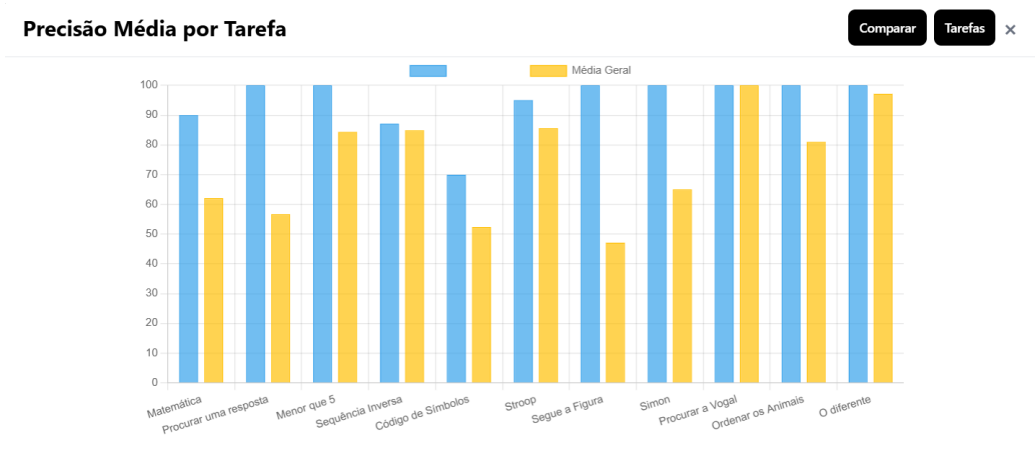
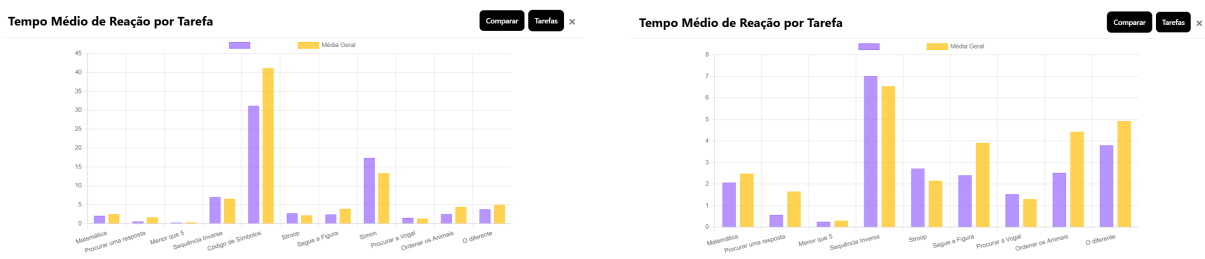


Figure 59: Overall accuracy per task for a 14-year-old patient

Figure 59 presents the average precision per task for a 14-year-old patient. When compared with the general population, this patient stands out with an above-average accuracy across most tasks. The ability to compare individual performance with platform-wide averages or specific age groups allows therapists to contextualize achievements and identify standout areas of competence. For instance, the patient performs notably well in tasks such as *Matemática* and *Stroop*, where their results exceed the general average by a significant margin. This highlights cognitive strengths, particularly in inhibition and arithmetic processing. Conversely, the patient’s performance on tasks like *Procurar uma Resposta* and *Segue a Figura* aligns with broader trends, as these tasks are generally more difficult across all users. This indicates that the patient’s challenges are not isolated, and such patterns can inform focused intervention strategies.



Without Task Filters Applied

With Task Filters Applied

Figure 60: Comparison of precision across tasks with and without filtering

The analysis of reaction times adds an important dimension to the understanding of user performance, particularly when precision alone may not tell the whole story. As shown in Figure 60, this patient maintains

high accuracy across most tasks; however, reaction time analysis reveals a more nuanced picture. For example, while the patient performs significantly above the general average in terms of accuracy in tasks like *Segue a Figura* and *Procurar uma Resposta*, their reaction times in these same tasks remain slightly above the global mean. This suggests that although the user is precise, the underlying cognitive processing speed or visual search efficiency may benefit from further improvement. Such insights highlight how accuracy and speed do not always evolve in tandem, and both should be considered when assessing cognitive performance. One particularly notable strength is observed in the *Código de Símbolos* task. Despite being one of the most time-consuming exercises for the general population, with an average reaction time exceeding 40 seconds as already mentioned, the patient completes it considerably faster than average. This strong result suggests an above-average ability in processing symbolic patterns and maintaining attention under repetitive demands. The large discrepancies in reaction time between tasks highlight just how essential the platform's dynamic filtering can be. By temporarily excluding exercises like *Simon* and *Código de Símbolos*, which naturally take longer due to their structure, therapists can reveal more meaningful differences across the remaining tasks. In this patient's case, once these longer tasks are filtered out (Figure 60, right), it becomes easier to detect subtle delays in otherwise strong performances, where accuracy is high but reaction time remains above average. This filtering functionality is not merely cosmetic; it allows a clearer interpretation of performance trends, helping professionals focus their attention on exercises that may require further training or cognitive support.

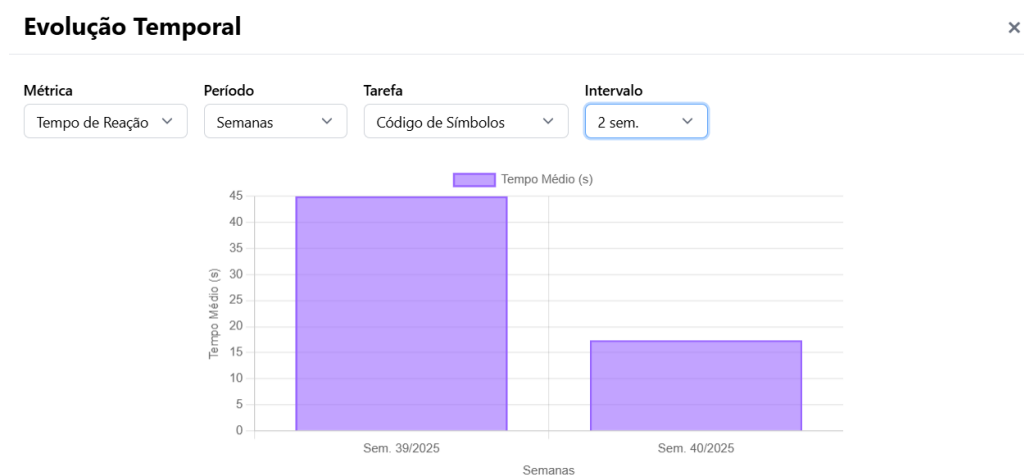


Figure 61: Temporal evolution of performance in the *Código de Símbolos* task

Moreover, the platform offers dedicated tools to visualize temporal evolution in both precision and reaction time, as illustrated in Figure 61. These visualizations enable therapists to track progress over various intervals, including days, weeks, or months, at both global and task-specific levels. In this case, a marked

reduction in the average reaction time for *Código de Símbolos* over a two week period provides strong evidence of the patient's improved processing speed and task familiarity which explains his lower average shown in Figure 60. Such evolution demonstrates the real impact of consistent training, and the importance of capturing longitudinal trends rather than relying solely on one-time snapshots. Ultimately, this multi-layered approach, combining performance comparison, filtering tools, and temporal analysis, offers therapists a comprehensive and adaptable framework to evaluate user progress in depth.

Overall, these tools demonstrate the platform's potential not just for evaluating cognitive performance, but also for guiding personalized interventions based on detailed, interpretable metrics.

The data collected through the platform so far offer a valuable overview of user performance across different cognitive dimensions. These results are based on a diverse and meaningful sample size, which allows for the extraction of insightful conclusions that, in many cases, align with established findings in cognitive research. However, the dataset still reflects a relatively short period of active use. As a result, there are not yet sufficient longitudinal data to evaluate the evolution of user performance over time, particularly regarding individual progression after several months of regular training. Such long-term tracking would not only help validate the platform's effectiveness but also provide therapists with individualized growth indicators. Although this chapter has already demonstrated how temporal visualizations can assist in tracking changes in precision or reaction time, truly robust longitudinal evaluation will only become possible as users accumulate more months of regular engagement. Despite this current limitation, early feedback indicates that users enjoyed the games they played, and it is reasonable to expect that their accuracy and performance would improve with continued use. In addition to the metrics already discussed, a **usability questionnaire** was made available to both therapists and patients using the platform (see Appendix 7.1). This allows them to provide evaluations, corrections, and general feedback. While this channel will be important for future qualitative insights, no detailed analysis of the responses is included here, as the number of submissions at the time of writing was not yet sufficient to draw meaningful conclusions.

Chapter 7

Conclusion

This document marks the conclusion of a Master's project that aimed to develop **BEAM**, a web-based platform designed to support attention training through interactive cognitive exercises. The platform is now fully implemented and live, offering therapists a wide range of tools for assigning, configuring, and monitoring training activities tailored to individual users.

In Chapter 1, the topic of this Master's work was introduced, the research hypothesis was defined, and the methodological approach was outlined. These elements established the foundation for the work, guiding both the theoretical exploration and the technical development that followed.

Chapter 2 provided an in-depth review of the project's domain, focusing on attention as a cognitive process. It examined different types of attention, associated disorders, and the increasing challenges posed by modern society. The chapter also explored the neurological underpinnings of attention and reviewed a variety of **Attention Training** methods and exercises. This literature review was essential in shaping the design decisions and ensuring that the platform would be based on sound cognitive principles.

Chapter 3 presented the initial system proposal. It detailed the application's domain model, use case diagrams, requirements, user interface mockups, system architecture, and the technologies selected for implementation. These components were informed by the research conducted earlier and served as a blueprint for the platform's development. The objectives and requirements defined at this stage were almost entirely fulfilled in the final implementation. The only feature proposed that was not carried forward was the ability to export results to PDF. After further consideration, this requirement was excluded, as it was determined to provide limited practical benefit and was unlikely to see significant usage by the intended audience.

Chapter 4 detailed the implementation process, while Chapter 5 served as a navigational map of the platform's various screens and features. Chapter 6 offered an in-depth look at how the platform was validated in real-world settings, with strong collaboration from therapists and clinical institutions. Over 160 exercises were completed by users across a broad age range, generating data that enabled the analysis of

accuracy by task, cognitive category, and demographic group. These evaluations not only confirmed the platform's usability and stability but also revealed performance patterns that align with known cognitive development stages, reinforcing the clinical relevance of the system.

The entire project followed a structured plan composed of five key stages: (1) bibliographic review and state-of-the-art analysis, (2) definition of the proposed solution and selection of exercises and technologies, (3) system design, (4) implementation of the system and its exercises, and (5) testing, refinement, and finalization. As shown in Figure 62, the initial GANTT chart served as a scheduling guide, and the work was completed successfully within the planned time frame.



Figure 62: Project Schedule

The resulting platform **BEAM** is now publicly accessible and continues to evolve based on user feedback. It has already been tested by therapists and patients in real-world clinical settings, and preliminary statistical data has been collected to evaluate user performance. With over 200 completed exercises at the time of writing, the system has begun to demonstrate its effectiveness and adaptability.

As a final product, **BEAM** addresses a clear need in cognitive training: a free, accessible, and customizable platform that can be used by therapists working with individuals across all age groups. The work developed through out this academic year devoted to the realization of the Master's Project demonstrates that it is possible to combine scientific knowledge, technical development, and clinical insight to build meaningful digital tools for attention enhancement.

Major Contributions. This Master's project resulted in three key contributions that represent its scientific, technical, and practical impact:

- **Development of the BEAM Platform:** a fully functional, web-based system designed to enhance attention through interactive exercises, including task personalization, therapist management tools, and performance analytics. The platform is publicly accessible at beam.epl.di.uminho.pt.
- **Creation of the BEAM Landing Page:** a public-facing landing page that summarizes the project's goals and showcases the platform's accessibility for therapists and institutions interested in exploring or adopting the solution.

- **Scientific Dissemination:** preparation and submission of the peer-reviewed article *BEAM: Applying Web Technologies for Training Attention* to the *AC'2025 Conference*¹, where the **BEAM** platform will be presented, contributing to the academic and professional dissemination of the project's results.

Thesis Statement. The research hypothesis stated in the introduction proposed that it is possible to effectively train attention through targeted cognitive challenges and exercises delivered in an engaging digital environment. This work has successfully demonstrated that hypothesis, confirming that a carefully designed web platform, integrating scientifically grounded exercises, user-centered design, and therapist-driven personalization, can indeed serve as an effective and accessible tool for cognitive training. This validation transforms the original hypothesis into the central thesis of this dissertation.

Overall, the project realized reflects a successful integration of theoretical research, clinical insight, and practical implementation. The platform was designed, developed, and validated within the planned time frame, and it now stands as a solid foundation for future improvements and broader clinical use. With encouraging initial results, strong interest from professionals, and several opportunities for growth, **BEAM** is well-positioned to continue evolving and making a meaningful impact in the field of cognitive training.

7.1 Future Work

Although great progress was made during the academic year, and the platform reached a stable and fully usable version, there are several promising directions for future development. Due to the limited time inherent to a one year Master's project, certain features and ideas were not pursued during this phase.

One such idea is the integration of eye-tracking technologies. During both the literature review and visits to clinics, this feature was identified as a potentially powerful tool for evaluating user focus and attention allocation. It could open new dimensions in the measurement of user engagement, particularly for tasks requiring visual scanning and sustained attention.

Another area with promising potential lies in the application of artificial intelligence. While not a current part of the platform, AI has been considered as a useful enhancement to assist therapists in interpreting results and identifying trends or anomalies in patient performance. This could eventually lead to more personalized recommendations and automated insights, making the platform even more effective for cognitive assessment and intervention.

¹ <https://www.computing-conf.org/>

The last point of future exploration involves long-term monitoring of users' cognitive progress. While the current dataset provides strong insights, the short duration of the testing phase limits the ability to measure individual growth over time. In particular, it would be beneficial to continue working with younger users to evaluate how consistent training impacts their performance across weeks or months. This longitudinal analysis could help therapists assess patient evolution more precisely and would provide stronger evidence of the platform's effectiveness as a training tool.

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Appendices

Usability Questionnaire

This appendix includes the full version of the usability questionnaire shared with platform users.

Questionário de Usabilidade da Plataforma BEAM

Este questionário tem como objetivo recolher a sua opinião sobre a experiência de utilização da plataforma **BEAM**.

As suas respostas são anónimas e ajudarão a melhorar a usabilidade e a qualidade dos exercícios e para também relatar na dissertação

* Indica uma pergunta obrigatória

1. É terapeuta ou paciente na plataforma? *

Marcar apenas uma oval.

☐ Terapeuta *Avançar para a pergunta 13*

☐ Paciente *Avançar para a pergunta 2*

Paciente

Esta secção destina-se aos utilizadores que realizaram os exercícios na plataforma BEAM.

O objetivo é compreender a sua experiência de utilização, incluindo a clareza das instruções, a facilidade de interação e a motivação sentida durante os exercícios.

As respostas ajudarão a melhorar a experiência do utilizador e a usabilidade geral da plataforma.

2. Foi fácil compreender como utilizar a plataforma na primeira vez que a experimentou? *

Marcar apenas uma oval.

1 2 3 4 5

Nada fácil ☐ ☐ ☐ ☐ ☐ Muito fácil

3. As instruções antes de cada exercício foram claras e suficientes para começar? *

Marcar apenas uma oval.

☐ Sim

☐ Não

☐ Às vezes

4. Comentários (opcional)

5. Como avalia a facilidade de navegação entre diferentes exercícios e resultados? *

Marcar apenas uma oval.

1 2 3 4 5

Muito Difícil ☐ ☐ ☐ ☐ ☐ Muito fácil

6. Quando conclui um exercício, recebeu feedback claro sobre o seu desempenho? *

Marcar apenas uma oval.

- ☐ Sim
- ☐ Não
- ☐ Às vezes

7. Comentários (opcional)

8. Sentiu-se confortável a interagir com os exercícios (ex: cliques, arrastar, tempo disponível)? *

Marcar apenas uma oval.

1 2 3 4 5

Muito Difícil 1 ☐ ☐ ☐ ☐ ☐ Muito fácil

9. Os exercícios pareceram-lhe motivadores e interessantes de realizar? *

Marcar apenas uma oval.

- ☐ Sim
☐ Não
☐ Às vezes

10. Sentiu que o tempo disponível em cada exercício era adequado? *

Marcar apenas uma oval.

- ☐ Sim
☐ Não
☐ Às vezes

11. Comentários (opcional)

12. Se pudesse alterar algo na plataforma BEAM para a tornar melhor para si, o que mudaria?

Terapeuta

Esta secção é direccionada aos terapeutas que utilizam a plataforma BEAM com os seus pacientes. Pretende recolher opiniões sobre a facilidade de uso, análise de resultados, acompanhamento de progresso e configuração de exercícios personalizados. As respostas permitirão otimizar as funcionalidades da plataforma e o apoio ao trabalho clínico.

13. Foi fácil compreender como utilizar a plataforma e atribuir tarefas aos pacientes? *

Marcar apenas uma oval.

1 2 3 4 5

Nada fácil ☐ ☐ ☐ ☐ ☐ Muito fácil

14. Comentários (opcional)

15. Considera que as opções disponíveis (nível de dificuldade, treino/teste, configurações personalizadas) são suficientes para o seu trabalho clínico? *

Marcar apenas uma oval.

- ☐ Sim
- ☐ Não
- ☐ Em parte

16. Se respondeu "Não" ou "Em parte" à pergunta anterior, sugira opções que podiam enriquecer a plataforma.

17. Como avalia a clareza da informação apresentada nos resultados dos pacientes? *

Marcar apenas uma oval.

1 2 3 4 5

Nada fácil ☐ ☐ ☐ ☐ ☐ Muito fácil

18. Acha que a plataforma facilita o acompanhamento da evolução dos pacientes ao longo do tempo? *

Marcar apenas uma oval.

- ☐ Sim
- ☐ Não
- ☐ Em parte

19. Comentários (opcional)

20. Como avalia a facilidade de utilização das funcionalidades de configuração de exercícios (ex: tempos, número de rondas, parâmetros de dificuldade)?

*

Marcar apenas uma oval.

1 2 3 4 5

Nada fácil ☐ ☐ ☐ ☐ ☐ Muito fácil

21. Considera que a criação e edição de configurações personalizadas é intuitiva para o planeamento terapêutico?

*

Marcar apenas uma oval.

1 2 3 4 5

Nada intuitiva ☐ ☐ ☐ ☐ ☐ Muito intuitiva

22. De modo geral, a plataforma facilita o seu trabalho terapêutico? *

Marcar apenas uma oval.

1 2 3 4 5

Nada ☐ ☐ ☐ ☐ ☐ Muito

23. Se pudesse alterar algo na plataforma BEAM para a tornar melhor para si, o que mudaria?
